# Advanced Microeconomics Theory

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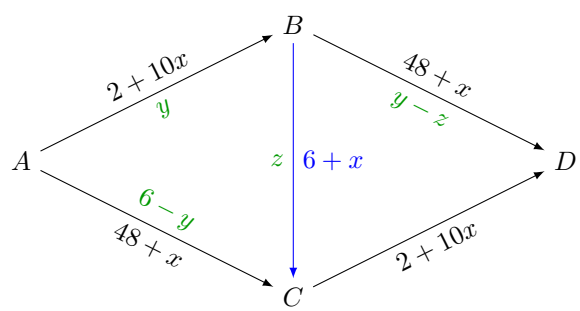
# Contents

# Chapter 1

# Introduction

## 1.1 Braess' Paradox




The core idea to solve for equilibrium is that, in equilibrium costs (the time spent on each possible route from $A$ to $D$) on all paths must be equal. So one can show that before the shortcut is connected, the equilibrium travel time for any driver is 83 minutes. After the shortcut is built, the new equilibrium travel time becomes 92, which is actually greater than the original one.

In the example, although the network capacity increased by adding the shortcut, the individual cost for every agent increased. This is due to the negative externality: when an agent switches to the shortcut, they reduce their own time marginally but impose a heavy congestion cost on the shared links $AB$ and $CD$. This cost is not internalized by the selfish agent.

**Remark.**

- In a multi-agent situation, having more choices can possibly make you (and even everyone) worse off.
- However in single-person situations, having more choices will make you at least as well off.

## 1.2 Coin Toss

Agent A announces her guess first, and then B announces her guess after observing A's guess. If both agents make the same guess, then both agents get 1. Otherwise, nature determines the winner by coin toss. Whoever's guess matches the coin toss wins 4 and the other gets 0.

| Coin | $A$ | $B$ | Payoff |
|---|---|---|---|
| $H$ | $H$ | $H$ | $(1,1)$ |
| $T$ | $H$ | $H$ | $(1,1)$ |
| $H$ | $T$ | $T$ | $(1,1)$ |
| $T$ | $T$ | $T$ | $(1,1)$ |
| $H$ | $H$ | $T$ | $(4,0)$ |
| $H$ | $T$ | $H$ | $(0,4)$ |
| $T$ | $H$ | $T$ | $(0,4)$ |
| $T$ | $T$ | $H$ | $(4,0)$ |

If the coin toss is unavailable to both of them, B will make a guess opposite to $A$'s as long as $B$ is not too risk averse. Hence the expected payoff is $(2,2)$.

If the coin toss is made available to $A$, and this rule is common knowledge, the best strategy for $A$ is to make a guess consistent with the actual coin toss, and $B$'s best response is to report the same as $A$. Hence both will get a certain payoff $(1,1)$.

**Remark.**

> Information can indeed have a negative value! In the example, if the revelation of coin toss to $A$ is made a common knowledge, this information can actually make $A$ worse off. However if the revelation is $A$'s private knowledge, then it must have nonnegative value.

## 1.3 Model of Herding

Suppose there are two restaurants in the town, $A$ and $B$. The common prior is that

$$\Pr(A > B) = p > \frac{1}{2},$$

where $A > B$ denotes the event that restaurant $A$ is better than $B$. This common prior states that everyone is inclined to choose $A$ if there is no other informative signal. Suppose there are two types of signals, $\alpha$ and $\beta$, such that

$$\Pr(\alpha | A > B) = q,$$
$$\Pr(\beta | B > A) = q,$$

where $q > \frac{1}{2}$ and $q > p$[1].

The $t$-th agent knows all $(t-1)$ prior agents' choices, but does not know their private signals.

Suppose the signal sequence is $(\alpha, \beta, \beta, \cdots, \beta)$.

[1] $q > p$ ensures that an agent observing a signal contrary to the prior will form a posterior belief that favors the signal over the prior.

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For Agent 1, she receives signal $\alpha$, which enhances the prior through Bayesian updating. Since $\Pr(A > B|\alpha) > \frac{1}{2}$, she chooses $A$.

For Agent 2, she deduces from Agent 1's choice to infer that Agent 1 must have received signal $\alpha$. But Agent 2's private signal is $\beta$. Nevertheless, the two signals cancel out[2]. Hence, the common prior dictates that Agent 2 should also choose $A$.

For Agent 3, she deduces that Agent 1 must have received signal $\alpha$ from Agent 1's choice. Regarding Agent 2, Agent 3 reasons as follows: if Agent 2 received $\alpha$, then Agent 2 would vote for $A$; if Agent 2 received $\beta$, then the two signals cancel out and Agent 2 still chooses $A$ by the prior. Therefore, Agent 2's choice is uninformative about the true state. Hence, Agent 3 only infers the informative signal $s_1 = \alpha$ from history, and combined with her own signal $s_3 = \beta$, the two signals cancel out, leading Agent 3 to also choose $A$.

The same logic applies to all the subsequent agents. Hence, all agents choose $A$ regardless of their private signals. This is the herding phenomenon.

**Remark.**

- The assumption that $q > p$ is important because it ensures that

$$\begin{aligned}\Pr(B > A|\beta) &= \frac{\Pr(\beta|B > A)\Pr(B > A)}{\Pr(\beta|B > A)\Pr(B > A) + \Pr(\beta|A > B)\Pr(A > B)} \\ &= \frac{q(1-p)}{q(1-p) + (1-q)p} > \frac{1}{2}.\end{aligned}$$

  Similarly, you can verify that $\Pr(A > B|\alpha) > \frac{1}{2}$.

  So this assumption implies that private signals are sufficiently informative to overturn the prior. Specifically, it ensures that an agent observing a signal contrary to the prior will form a posterior belief that favors the signal over the prior. Otherwise, the agent will be "blinded" by the prior.

- Even though the signal is strong enough to overturn the prior, the information cascade is not perfect. Starting from Agent 2, the information cascade breaks.

- Agent 2 only cares about maximizing their own payoff. If Agent 2 also cared about social welfare, they would choose according to their private signal (choose $B$ if they get $\beta$) to reveal their information to Agent 3 and subsequent agents. However, there is no individual incentive to do so, making this an information externality that leads to inefficiency.

[2]This can be rigorously verified using Bayes' rule:

$$\begin{aligned}\Pr(A > B|\alpha, \beta) &= \frac{\Pr(\alpha, \beta|A > B)\Pr(A > B)}{\Pr(\alpha, \beta|A > B)\Pr(A > B) + \Pr(\alpha, \beta|B > A)\Pr(B > A)} \\ &= \frac{q(1-q)\cdot p}{q(1-q)\cdot p + (1-q)q\cdot(1-p)} \\ &= p > \frac{1}{2}.\end{aligned}$$

Thus, observing signals $\alpha$ (inferred from Agent 1) and $\beta$ (private) leaves Agent 2's posterior identical to the prior, as if she had received no informative evidence.

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# Chapter 2

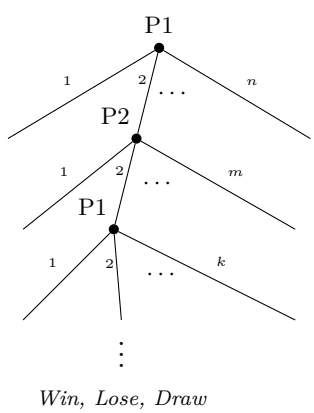
# Representing a Game

Typically we have two ways to represent a game:

- Extensive form (game tree)
- Normal form (payoff matrix)

## 2.1 Game Tree

**Example (Chess).**

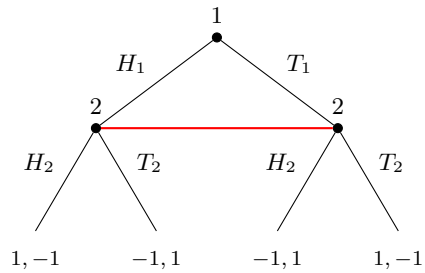



$$\begin{cases} \text{Nodes} \\ \text{Players} \\ \text{Choices at every node} \\ \text{Terminal nodes (outcome)} \\ \text{Information sets} \end{cases}$$

**Definition 2.1: Game with Perfect / Imperfect Information**

- A game with perfect information is a game where a player at node $n$ knows all the moves that lead to $n$.
- A game with imperfect information is a game where a player at node $n$ only knows some moves that lead to $n$.

**Example (Matching Pennies).**



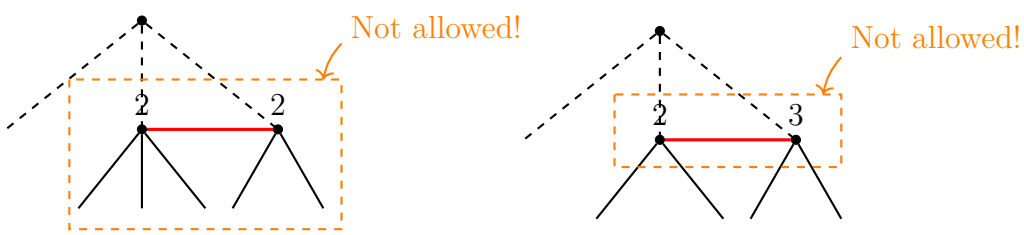

## 2.2 Information Sets

All the nodes of a game tree are partitioned into information sets.

### Rules for Information Sets

- Each node in an information set must belong to the same player.
- Each node in an information set must have the same choices.

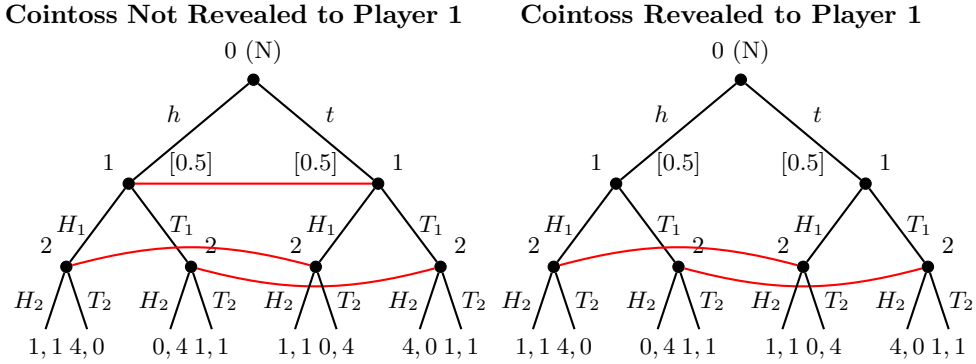
**Example.**




The first tree is invalid because the two nodes in the information set have different choices. The second tree is invalid because the two nodes in the information set belong to different players.

**Example (Coin Toss Revisited).**

**Cointoss Not Revealed to Player 1** **Cointoss Revealed to Player 1**

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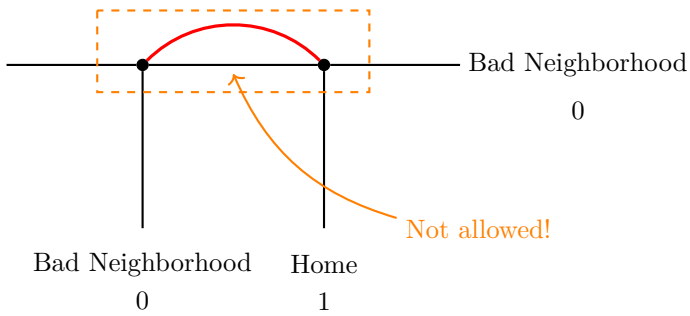
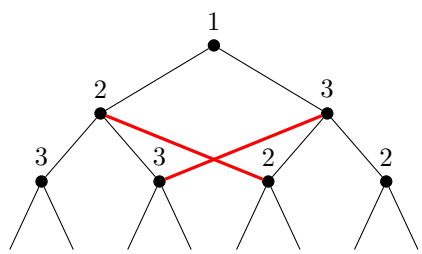
**Assumption 2.2**

Each path from the root to a terminal node intersects each information set at most once.

**Example (Absent-Minded Driver).**




**Remark ("No Good Timing").**




In this game, there is no good notion of "timing" or "stages". So the takeaway is that, the timing of players' moves is not important. What matters is the information structure, i.e., who knows what.

## 2.3 Normal Form

**Definition 2.3: Game**

A game $G$ of $n$ players consists of the set of strategies $S_i$ and utility functions $u_i : S_1 \times S_2 \times \cdots \times S_n \to \mathbb{R}$ for $i = 1, 2, \cdots, n$:

$$G = (S_1, S_2, \cdots, S_n; u_1, u_2, \cdots, u_n) \equiv (S_i, u_i)_{i=1}^n .$$

**Definition 2.4: Belief**

A belief of player $i$, denoted by $\mu_i \in \Delta(S_{-i})$, is a probability distribution over $S_1 \times \cdots \times S_{i-1} \times S_{i+1} \times \cdots \times S_n \equiv S_{-i}$.

**Remark.**

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A belief is a probability distribution over $S_{-i}$, i.e., $\mu_i \in \Delta_{-i}$. But typically,

$$\Delta(S_i \times S_j) \neq \Delta(S_i) \times \Delta(S_j).$$

**Example.**

Consider a game of 3 players where Player 2 has strategies $s_2$ and $s_2'$ to choose from, and Player 3 has $s_3$ and $s_3'$. From the Player 1's perspective, she can theoretically form the following two "beliefs":

| | $s_3$ | $s_3'$ |
|---|---|---|
| $s_2$ | 0 | 1/4 |
| $s_2'$ | 1/4 | 1/2 |

| | $s_3$ | $s_3'$ |
|---|---|---|
| $s_2$ | 1/9 | 2/9 |
| $s_2'$ | 4/9 | 4/9 |

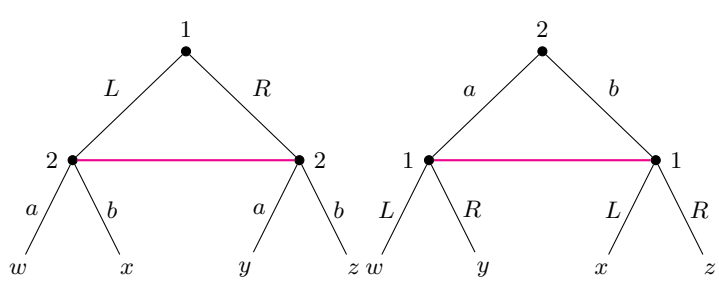
Note that the first belief belongs to $\Delta(S_2 \times S_3)$ but not $\Delta(S_2) \times \Delta(S_3)$, while the second belief belongs to both $\Delta(S_2 \times S_3)$ and $\Delta(S_2) \times \Delta(S_3)$.

Consider the game where Player 1 has strategy space $S_1 = \{s_1, s_1', s_1''\}$, and Player 2 has strategy space $S_2 = \{s_2, s_2'\}$. The normal form of this game is mapping from the strategy profiles to the payoffs of the players. In this example, it can be illustrated by a $3 \times 2$ matrix:

| | $s_2$ | $s_2'$ |
|---|---|---|
| $s_1$ | $\times$ | $\times$ |
| $s_1'$ | $\times$ | $\times$ |
| $s_1''$ | $\times$ | $\times$ |

**Remark (Relationship Between Game Trees and Normal Forms).**

- Every tree leads to a unique normal form.
- However, a normal form can lead to multiple game trees that are "equivalent".




| | $a$ | $b$ |
|---|---|---|
| $l$ | $w$ | $x$ |
| $r$ | $y$ | $z$ |

- Many game trees can lead to the same normal form.

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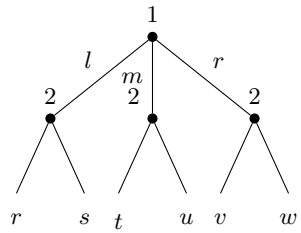
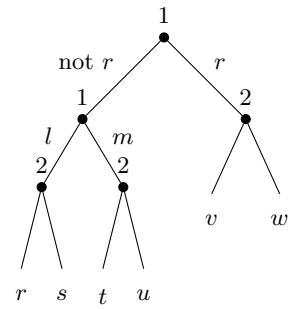


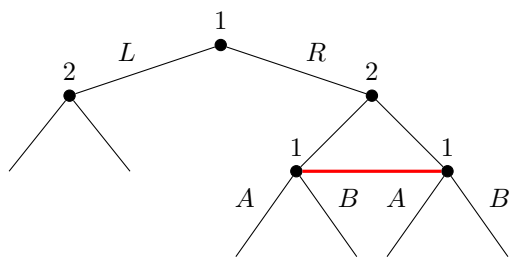

*Open Question: Do we lose some important information of the game by looking only at the normal form?*

## 2.4 Strategies

**Definition 2.5: Strategy**

The strategy for player $i$, denoted by $s_i$, is a complete plan of which choices to make at every one of her information sets.

**Example.**




In this game, Player 1's strategies are $LA$, $LB$, $RA$, and $RB$, where $LA$ means that Player 1 chooses $L$ at the first information set belonging to her, and chooses $B$ at the second. The strategies at all information sets will lead to a unique outcome.

*Intuition: A player's strategy is a book with a page for each information set assigned to that player, specifying the choice there.*

**Definition 2.6: Mixed Strategy**

A mixed (or, randomized) strategy for player $i$, denoted by $\sigma_i$, is a probability distribution over $S_i$, i.e., $\sigma_i \in \Delta(S_i)$.

*Pure strategy is a special case of mixed strategy, with probability one choosing a specific action.*

Rather than asking what is an optimal strategy, we think about it in a reverse way: what are the suboptimal strategies that a rational player would never choose, so that we can rule out in the analysis of equilibrium?

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**Definition 2.7: Strcitly Dominated Strategy (PS Notion)**

A pure strategy $s_i$ is strictly dominated if there is another strategy $s_i'$ such that

$$u_i(s_i', s_{-i}) > u_i(s_i, s_{-i}), \quad \forall s_{-i} \in S_{-i}.$$

This notion can be equivalently defined by the domination of mixed strategies.

**Definition 2.8: Strictly Dominated Strategy (MS Notion)**

A pure strategy $s_i \in S_i$ is strictly dominated if there exists a mixed strategy $\sigma_i \in \Delta(S_i)$ such that

$$u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}), \quad \forall s_{-i} \in S_{-i},$$

where $u_i(\sigma_i, s_{-i})$ is defined by

$$u_i(\sigma_i, s_{-i}) = \sum_{s_i' \in S_i} \sigma_i(s_i') u_i(s_i', s_{-i}).$$

**Definition 2.9: Best Response**

A pure strategy $s_i$ is a best response to the belief $\mu_i$ if

$$u_i(s_i \mu_i) \geq u_i(s_i', \mu_i), \quad \forall s_i' \in S_i,$$

where $u_i(s_i, \mu_i)$ is defined by

$$u_i(s_i, \mu_i) = \sum_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \mu_i(s_{-i}).$$

**Definition 2.10: Never A Best Response (NBR)**

A pure strategy $s_i$ is never a best response if for all beliefs $\mu_i \in \Delta(S_{-i})$, there exists another pure strategy $s_i' \in S_i$ such that

$$u_i(s_i', \mu_i) > u_i(s_i, \mu_i).$$

**Theorem 2.11**

$s_i$ is never a best response $\iff$ $s_i$ is strictly dominated.

Therefore, the two notions of inferior or suboptimal strategies are equivalent. We can revisit the concept of dominated strategies from the perspective of beliefs.

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**Definition 2.12: Strictly Dominated Strategy (Belief Notion)**

A pure strategy $s_i$ is strictly dominated if there exists another pure strategy $s_i' \in S_i$ such that

$$u_i(s_i', \mu_i) > u_i(s_i, \mu_i), \quad \forall \mu_i \in \Delta(S_i).$$

**Definition 2.13: Weakly Dominated Strategy (Belief Notion)**

A pure strategy $s_i$ is weakly dominated if there exists another pure strategy $s_i' \in S_i$ such that

$$u_i(s_i', \mu_i) \geq u_i(s_i, \mu_i), \quad \forall \mu_i \in \Delta(S_i),$$

and there exists some belief $\mu_i^* \in \Delta(S_{-i})$ such that

$$u_i(s_i', \mu_i^*) > u_i(s_i, \mu^*).$$

## 2.5 Iterated Elimination of Dominated Strategies

The previous theorem states that donimated strategy is a belief-free notion. This allows us to develop the concept of iterated elimination of dominated strategies (IESDS), which is a powerful tool to analyze games without having to worry about players' beliefs.

The strategies that survive the IESDS are called *rationalizable strategies*, which are the strategies that can be rationalized by some beliefs about other players' strategies. Rationalizability is a weaker notion than Nash equilibrium, as it does not require mutual consistency of beliefs. However, it is still a useful concept to understand the strategic behavior of players in a game.

Notably, IESDS's outcome is independent of the order of elimination, which is a desirable property. This is because the set of dominated strategies is well-defined and does not depend on the sequence in which they are removed. As a result, we can confidently apply IESDS to simplify games and analyze the strategic interactions without worrying about the order of elimination. This is *not* the case with *IEWDS* (iterated elimination of weakly dominated strategies), where the order of elimination matters.

**Example (IEWDS and Order of Elimination).**

Consider the following 2-player game and IEWDS.

| | $L$ | $R$ |
|---|---|---|
| $U$ | $1,1$ | $0,0$ |
| $M$ | $1,1$ | $2,1$ |
| $D$ | $0,0$ | $2,1$ |

- If Player 1 eliminates $U$ and $D$, then Player 2 can eliminate nothing, and the process ends. The survived strategies are $M$ for Player 1, and $L$ and $R$ for Player 2.
- If Player 1 eliminates only $U$, then Player 2 can eliminate $L$, and the process ends. The survived strategies are $M$ and $D$ for Player 1, and $R$ for Player 2.

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- If Player 1 eliminates only $D$, then Player 2 can eliminate $R$, and the process ends. The survived strategies are $U$ and $M$ for Player 1, and $L$ for Player 2.

From this example, we can clearly see the "outcome" of IEWDS depends on the order of elimination.

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# Chapter 3

# Nash Equilibrium

## 3.1 Nash Equilibrium and Nash's Theorem

**Definition 3.1: Nash Equilibrium**

A strategy profile $(\bar{\sigma}_1, \bar{\sigma}_2, \cdots, \bar{\sigma}_n)$ (where $\bar{\sigma}_i \in \Delta(S_i)$) is a Nash equilibrium of the game $G = (S_i, u_i)_{i=1}^n$ if for any player $i$,

$$u_i(\bar{\sigma}_i, \bar{\sigma}_{-i}) \geq u_i(s_i, \bar{\sigma}_{-i}), \quad \forall s_i \in S_i.$$

**Remark.**

Equivalently we can define Nash equilibrium by

$$u_i(\bar{\sigma}_i, \bar{\sigma}_{-i}) \geq u_i(\sigma_i, \bar{\sigma}_{-i}), \quad \forall \sigma_i \in \Delta(S_i).$$

But it suffices to only check pure strategies $s_i$ because mixed strategies are just probability distributions over pure strategies, and the expected utility of a mixed strategy is a convex combination of the utilities of the pure strategies.

**Theorem 3.2: Nash**

Every finite game has a Nash equilibrium in mixed strategies.

***Proof for Theorem***

Here we apply the Kakutani's Fixed Point Theorem.

**Theorem 3.3: Kakutani's Fixed Point Theorem**

Suppose $F : X \rightrightarrows X$ (i.e., $F(x) \subseteq X$), and

- $X$ is compact and convex;
- $F(x)$ is convex for all $x \in X$;
- $F$ has a closed graph, i.e., the set $\{(x, y) | f \in F(x)\}$ is closed.

Then, there exists $x^* \in X$ such that $x^* \in F(x^*)$.

Let $X_i \equiv \Delta(S_i)$, and $X = X_1 \times X_2 \times \cdots \times X_n = \prod_{i=1}^n \Delta(S_i)$. Define the best response correspondence $B_i : \prod_{j \neq i} \Delta(S_j) \rightrightarrows \Delta(S_i)$ by

$$\sigma_i \in B_i(\sigma_{-i}) \iff u_i(\sigma_i, \sigma_{-i}) \geq u_i(\sigma_i', \sigma_{-i}), \quad \forall \sigma_i' \in \Delta(S_i).$$

And we define $B : X \rightrightarrows X$ by

$$B(\sigma) = \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{pmatrix} (\sigma).$$

It is easy to verify that $B(\sigma)$ is convex for all $\sigma \in X$, and $B$ has a closed graph. Therefore, by Kakutani's theorem, there exists $\bar{\sigma} \in X$ such that $\bar{\sigma} \in B(\bar{\sigma})$, which means that $\bar{\sigma}$ is a Nash equilibrium.

**Remark.**

- "Mixed strategies" is important for Nash's Theorem. Finite games may not have NE in pure strategies.
- If $\bar{\sigma}$ is a NE, and $\bar{\sigma}_i(s_i) > 0$, then $s_i$ must be a best response to $\bar{\sigma}_{-i}$, and thus $u_i(s_i, \bar{\sigma}_{-i}) = u_i(\bar{\sigma}_i, \bar{\sigma}_{-i})$. In other words, if a pure strategy is played with positive probability in a mixed strategy NE (i.e., in the *support* of the mixed strategy $\bar{\sigma}$), then it must yield the same expected payoff as the mixed strategy itself.
- Then why bother mixing? Because each player mixes to make other players mix. If a player chooses a pure strategy, then other players will have no incentive to mix, and thus the first player will have no incentive to mix either. Therefore, mixing can be a strategic move to induce other players to mix, which can lead to a more favorable outcome for the player.

## 3.2 Potential Games

One of the central questions in game theory is whether a game has a *pure strategy Nash equilibrium* (PSNE). Some games, like matching penny, have no PSNE. However, certain classes of games are guaranteed to have PSNE. Two important classes are:

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- Potential games
- Supermodular games

In this section we will focus on potential games, which are a broad class of games that include many interesting examples, such as congestion games, coordination games, and certain types of auctions. Potential games have a special structure that allows us to analyze them using a potential function, which captures the incentives of all players in a single function.

### 3.2.1 Definition

**Definition 3.4: Potential Game**

A finite game $G = (S_i, u_i)_{i=1}^n$ is called a *potential game* if there exists a function $P : S \to \mathbb{R}$ (where $S = S_1 \times S_2 \times \cdots \times S_n$) such that for every player $i$ and every pair of strategies $s_i, s_i' \in S_i$:

$$u_i(s_i, s_{-i}) - u_i(s_i', s_{-i}) = P(s_i, s_{-i}) - P(s_i', s_{-i}), \quad s_{-i} \in S_{-i}.$$

The function $P$ is called the *potential function*.

**Example (Congestion Game).**

Consider a road network where $n$ drivers choose routes. Let $s_i$ be the route chosen by driver $i$. The payoff to driver $i$ is:

$$u_i(s_i, s_{-i}) = -c(s_i, n_i)$$

where $c(r, n_r)$ is the congestion cost on route $r$ when $n_r$ drivers use it. Define the potential function as:

$$P(s) = -\sum_r \int_0^{n_r(s)} c(r, x)dx$$

This is indeed a potential function because the private incentive to switch routes equals the change in total cost (up to a sign).

**Remark.**

- Potential games can be thought of as games where all players have identical payoffs.
- Potential games arise naturally when:
  - Players care about a common objective (like aggregate welfare)
  - Individual incentives are aligned with some global measure
  - Strategic interactions are "symmetric" in the sense of cross-partial derivatives
- The potential function $P$ captures the payoff differences for individual players when they deviate from one strategy to another. The key insight is that a player's incentive

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to deviate depends only on how the potential function changes, not on the baseline payoff levels.

- In the continuous case with smooth payoff functions, a game is a potential game if and only if:

$$\frac{\partial P}{\partial s_i} = \frac{\partial u_i}{\partial s_i} \quad \text{for all players } i$$

This leads to the following necessary condition, for all pairs of players $i$ and $j$:

$$\frac{\partial^2 u_i}{\partial s_j \partial s_k} = \frac{\partial^2 u_j}{\partial s_k \partial s_j} \quad \text{whenever these derivatives exist.}$$

### 3.2.2 Existence of PSNE in Potential Games

**Theorem 3.5: Existence of PSNE in Potential Games**

Every potential game has at least one pure strategy Nash equilibrium.

***Proof for Theorem***

A PSNE of game $G$ is a maximizer of the potential function $P$. Since $S$ is finite (in the discrete case), $P$ attains its maximum. At such a maximum point $s^*$, if player $i$ deviates to any other strategy $s_i \neq s_i^*$, then:

$$u_i(s_i^*, s_{-i}^*) - u_i(s_i, s_{-i}^*) = P(s_i^*, s_{-i}^*) - P(s_i, s_{-i}^*) \geq 0$$

Thus no player wants to deviate, so $s^*$ is a PSNE. ∎

**Remark.**

Different information structures (simultaneous move vs. sequential move) can lead to different equilibria in general games. However, in potential games, the set of pure strategy Nash equilibria is often robust across different information structures. This is because the potential function captures the incentives of all players in a way that is independent of the timing of moves. As a result, the same strategy profiles that maximize the potential function will be equilibria regardless of whether players move simultaneously or sequentially.

## 3.3 Supermodular Games

While potential games ensure the existence of PSNE through a global potential function, supermodular games approach the problem differently: they rely on the structure of strategic complementarities between players' actions.

The key insight is *strategic complementarity*: if one player's action increases, it becomes more attractive for other players to increase their actions as well. This creates a natural coordination mechanism that leads to equilibrium. Unlike potential games, which require

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a global objective function, supermodular games only require local complementarity conditions on payoffs.

Examples of strategic complementarity abound:

- In a price competition setting, if a competitor raises prices, your prices become more attractive to consumers, so you want to raise your price too.
- In technology adoption, if more people adopt a technology, the benefits of adoption increase for you (network effects).
- In public goods games, if others contribute more, your optimal contribution may also increase.

### 3.3.1 Definitions

We need lattice-theoretic concepts to formalize strategic complementarity.

**Definition 3.6: Meet & Join**

Let $x$ and $y$ be two vectors of $\mathbb{R}^n$. The *meet* of $x$ and $y$, denoted by $x \wedge y$, is the coordinate-wise minimum:

$$x \wedge y = \begin{pmatrix} \min(x_1, y_1) \\ \min(x_2, y_2) \\ \vdots \\ \min(x_n, y_n) \end{pmatrix}$$

The *join* of $x$ and $y$, denoted by $x \vee y$, is the coordinate-wise maximum:

$$x \vee y = \begin{pmatrix} \max(x_1, y_1) \\ \max(x_2, y_2) \\ \vdots \\ \max(x_n, y_n) \end{pmatrix}$$

**Remark.**

Meet and join decompose a pair of strategy profiles into their "lower bound" and "upper bound" in each dimension. Later we will see, if a payoff function is supermodular, then being at both extremes (the largest and smallest actions) together is at least as good as being at the two middle ground positions.

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**Definition 3.7: Supermodular Function**

A function $f : \mathbb{R}^n \to \mathbb{R}$ is *supermodular* if for all $x, y \in \mathbb{R}^n$:

$$f(x \wedge y) + f(x \vee y) \geq f(x) + f(y)$$

This is called the *supermodularity inequality.*

Intuitively, the supermodularity inequality says: having some coordinates large and others small is better than having all coordinates at intermediate levels.

In a game setting, we say player $i$'s payoff is supermodular if:

$$u_i(s) + u_i(s') \leq u_i(s \wedge s') + u_i(s \vee s') \quad \forall s, s' \in S$$

This means: player $i$'s payoff is supermodular in their own strategy relative to opponents' strategies.

**Remark.**

When $f$ is twice continuously differentiable, supermodularity is equivalent to:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} \geq 0 \quad \text{for all } i \neq j.$$

This means: if you increase $x_i$, the marginal benefit of increasing $x_j$ also increases (*strategic complementarity*).

### 3.3.2 Existence of PSNE in Supermodular Games

The next two lemmas establish that best response correspondences have monotonicity properties under supermodularity, which enables us to use fixed point theorems.

**Lemma 3.8: Monotonicity of Best Responses**

Suppose $u : [0,1]^p \times [0,1]^q \to \mathbb{R}$ is supermodular. Suppose $y', y'' \in [0,1]^q$ with $y'' \geq y'$. Let $x' \in [0,1]^p$ be a maximizer of $u(\cdot, y')$, and $x'' \in [0,1]^p$ be a maximizer of $u(\cdot, y'')$. Then $x' \wedge x''$ is a maximizer of $u(\cdot, y')$, and $x' \vee x''$ is a maximizer of $u(\cdot, y'')$.

***Proof for Lemma***

By construction, since $x'$ maximizes $u(\cdot, y')$ and $x''$ maximizes $u(\cdot, y'')$:

$$u(x', y') \geq u(x' \wedge x'', y')$$

Also note that $u(x' \wedge x'', y') = u(x' \wedge x'', y' \wedge y'')$ since $y' \wedge y'' = y'$ (as $y' \leq y''$).

Similarly:

$$u(x'', y'') \geq u(x' \vee x'', y'') = u(x' \vee x'', y' \vee y'')$$

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By supermodularity:

$$u(x', y') + u(x'', y'') \geq u(x' \wedge x'', y' \wedge y'') + u(x' \vee x'', y' \vee y'')$$

Suppose $u(x', y') > u(x' \wedge x'', y' \wedge y'')$. Then:

$$u(x', y') + u(x'', y'') > u(x' \wedge x'', y' \wedge y'') + u(x' \vee x'', y' \vee y'')$$

which violates supermodularity.

Therefore, we must have:

$$\begin{aligned} u(x', y') &= u(x' \wedge x'', y' \wedge y'') \\ u(x'', y'') &= u(x' \vee x'', y' \vee y'') \end{aligned}$$

This means both $x' \wedge x''$ and $x' \vee x''$ are maximizers at the respective points. ■

**Lemma 3.9: Monotonicity of Largest Maximizers**

Suppose $u : [0,1]^p \times [0,1]^q \to \mathbb{R}$ is supermodular. Define:

$$h(y) = \max\{x : x \text{ maximizes } u(\cdot, y)\}$$

(the largest maximizer). Then $h$ is non-decreasing: if $y'' \geq y'$, then $h(y'') \geq h(y')$.

***Proof for Lemma***

Suppose $y'' \geq y'$. Let $x' = h(y')$ and $x'' = h(y'')$ be the largest maximizers at $y'$ and $y''$ respectively. By the previous lemma, $x' \wedge x''$ maximizes $u(\cdot, y')$ and $x' \vee x''$ maximizes $u(\cdot, y'')$.

Since $x' \vee x''$ is a maximizer at $y''$ and $x''$ is the largest maximizer at $y''$, we have $x'' \geq x' \vee x''$. By definition, $x' \vee x'' \geq x'$. So $x'' \geq x'$. Therefore, $h(y'') = x'' \geq x' = h(y')$. ■

**Theorem 3.10: Existence of PSNE in Supermodular Games**

Every finite supermodular game has at least one pure strategy Nash equilibrium.

***Proof for Theorem***

This is a direct result of Tarski's Fixed Point Theorem.

**Theorem 3.11: Tarski's Fixed Point Theorem**

If $f : L \to L$ is a non-decreasing function on a complete lattice $L$, then $f$ has a fixed point.

For each player $i$, define $b_i(s_{-i})$ as the largest maximizer of $u_i(\cdot, s_{-i})$. By the lemmas above, $b_i$ is non-decreasing in $s_{-i}$.

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Define the best response mapping $b : S \to S$ by:

$$b(s) = (b_1(s_{-1}), b_2(s_{-2}), \ldots, b_n(s_{-n}))$$

Since $b$ is component-wise non-decreasing and $S$ is a finite lattice, by *Tarski's Fixed Point Theorem*, $b$ has a fixed point $\bar{s}$, i.e., $b(\bar{s}) = \bar{s}$. This fixed point is a Nash equilibrium.

**Example (Bertrand Duopoly with Differentiated Products).**

Consider two firms competing in prices. Let $D_i(p_1, p_2)$ be firm $i$'s demand, which is decreasing in its own price $p_i$ and increasing in the competitor's price $p_j$. With constant marginal cost $c$, firm $i$'s profit is:

$$\pi_i(p_1, p_2) = (p_i - c)D_i(p_1, p_2)$$

The key observation is:

$$\frac{\partial^2 \pi_i}{\partial p_i \partial p_j} = \frac{\partial}{\partial p_j}\left[\frac{\partial \pi_i}{\partial p_i}\right] > 0$$

This is because when the competitor's price increases, the marginal profit from raising your own price increases (since demand becomes more favorable). Thus, firm $i$'s profit is supermodular in $(p_1, p_2)$, and the duopoly game has a PSNE.

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# Chapter 4

# Subgame Perfect Nash Equilibrium

## 4.1 Definitions and Existence

**Definition 4.1: Subgame**

Given an extensive form $T$, a subgame $T'$ is some node $r \in T$ and all its succedding nodes such that if an information set $I$ intersects $T'$, then all nodes in $I$ are also in $T'$.

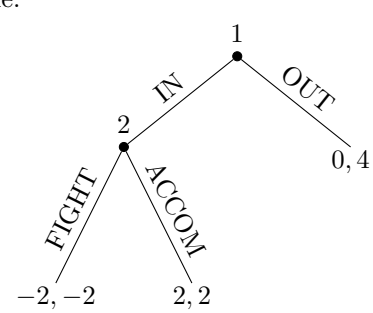
**Definition 4.2: Subgame Perfect Nash Equilibrium**

$\sigma$ is a *subgame perfect equilibrium* if

- $\sigma$ is a Nash equilibrium.
- $\sigma$ induces a Nash equilibrium in every subgame $T' \subseteq T$.

**Example (Entry Deterrence Game).**

Consider the following game:




From its normal form (payoff matrix) below, we know that there are two Nash equilibria: (OUT, FIGHT) and (IN, ACCOM).

| | FIGHT | ACCOM |
|---|---|---|
| OUT | $0, 5$ | $0, 5$ |
| IN | $-2, -2$ | $2, 2$ |

However, only (IN, ACCOM) is a subgame perfect equilibrium, because if Player 1 chooses IN, then Player 2's best response is to choose ACCOM (so to choose FIGHT is not a credible threat). Therefore, the unique subgame perfect equilibrium is (IN, ACCOM).

Suppose now the game is repeatedly played for $T = 20$ times. At $t = 1$, Player 1 chooses IN or OUT. If Player 1 chooses OUT, the game ends. If Player 1 chooses IN, then in any period $t \geq 1$, the Player 1 can choose to stay (IN) or exit (OUT). If OUT is ever chosen by Player 1, then OUT for all the periods onwards.

Clearly, using backward induction, we know that the unique subgame perfect equilibrium is for Player 1 to choose IN and Player 2 to choose ACCOM in all periods.

However, if we add one additional constraint: Player 1 can only fight for at most 15 periods, then the analysis changes.

**Claim**

The unique SPE with the constraint is that

- Player 1 never enters (IN), and
- Player 2 chooses FIGHT if Player chooses IN.

***Proof for Claim.***

Suppose that Player 1 has entered and been fought for the previous 14 periods. In period $t = 15$, if Player 1 chooses IN, Player 2 compares the payoffs of FIGHT and ACCOM:

- If Player 2 chooses FIGHT: Player 1's capacity is exhausted, forcing them to choose OUT from $t = 16$ to $t = 20$. Player 2's total remaining payoff is $-2(\text{at } t = 15) + 5 \times 4(\text{monopoly profit}) = 18$.
- If Player 2 chooses ACCOM: Player 1 stays in the market. Player 2's total remaining payoff is $2(\text{at } t = 15) + 5 \times 2(\text{duopoly profit}) = 12$.

Since $18 > 12$, FIGHT becomes a *credible threat* at $t = 15$. Anticipating this, Player 1 will choose OUT at $t = 15$ to avoid the $-2$ payoff.

Consider period $t = 14$. Player 1 anticipates that choosing IN will result in Player 2 choosing FIGHT (to trigger the exit at $t = 15$ and secure future monopoly profits). Choosing IN at $t = 14$ followed by exit at $t = 15$ yields a lower payoff than choosing OUT immediately at $t = 14$.

Following this logic backward to $t = 1$, consequently, the only rational choice for Player 1 is to never enter (OUT), and for Player 2 to threaten to FIGHT at any entry node.

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**Theorem 4.3**

Every finite game in extensive form has at least one subgame perfect equilibrium.

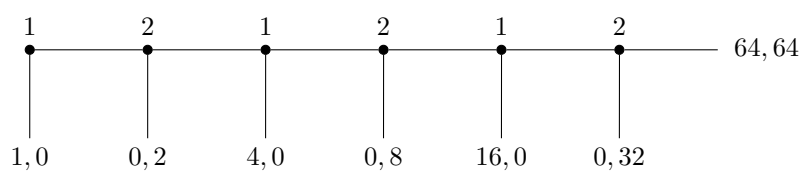
***Proof for Theorem***

The proof is to find all subgames that do not have any proper subgames. In each such subgame, find a Nash equilibrium of the subgame, and replace each node by expected payoffs from any Nash equilibrium of the subgame.

**Remark.**

The notion of subgame perfect equilibrium is to generalize backward induction to games with imperfect information.

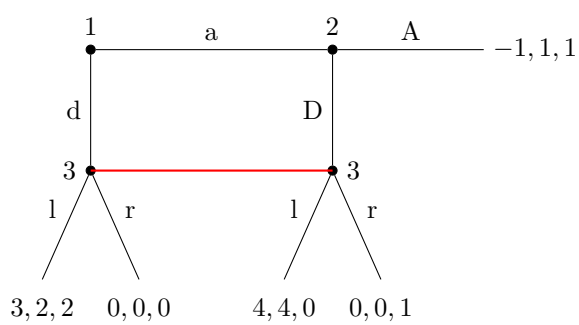
**Example (Centipede Game).**




The unique subgame perfect equilibrium is for each player to exit immediately at their first decision node, yielding payoff $(1, 0)$. However, the backward induction outcome $(1, 0)$ is Pareto dominated by the outcome $(64, 64)$ at the end. Both players would be better off if the game continued to the end, but the equilibrium logic prevents this.

## 4.2 Subgame Perfection

**Example (Selten's Horse).**




There are two pure strategy Nash equilibria: $(d, A, l)$ and $(a, A, r)$. Since this game has no proper subgames, both equilibria are subgame perfect.

However, the SPE $(d, A, l)$ appears problematic: if Player 1 accidentally chooses $a$ instead of $d$, Player 2 would prefer to choose $D$ rather than $A$. This suggests the equilibrium is not robust to small mistakes.

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Recall that $\sigma$ is a Nash equilibrium if and only if for each player $i$ and any pure strategy $s_i'$: $u_i(\sigma_i, \sigma_{-i}) \geq u_i(s_i', \sigma_{-i})$, and for all the pure strategies that are in the support of $\sigma_i$, the expected payoff is the same: $u_i(s_i, \sigma_{-i}) = u_i(\sigma_i, \sigma_{-i})$ for all $s_i$ such that $\sigma_i(s_i) > 0$. Hence, if for some pure strategy $s_i'$, we have $u_i(\sigma_i, \sigma_{-i}) > u_i(s_i', \sigma_{-i})$, then $\sigma_i(s_i') = 0$.

To formalize the intuition from the previous example of Selten's Horse, suppose all players' choices are subject to small trembles (mistakes).

**Definition 4.4: $\varepsilon$-Equilibrium**

A mixed strategy profile $\sigma^\varepsilon$ is an *$\varepsilon$-equilibrium* (for small $\varepsilon > 0$) if:

- Every pure strategy has positive probability: $\sigma_i^\varepsilon(s_i) > 0$ for all $i$ and $s_i$.
- Strategies that are not a best response are played with low probability: if $u_i(s_i', \sigma_{-i}^\varepsilon) > u_i(s_i, \sigma_{-i}^\varepsilon)$, then $\sigma_i^\varepsilon(s_i') < \varepsilon$.

**Definition 4.5: Trembling Hand Perfect Equilibrium**

A strategy profile $\sigma$ is a *trembling hand perfect equilibrium* (THPE) if it is the limit of a sequence of $\varepsilon$-equilibria $\sigma^\varepsilon$ as $\varepsilon \to 0$.

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# Chapter 5

# Two-Person bargaining

## 5.1 Nash Bargaining Solution

Let $X$ be the set of possible outcomes. Let $U$ be the set of feasible expected utilities from $X$, which is assumed to be a compact and convex subset of $\mathbb{R}^2$. Let $d \in U$ be the disagreement point, which is the outcome if the two parties fail to reach an agreement. We assume that $\exists u \in U$ such that $u \gg d$ (i.e., there is some outcome that is strictly better than the disagreement point for both parties). A bargaining solution is a function $F$ that maps $(U, d)$ to a point $F(U, d) \in U$.

**Requirement: Scale Invariance**

Consider the transformation $(U, d) \mapsto (\bar{U}, \bar{d})$ such that

$$\bar{U} = \left\{ (\bar{u}_1, \bar{u}_2) \middle| \bar{u}_i = \alpha_i u_i + \beta_i \text{ for some } \alpha_i > 0, \quad (u_1, u_2) \in U \right\},$$
$$\bar{d}_i = \alpha_i d_i + \beta_i.$$

Then,

$$F_i(\bar{U}, \bar{d}) = \alpha_i F_i(U, d) + \beta_i.$$

**Requirement: Efficiency**

$\nexists u \in U$ such that $u \gg F(U, d)$.

**Requirement: Symmetry**

If $U$ is symmetric (i.e., $(u_1, u_2) \in U \iff (u_2, u_1) \in U$) and $d_1 = d_2$, then

$$F_1(U, d) = F_2(U, d).$$

**Requirement: Independence of Irrelevant Alternatives (IIA)**

Consider $(U, d)$ and $(\bar{U}, d)$ such that $U \subseteq \bar{U}$. If $F(\bar{U}, d) \in U$, then,

$$F(\bar{U}, d) = F(U, d).$$

**Theorem 5.1**

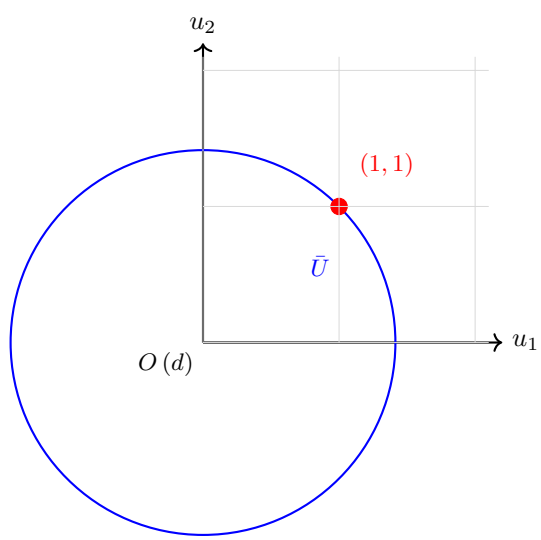
There is one and only one $F$ that satisfies the above requirements. $F$ is given by:

$$F^N(U, d) = \arg\max_{u \in U, u \geq d} (u_1 - d_1)(u_2 - d_2).$$

***Proof for Theorem***

Firstly, it is easy to check that $F^N$ is well-defined in the sense that $F^N$ satisfies the above requirements.

Let $(u_1^*, u_2^*) = F^N(U, d)$. We can always rescale the utilities such that $(u_1^*, u_2^*) \mapsto (1, 1)$ and $(d_1, d_2) \mapsto (0, 0)$, without loss of generality. Denote the new space as $(\bar{U}, \bar{d})$.




**Claim**

If we can show that $F = F^N$ for any $F$ in the scaled space $(\bar{U}, \bar{d})$, then we must have $F = F^N$ for any $(U, d)$.

***Proof for Claim.***

This is true by scale invariance.

**Claim**

$F^N(\bar{U}, d) = (1, 1)$.

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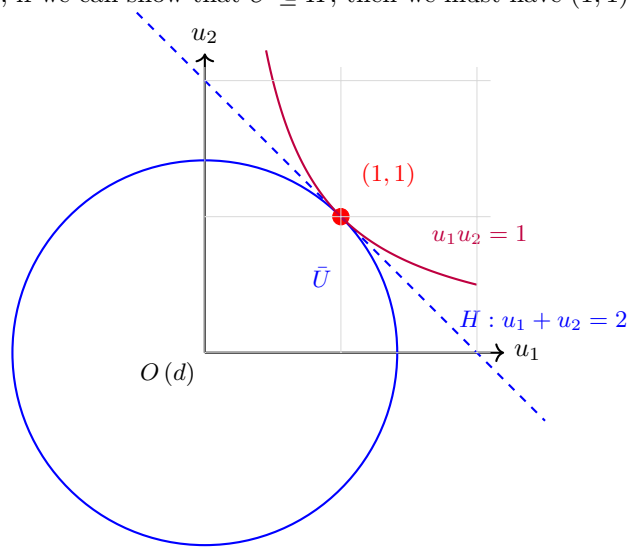
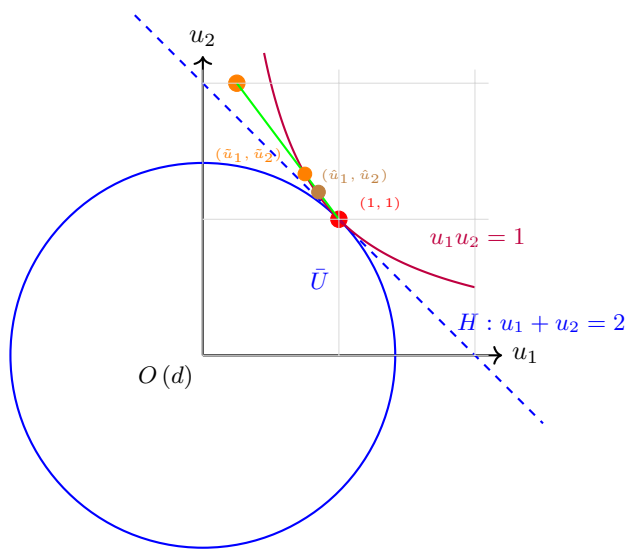
***Proof for Claim.***

Define the set $H = \{(u_1, u_2) \,|\, u_1 + u_2 \leq 2\}$. Note that the boundary of $H$ is the tangent line to the hyperbola $u_1 u_2 = 1$ through $(1, 1)$ with slope -1. Since $(1, 1) \in \bar{U}$ and $(1, 1) \in H$, if we can show that $\bar{U} \subseteq H$, then we must have $(1, 1)$ as the solution.




To show that $\bar{U} \subseteq H$, suppose for contradiction that $\bar{U}$ contains some point $(u_1, u_2)$ such that $u_1 + u_2 > 2$. The line segment connecting $(1, 1)$ and $(u_1, u_2)$ must intersect the hypobola $u_1 u_2 = 1$ at some point $(\tilde{u}_1, \tilde{u}_2)$. So the line segment joining $(\tilde{u}_1, \tilde{u}_2)$ and $(1, 1)$ must be above the hypobola $u_1 u_2 = 1$. Since $\bar{U}$ is convex, we can find a convex combination of $(u_1, u_2)$ and $(1, 1)$ that falls within the line segment joining $(\tilde{u}_1, \tilde{u}_2)$ and $(1, 1)$, denoted by $(\hat{u}_1, \hat{u}_2)$ such that $(\hat{u}_1, \hat{u}_2) \in \bar{U}$ and $\hat{u}_1 \hat{u}_2 > 1$. This contradicts the assumption that $F^N(\bar{U}, d) = (1, 1)$.

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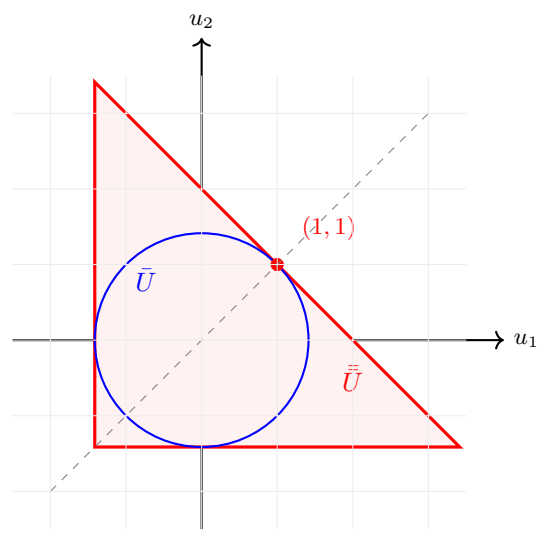
**Claim**

For any $F$, the solution must be $(1, 1)$.

***Proof for Claim.***

We can always include the $\bar{U}$ using a symmetric right triangle whose hypotenuse is on the boundary of $H$ and whose right angle sits on the line of $u_1 = u_2$. Denote the new triangle by $\bar{\bar{U}}$.




By efficiency of $F$, the solution must lie on the boundary of $\bar{\bar{U}}$. By symmetry of $F$, the solution must be $(1, 1)$. Additionally, note that $\bar{U} \subseteq \bar{\bar{U}}$, and $F(\bar{\bar{U}}, \bar{d}) \in \bar{U}$. By IIA, we must have $F(\bar{\bar{U}}, \bar{d}) = F(\bar{U}, \bar{d})$. Therefore, $F(\bar{U}, \bar{d}) = (1, 1)$.

From all the results above, we conclude that $F = F^N$ for any $(U, d)$.

**Remark.**

This is actually a *normative* result in the sense that we require the bargaining solution to satisfy certain axioms. The Nash bargaining solution is the unique solution that satisfies these axioms. However, it is not necessarily a positive result in the sense that we do not have a game-theoretic model that can generate the Nash bargaining solution as an equilibrium outcome.

## 5.2 Dividing a Dollar

Let $X = [0, 1]$. Suppose utility functions for the two players are $u_1 : X \to \mathbb{R}$ and $u_2 : X \to \mathbb{R}$, which are continuous, strictly increasing, and strictly concave. The problem is given by

$$\max_{x \in [0,1]} u_1(x) u_2(1 - x).$$

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The FOC is:

$$\begin{aligned} & u_1'(x^*)u_2(1-x^*) - u_1(x^*)u_2'(1-x^*) = 0 \\ \implies & \frac{u_1'(x^*)}{u_1(x^*)} = \frac{u_2'(1-x^*)}{u_2(1-x^*)}. \end{aligned}$$

Suppose in addition that Player 1 becomes more risk averse than Player 2:

$$\begin{aligned} & v_1 : X \to \mathbb{R}, \text{ where } v_1 = \phi(u_1), \quad \phi' > 0, \quad \phi'' < 0, \\ & u_2 : X \to \mathbb{R}. \end{aligned}$$

So the problem becomes:

$$\max_{x\in[0,1]} v_1(x)u_2(1-x) = \max_{x\in[0,1]} \phi(u_1(x))u_2(1-x).$$

The FOC is:

$$\begin{aligned} & \phi'(u_1(x^{**}))u_1'(x^{**})u_2(1-x^{**}) - \phi(u_1(x^{**}))u_2'(1-x^{**}) = 0 \\ \implies & \frac{\phi'(u_1(x^{**}))u_1'(x^{**})}{\phi(u_1(x^{**}))} = \frac{u_2'(1-x^{**})}{u_2(1-x^{**})}. \end{aligned}$$

Since $\phi$ is a concave function,

$$\phi'(t) < \frac{\phi(t)}{t} \implies \frac{\phi'(t)}{\phi(t)} < \frac{1}{t}.$$

Thus,

$$\frac{u_1'(x^*)}{u_1(x^*)} > \frac{\phi'(u_1(x^{**}))u_1'(x^{**})}{\phi(u_1(x^{**}))} = \frac{u_2'(1-x^{**})}{u_2(1-x^{**})}.$$

So in comparison,

$$\begin{cases} \frac{u_1'(x^*)}{u_1(x^*)} = \frac{u_2'(1-x^*)}{u_2(1-x^*)} \\ \frac{u_1'(x^{**})}{u_1(x^{**})} > \frac{u_2'(1-x^{**})}{u_2(1-x^{**})} \end{cases} \implies x^{**} < x^*.$$

## 5.3 Alternating Offer Model

Consider the problem of dividing a dollar: Two players will propose a division of a dollar in turns. Without loss of generality, suppose Player 1 proposes first. If Player 1 proposes to give $x$ to Player 2, then Player 2 can either accept or reject. If Player 2 accepts, then the payoffs are $(1-x, x)$. If Player 2 rejects, then the game moves to the next period, where Player 2 proposes a division of a dollar and Player 1 can accept or reject. If Player 1 accepts, then the payoffs are $(x', 1-x')$. If Player 1 rejects, then the game continues to the next period, and so on. The game ends when one proposal is accepted. The discount factor for both players is $\delta \in (0,1)$.

**Claim**

Any split $(x, 1-x)$ where $x \in (0,1)$ is a NE outcome.

***Proof for Claim.***

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This split can be supported by the strategies where Player 1 only accepts proposals that give them at least $x$, and Player 2 only accepts proposals that give them at least $1-x$.

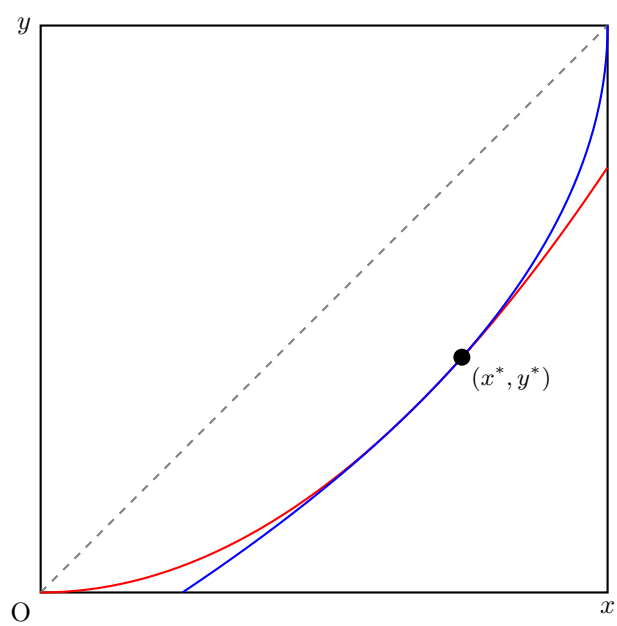
**Theorem 5.2: Rubinstein**

Let $(x^*, y^*)$ be the solution to

$$u_1(y) = \delta u_1(x)$$
$$u_2(1-x) = \delta u_2(1-y)$$

Then there is a unique subgame perfect equilibrium in which in $G_1$, Player 1 asks for $x^*$ and accepts iff $y \geq x^*$, and in $G_2$, Player 2 offers $y^*$ and Player 1 accepts iff $x \leq y^*$.

***Proof for Theorem***




Let $(x^*, y^*)$ be the solution to

$$u_1(y) = \delta u_1(x)$$
$$u_2(1-x) = \delta u_2(1-y)$$

Player 1 always offers $x^*$, and rejects Player 2's offer if $y < x^*$. Player 2 always offers $y^*$, and rejects Player 1's offer if $x > y^*$.

$(z, t)$ is an outcome if a split of $(z, 1-z)$ is accepted in period $t$.

Define $v_1(z, t)$ as the dollar amount such that Player 1 is indifferent between receiving $v_1(z, t)$ today and receiving $z$ in period $t$.

$$u_1(v_1(z, t)) = \delta^{t-1} u_1(z).$$

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Define $v_2(z,t)$ as the dollar amount such that Player 2 is indifferent between giving Player 1 $v_2(z,t)$ today and giving Player 1 $z$ in period $t$.

$$u_2(1 - v_2(z,t)) = \delta^{t-1} u_2(1-z).$$

Define

$$M_i = \sup \{v_i(z,t) : (z,t) \text{ is a SPE of } G_i\},$$
$$m_i = \inf \{v_i(z,t) : (z,t) \text{ is a SPE of } G_i\},$$

where $G_i$ is the subgame starting with Player $i$'s proposal.

**Claim**

$$u_1(M_2) \leq \delta u_1(M_1).$$

***Proof for Claim.***

If Player 2 offers $y$ such that $u_1(y) > \delta u_1(M_1)$, then Player 1 would accept the offer, because the right hand side $\delta u_1(M_1)$ is the most that Player 1 can get in the subsequent $G_1$. So Player 2 would never offer such $y$. Hence, we must have $u_1(M_2) \leq \delta u_1(M_1)$.

**Claim**

$$u_1(M_2) \geq \delta u_1(M_1).$$

***Proof for Claim.***

If Player 2 offers $y$ such that $u_1(y) < \delta u_1(M_1)$, then Player 1 will reject the offer, because by rejecting and entering $G_1$, Player 1 can get at most $M_1$ (with current payoff equivalent $\delta u_1(M_1)$). Therefore, any successful offer in $G_2$ must satisfy $u_1(y) \geq \delta u_1(M_1)$. Taking the supremum over all $y$ in $G_2$ equilibria, we have $u_1(M_2) \geq \delta u_1(M_1)$.

**Claim**

$$u_2(1 - M_1) \geq \delta u_2(1 - M_2).$$

***Proof for Claim.***

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In $G_1$, if Player 1 demands more than $v_2(M_2, 2)$, then Player 2 will reject. This implies

$$M_1 \leq v_2(M_2, 2).$$

And this implies

$$u_2(1 - M_1) \geq u_2(1 - v_2(M_2, 2)) = \delta u_2(1 - M_2).$$

## 5.4 Unequal Discounting

Suppose now Player 1 has utility function $w_1$ and discount factor $\delta_1$, and Player 2 has utility function $w_2$ and discount factor $\delta_2$, with $\delta_1 < \delta_2$. We make the following transformation of the original problem:

- Player 2 has $u_2 = w_2$ and $\delta = \delta_2$.
- Player 1 has $u_1 = [w_1(x)]^{\frac{\ln \delta_2}{\ln \delta_1}}$ and $\delta = \delta_2$. (Note that $\frac{\ln \delta_2}{\ln \delta_1} < 1$ so that $u_1$ is concave, and is even "more concave" than $w_1$.)

Without loss of generality, assume that

$$\begin{aligned}
& \delta_1^{t-1} w_1(z) > \delta_1^{t'-1} w_1(z') \\
\iff & \left[\delta_1^{t-1} w_1(z)\right]^{\frac{\ln \delta_2}{\ln \delta_1}} > \left[\delta_1^{t'-1} w_1(z')\right]^{\frac{\ln \delta_2}{\ln \delta_1}} \\
\iff & \frac{\ln \delta_2}{\ln \delta_1} \left[(t-1) \ln \delta_1 + \ln w_1(z)\right] > \frac{\ln \delta_2}{\ln \delta_1} \left[(t'-1) \ln \delta_1 + \ln w_1(z')\right] \\
\iff & (t-1) \ln \delta_2 + \frac{\ln \delta_2}{\ln \delta_1} \ln w_1(z) > (t'-1) \ln \delta_2 + \frac{\ln \delta_2}{\ln \delta_1} \ln w_1(z') \\
\iff & \ln \delta_2^{t-1} + \ln \left[w_1(z)\right]^{\frac{\ln \delta_2}{\ln \delta_1}} > \ln \delta_2^{t'-1} + \ln \left[w_1(z')\right]^{\frac{\ln \delta_2}{\ln \delta_1}} \\
\iff & \delta_2^{t-1} u_1(z) > \delta_2^{t'-1} u_1(z').
\end{aligned}$$

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# Chapter 6

# Private Value Auctions

Under the private value setting, we have $n$ bidders and a single object. Each bidder draws a value $X_i \sim F$ independently. Only bidder $i$ knows their own value $X_i = x_i$, and the distribution $F$ is common knowledge. Each bidder submits a bid $b_i$ to the auctioneer, and the auctioneer determines the winner and the price to be paid based on the bids.

## 6.1 Second Price Sealed Bid Auction (SPA)

In a second price sealed bid auction, each bidder submits a bid without knowing the bids of others. The highest bidder wins the item, but pays the second-highest bid.

**Claim**

It is a weakly dominant strategy for each bidder to bid their true value in a SPA, i.e., $b_i = x_i$.

***Proof for Claim.***

Graphical proof:

Suppose bidder 1 has value $x_1$. We want to show that bidding $b_1 = x_1$ is a weakly dominant strategy for bidder 1. Let $x_{-1}$ be the highest competing bid among the other bidders. The profit for bidder 1 is given by:

$$\pi_1(b_1, x_{-1}) = \begin{cases} x_1 - x_{-1} & \text{if } b_1 > x_{-1}, \\ 0 & \text{if } b_1 < x_{-1}. \end{cases}$$

This correspondes to the **black** line in the figure below.

If instead of bidding $b_1 = x_1$, bidder 1 bids $b_1^+ > x_1$, then the profit is given by:

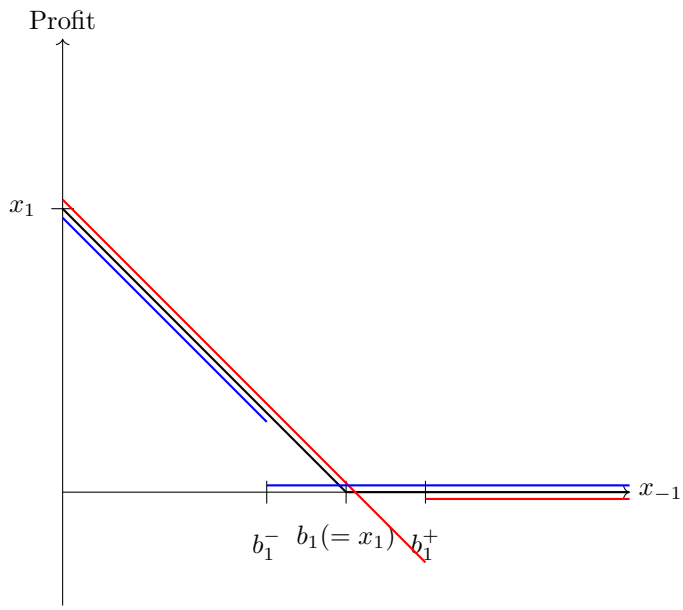
$$\pi_1(b_1^+, x_{-1}) = \begin{cases} x_1 - x_{-1} & \text{if } b_1^+ > x_{-1}, \\ 0 & \text{if } b_1^+ < x_{-1}. \end{cases}$$

This correspondes to the **red** line in the figure below.

On the other hand, if bidder 1 bids $b_1^- < x_1$, then the profit is given by:

$$\pi_1(b_1^-, x_{-1}) = \begin{cases} x_1 - x_{-1} & \text{if } b_1^- > x_{-1}, \\ 0 & \text{if } b_1^- < x_{-1}. \end{cases}$$

This correspondes to the **blue** line in the figure below.




The graph completes the proof.

## 6.2 First Price Sealed Bid Auction (FPA)

In a first price sealed bid auction, each bidder submits a bid without knowing the bids of others. The highest bidder wins the item and pays their own bid.

**Example (Motivating Example of Two Bidders).**

Suppose the true value $X_i$ for bidder $i$ is drawn from a uniform distribution on $[0, 1]$. Suppose the belief of Player 1 is that Player 2 will bid $B_2 = kX_2$ for some $\frac{1}{2} \leq k < 1$. Suppose Player 1 bids $b_1 \leq k$. Then the expected profit for Player 1 is given by

$$\begin{aligned} \mathbb{E}\left(\pi_1(b_1)\right) &= \Pr(kX_2 < b_1)\,(x_1 - b_1) \\ &= \Pr\left(X_2 < \frac{b_1}{k}\right)(x_1 - b_1) \\ &= \frac{b_1}{k}(x_1 - b_1). \end{aligned}$$

Then the best response of Player 1 to her belief $B_2 = kX_2$ with $\frac{1}{2} \leq k < 1$ is to bid $b_1 = \frac{x_1}{2}$.

Similarly, we can show that it is a symmetric equilibrium for both players to bid $b_i = \frac{x_i}{2}$ in a FPA.

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**Remark (Why $k \geq 1/2$?).**

The condition $k \geq 1/2$ is strictly necessary to ensure that the unconstrained interior maximizer $b_1^* = \frac{x_1}{2}$ is globally valid for all possible valuations $x_1 \in [0, 1]$.

In the derivation, Player 1's expected profit function $\mathbb{E}(\pi_1(b_1)) = \frac{b_1}{k}(x_1 - b_1)$ is predicated on the assumption that $b_1 \leq k$, since Player 2's maximum possible bid is essentially $k \cdot 1 = k$. To guarantee that the optimal bid always falls within this valid domain (i.e., $\frac{x_1}{2} \leq k$), this inequality must hold for the supremum of $x_1$. Since $x_1 \leq 1$, this immediately requires $k \geq \frac{1}{2}$.

If we were to relax this condition such that $k < 1/2$, a bidder with a sufficiently high valuation ($x_1 > 2k$) would find her unconstrained best response to be $\frac{x_1}{2} > k$. However, since bidding exactly $k$ (or marginally above it) already guarantees a 100% probability of winning, bidding $\frac{x_1}{2}$ would strictly decrease her profit without any marginal gain in win rate. Consequently, the best response for high-valuation types would collapse into a corner solution $b_1 = k$. Thus, $k \geq 1/2$ rules out this corner solution and preserves the symmetric linear strategy over the entire support.

### 6.2.1 Symmetric Equilibrium of FPA

Now we extend the above example to the case with $n$ bidders whose private values are drawn from an arbitrary distribution function $F$. We start the analysis by assuming there is a symmetric equilibrium where each bidder bids $b_i = \beta(X_i)$ for some strictly increasing function $\beta$.

Without loss of generality, suppose bidder 1 has value $x_1$ and bids $b_1$, and all other bidders follow the symmetric strategy $\beta$. Then the probability that bidder 1 wins the auction is given by

$$\begin{aligned}
\Pr\left(b_1 > \max_{i \neq 1} b_i\right) &= \Pr\left(b_1 > \max_{i \neq 1} \beta(X_i)\right) \\
&= \Pr\left(b_1 > \beta\left(\max_{i \neq 1} X_i\right)\right) \\
&= \Pr\left(\beta^{-1}(b_1) > \max_{i \neq 1} X_i\right) \\
&= \Pr\big(X_i < \beta^{-1}(b_1), \forall i \neq 1\big) \\
&= \left[F(\beta^{-1}(b_1))\right]^{n-1}.
\end{aligned}$$

For simplicity, we define

$$Y_1 = \max_{i \neq 1} X_i.$$

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Let $G$ be the distribution function of $Y_1$. Then we have

$$\begin{aligned} G(y) &= \Pr(Y_1 \leq y) \\ &= \Pr\left(\max_{i \neq 1} X_i \leq y\right) \\ &= \Pr(X_i \leq y, \forall i \neq 1) \\ &= [F(y)]^{n-1}. \end{aligned}$$

Then the probability that bidder 1 wins the auction can be rewritten as

$$\Pr\left(b_1 > \max_{i \neq 1} \beta(X_i)\right) = \Pr\big(Y_1 < \beta^{-1}(b_1)\big) = G(\beta^{-1}(b_1)).$$

The expected profit for bidder 1 is then given by

$$\begin{aligned} \mathbb{E}\left(\pi_1(b_1)\right) &= \Pr\left(b_1 > \max_{i \neq 1} \beta(X_i)\right)(x_1 - b_1) \\ &= G(\beta^{-1}(b_1))(x_1 - b_1). \end{aligned}$$

Since $b_1$ is the best response to the symmetric strategy $\beta$, we have

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}b_1}\mathbb{E}\left(\pi_1(b_1)\right) &= \frac{\mathrm{d}}{\mathrm{d}b_1} G(\beta^{-1}(b_1))(x_1 - b_1) \\ &= G'(\beta^{-1}(b_1)) \cdot \frac{\mathrm{d}}{\mathrm{d}b_1}\beta^{-1}(b_1) \cdot (x_1 - b_1) - G(\beta^{-1}(b_1)) \\ &= 0. \end{aligned}$$

Note that $\frac{\mathrm{d}}{\mathrm{d}b_1}\beta^{-1}(b_1) = \frac{1}{\beta'(\beta^{-1}(b_1))}$.[1] And by the optimality of $\beta$, we have $b_1 = \beta(x_1)$ and thus $\beta^{-1}(b_1) = x_1$. So the FOC can be rewritten as

$$\begin{aligned} &\frac{g(x_1)}{\beta'(x_1)}\left(x_1 - \beta(x_1)\right) - G(x_1) = 0 \\ \Longrightarrow\ &\beta'(x_1)G(x_1) + \beta(x_1)g(x_1) = x_1 g(x_1) \\ \Longrightarrow\ &\frac{\mathrm{d}}{\mathrm{d}x_1}\left(\beta(x_1)G(x_1)\right) = x_1 g(x_1) \\ \Longrightarrow\ &\beta(x_1)G(x_1) - \beta(0)G(0) = \int_0^{x_1} y g(y)\mathrm{d}y \\ \Longrightarrow\ &\beta(x_1)G(x_1) = \int_0^{x_1} y g(y)\mathrm{d}y \\ \Longrightarrow\ &\beta(x_1) = \frac{\int_0^{x_1} y g(y)\mathrm{d}y}{G(x_1)} \\ \Longrightarrow\ &\beta(x_1) = \mathbb{E}\left[Y_1 | Y_1 < x_1\right]. \end{aligned}$$

Note that a bidder with the lowest possible valuation $x = 0$ has zero probability of winning anything of value, so they will bid zero to avoid paying a positive amount. Therefore, $\beta(0) = 0$.

[1] This is true because $f(f^{-1}(x)) = x$ for any strictly increasing function $f$. Differentiating both sides with respect to $x$, we have $f'(f^{-1}(x)) \cdot \frac{\mathrm{d}}{\mathrm{d}x} f^{-1}(x) = 1$. Hence, $\frac{\mathrm{d}}{\mathrm{d}x} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$.

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**Remark.**

- **Why $\frac{\int_0^{x_1} yg(y)\mathbf{d}y}{G(x_1)} = \mathbb{E}\left[Y_1|Y_1 < x_1\right]$?**

  This equality follows directly from the definition of conditional expectation.

  First, we determine the conditional probability density function of the highest competing valuation $Y_1$, given the condition that you win the auction (i.e., $Y_1 < x_1$). Because we are restricting the possible values of $Y_1$ to the interval $[0, x_1]$, we must normalize the original PDF $g(y)$ by dividing it by the total probability of this condition occurring, which is $\Pr(Y_1 < x_1) = G(x_1)$. Thus, the conditional PDF is given by:

  $$f(y \mid Y_1 < x_1) = \frac{g(y)}{G(x_1)} \quad \text{for } 0 \leq y < x_1.$$

  By the standard definition of expected value, we integrate the variable $y$ multiplied by its conditional PDF over the restricted support:

  $$\begin{aligned}\mathbb{E}\left[Y_1|Y_1 < x_1\right] &= \int_0^{x_1} y \cdot f(y \mid Y_1 < x_1)\mathrm{d}y \\ &= \int_0^{x_1} y \left(\frac{g(y)}{G(x_1)}\right)\mathrm{d}y.\end{aligned}$$

  Since $G(x_1)$ only depends on $x_1$ and acts as a constant with respect to the integration variable $y$, we can factor it out of the integral:

  $$\mathbb{E}\left[Y_1|Y_1 < x_1\right] = \frac{1}{G(x_1)} \int_0^{x_1} yg(y)\mathrm{d}y.$$

- **Economic Intuition of FPA's Bidding Strategy**: The result $\beta(x) = \mathbb{E}\left[Y_1|Y_1 < x\right]$ reveals a profound economic intuition that elegantly connects the First-Price Auction (FPA) with the Second-Price Auction (SPA).

  In an SPA, a winning bidder pays exactly the highest competing valuation, $Y_1$. This payment is determined *ex-post* (after the bids are submitted).

  In an FPA, however, a winning bidder pays their own bid. Since you must determine your payment *ex-ante* (without knowing $Y_1$), how do you optimize your bid? To maximize your payoff while securing the item when you have the highest valuation, you essentially "simulate" the SPA in your mind.

  Conditional on winning (which implies $Y_1 < x$), your best unbiased estimate of what you *would* have paid in an SPA is exactly the expected value of the highest competing opponent, $\mathbb{E}\left[Y_1|Y_1 < x\right]$. Thus, the optimal strategy in an FPA is to simply bid your expected SPA payment.

In a FPA, you bid $\beta(x) = \mathbb{E}\left[Y_1|Y_1 < x\right]$, and you pay your own bid if you win the auction. You win the auction if $Y_1 < x$, which happens with probability $G(x)$. So the

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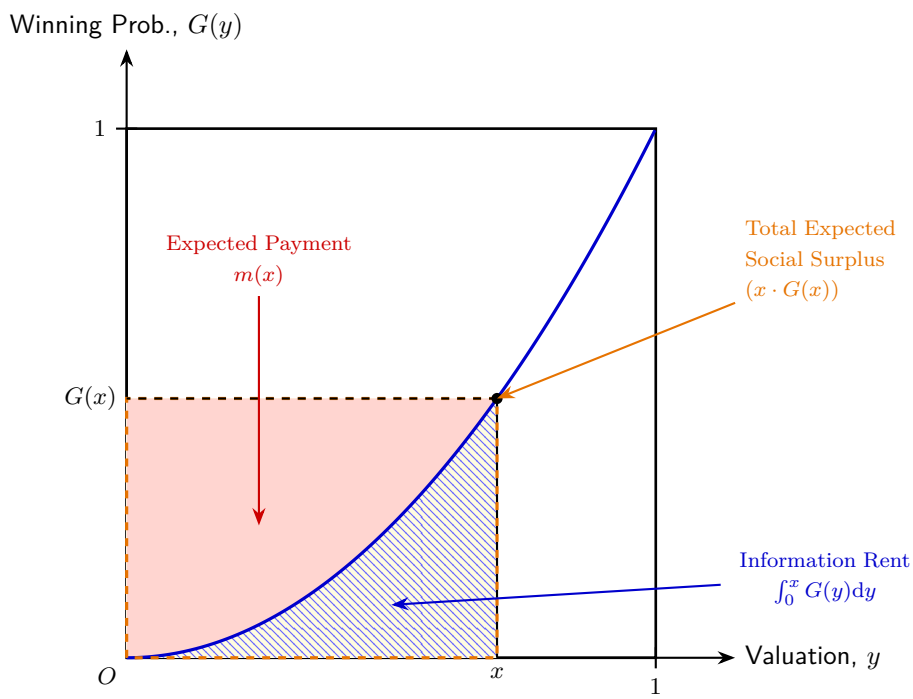
expected payment in a FPA for a bidder with valuation $x$ (denoted by $m(x)$) is:

$$m(x) = G(x) \cdot \mathbb{E}\left[Y_1 | Y_1 < x\right] = \int_0^x y g(y) \mathrm{d}y.$$

Furthermore, by integration by parts,

$$\begin{aligned} m(x) &= \beta(x) G(x) \\ &= \int_0^x y g(y) \mathrm{d}y \\ &= \int_0^x y \mathrm{d}G(y) \\ &= \left[y G(y)\right]\Big|_0^x - \int_0^x G(y) \mathrm{d}y \\ &= x G(x) - \int_0^x G(y) \mathrm{d}y. \end{aligned}$$

This derived identity, $m(x) = xG(x) - \int_0^x G(y)\mathrm{d}y$, allows for a beautiful geometric decomposition of the total social surplus generated by the auction. This surplus-splitting perspective is visualized in the figure below:




**Remark (Surplus Splitting and Information Rent).**

The identity $m(x) = xG(x) - \int_0^x G(y)\mathrm{d}y$ provides a powerful geometric and economic interpretation of the auction's outcome, as illustrated in the figure above:

- **Total Expected Surplus ($xG(x)$):** Represented by the entire rectangle defined by $(x, G(x))$, this is the total social value created when an object is allocated to a bidder with valuation $x$ (who wins with probability $G(x)$).

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- **Information Rent ($\int_0^x G(y)\mathrm{d}y$):** This corresponds to the area *under* the curve $G(y)$. Because the seller cannot observe your private valuation $x$, you are able to extract a surplus from the mechanism. This "rent" is the cost the seller must pay to induce you to reveal your true type. Note that it increases with your valuation, reflecting the higher bargaining power of high-value types.

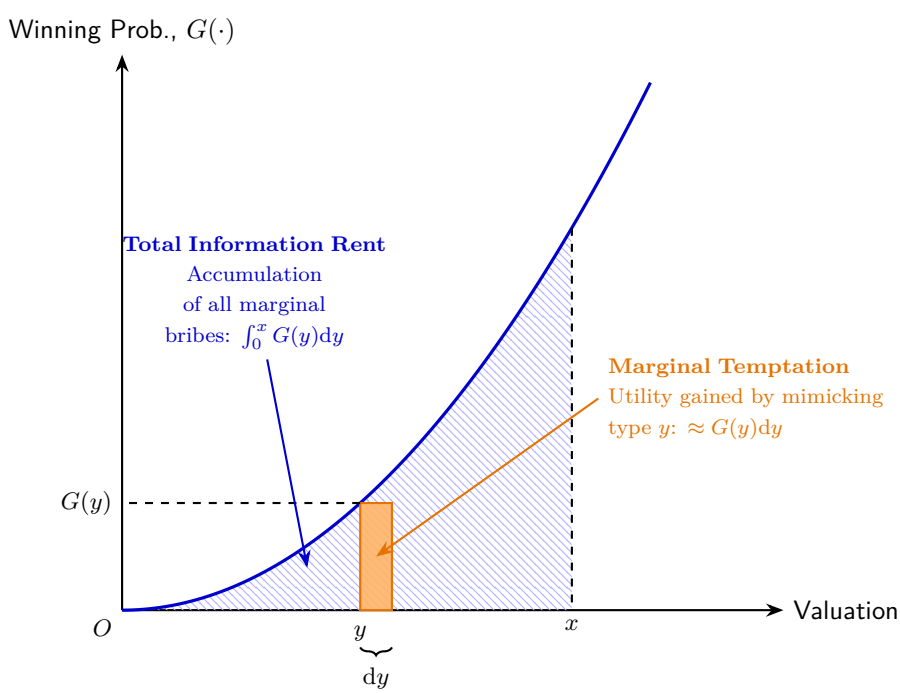
- **Expected Payment ($m(x)$):** This is the area *above* the curve but within the $x \times G(x)$ rectangle. It is the portion of the total surplus that the seller successfully captures as revenue.

From this perspective, the Revenue Equivalence Theorem (shown later) holds because, in any efficient auction, the *information rent* is uniquely determined by the allocation rule $G(x)$.

**Remark (Intuition and Derivation of the Information Rent).**

- **Why the Information Rent is Equal to $\int_0^x G(y)\mathrm{d}y$?**
  In any auction, a high-value bidder naturally wants to under-report their valuation to secure a lower price. Suppose your true valuation is $y + \mathrm{d}y$, but you consider pretending to be type $y$. If you do, you will win the object with probability $G(y)$. Because your actual valuation is higher by $\mathrm{d}y$, you strictly gain an extra marginal utility of $G(y)\mathrm{d}y$ just by lying.

  To achieve *Incentive Compatibility (IC)*, the seller must "bribe" you not to lie. To prevent a bidder with valuation $x$ from mimicking *any* lower type all the way down to 0, the mechanism must compensate them for all these accumulated marginal temptations. Integrating these marginal rents from 0 to $x$ yields the total required information rent: $\int_0^x G(y)\mathrm{d}y$.

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- **Mathematical Derivation (The Envelope Theorem):**
  Let $U(z|x) = xG(z) - m(z)$ be the expected utility of a bidder with true value $x$ reporting value $z$. Under truth-telling (which is optimal by design), the equilibrium utility—which is exactly the information rent—is defined as $V(x) = U(x|x)$.

  By the *Envelope Theorem*, the total derivative of $V(x)$ with respect to $x$ is simply the partial derivative of the objective function $U(z|x)$ with respect to the parameter $x$, evaluated at the optimal choice $z = x$:

  $$V'(x) = \left.\frac{\partial U(z|x)}{\partial x}\right|_{z=x} = G(x)$$

  This differential equation reveals that the marginal growth rate of the information rent is exactly the winning probability. Integrating both sides from 0 to $x$, and assuming the lowest type gets zero rent to satisfy Individual Rationality ($V(0) = 0$), we immediately obtain:

  $$V(x) = \int_0^x G(y)\mathrm{d}y$$

**Remark (An Alternative, Intuitive Form of the Bidding Strategy in a FPA).**

Recall that for a bidder with true value $x$, the bidding strategy is

$$\beta(x) = \frac{\int_0^x yg(y)\mathrm{d}y}{G(x)}.$$

By integration by parts,

$$\int_0^x yg(y)\mathrm{d}y = xG(x) - 0 \cdot G(0) - \int_0^x G(y)\mathrm{d}y.$$

We can rewrite the bidding strategy in a more intuitive form:

$$\beta(x) = x - \frac{\int_0^x G(y)\mathrm{d}y}{G(x)}.$$

This formulation perfectly captures the essence of strategic behavior in a FPA. A rational bidder will never bid their true valuation $x$ (which would yield zero surplus); instead, they "shade" their bid downwards. This formula tells us exactly *how much* they shade:

- $x$ is the bidder's true valuation (their absolute maximum willingness to pay).
- $\int_0^x G(y)\mathrm{d}y$ is the expected *information rent* the bidder is theoretically entitled to because their valuation is private.
- Dividing this expected rent by $G(x)$ converts it into a *conditional rent*—the actual rent the bidder secures *conditional on winning the auction.*

  In an FPA, if you win, you pay exactly your bid $\beta(x)$. This means your actual, realized profit *when you win* is a deterministic amount: $x - \beta(x)$. However, you only win with probability $G(x)$. To achieve your required *expected* information rent (the numerator

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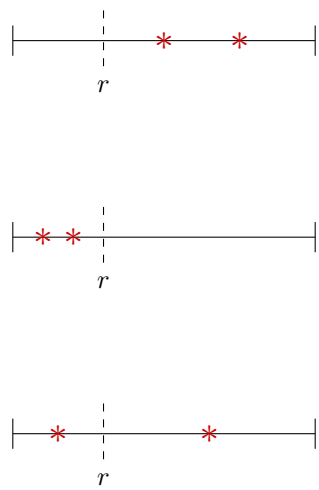
> $\int_0^x G(t)\mathrm{d}t$) unconditionally, your profit on the days you *actually win* must be scaled up by dividing by $G(x)$. Thus, equating your realized profit to the conditional rent gives $x - \beta(x) = \frac{\int_0^x G(t)\mathrm{d}t}{G(x)}$, perfectly explaining the mechanics of bid shading.

## 6.3 Reserve Prices

### 6.3.1 SPA with $n = 2$

In a SPA, the seller can set a reserve price $r$, meaning the item is only sold if the highest bid is no less than $r$. With the reserve price, the payment is then the maximum between the payment determined by the auction rule (namely the second-highest bid) and the reserve price. A natural question arises: *Is introducing a reserve price strictly profitable for the seller?*

To answer this intuitively, let's assume there are $n = 2$ bidders with valuations drawn from a standard Uniform distribution, $F \sim U[0, 1]$. We can perform a marginal analysis by comparing the seller's revenue with no reserve price ($r = 0$) to a scenario with a very small positive reserve price ($r \approx 0$). We can divide the possible valuation realizations into three distinct cases:




- **Case 1: Both valuations are greater than $r$.** The reserve price has no effect on the outcome. The bidder with the highest valuation wins and pays the second-highest valuation. The expected revenue is exactly the same as if there were no reserve price.
- **Case 2: Both valuations are below the reserve price $r$.** In this case, the item is not sold, yielding a revenue of zero. If there were no reserve price, the seller would have received the lower of the two valuations. Since both are drawn from $U[0, r]$, the conditional expected payment would be $r/3$. This case happens with probability $r^2$. Therefore, the unconditional expected loss for the seller is:

$$\text{Expected Loss} = r^2 \cdot \frac{r}{3} = \frac{r^3}{3}$$

- **Case 3: One valuation is above $r$, and the other is below $r$.** The bidder with the higher valuation wins. Without a reserve price, they would only pay the losing bidder's

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valuation (expected to be $r/2$). With the reserve price, the winner is forced to pay $r$. The probability of exactly one bidder drawing a value above $r$ and one below $r$ is $2r(1-r)$. Therefore, the expected gain for the seller is:

$$\text{Expected Gain} = 2r(1-r) \cdot \left(r - \frac{r}{2}\right) = 2r(1-r) \cdot \frac{r}{2} = r^2 - r^3$$

**Claim**

$r = 0$ is never optimal for a revenue-maximizing seller.

***Proof for Claim.***

We evaluate the net change in expected revenue when introducing a small positive reserve price ($r > 0$) compared to not having one ($r = 0$). The net change is the expected gain from Case 3 minus the expected loss from Case 2:

$$\Delta\text{Revenue} = \text{Expected Gain} - \text{Expected Loss} = (r^2 - r^3) - \frac{r^3}{3} = r^2 - \frac{4}{3}r^3$$

For a sufficiently small $r$ (specifically, $0 < r < \frac{3}{4}$), the $r^2$ term strictly dominates the $r^3$ term, meaning $r^2 > \frac{4}{3}r^3$. Consequently, $\Delta\text{Revenue} > 0$. Since setting a tiny positive reserve price strictly increases expected revenue compared to setting $r = 0$, a reserve price of zero is never optimal.

### 6.3.2 Optimal Reserve Price in a FPA

**Claim**

In a FPA with a reserve price $r$, the symmetric bidding strategy for a bidder with valuation $x$ is given by:

$$\beta^r(x) = \begin{cases} \frac{rG(r) + \int_r^x tg(t)\mathrm{d}t}{G(x)} & \text{if } x > r, \\ 0 & \text{if } x \leq r. \end{cases}$$

where $G(x) = F(x)^{n-1}$ is the distribution function of the highest valuation among the $n-1$ opposing bidders, and $g(x)$ is its corresponding probability density function.

***Proof for Claim.***

- **The Conditional Expectation (Intuitive Approach)**

  Without a reserve price, a bidder's optimal strategy in a standard FPA is to bid the expected value of the highest competing valuation, conditional on winning: $\beta(x) = \mathbb{E}[Y_1 | Y_1 < x]$, where $Y_1$ is the highest valuation among opponents.

  When a reserve price $r$ is introduced, the seller acts as a "ghost bidder" who steadfastly bids $r$. To win, a bidder must defeat both the actual opponents ($Y_1 < x$) and the ghost bidder. Thus, the *effective* highest competing valuation becomes $\max(Y_1, r)$.

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The optimal bid is the conditional expectation of this effective opponent:

$$\beta^r(x) = \mathbb{E}[\max(Y_1, r) \mid Y_1 < x]$$

By definition of conditional expectation, we integrate over the distribution of $Y_1$ up to $x$:

$$\beta^r(x) = \int_0^x \max(y, r) \left(\frac{g(y)}{G(x)}\right) \mathrm{d}y$$

We can split this integral at $t = r$:

$$\beta^r(x) = \frac{1}{G(x)} \left(\int_0^r rg(y)\mathrm{d}y + \int_r^x yg(y)\mathrm{d}y\right)$$

Since $\int_0^r g(y)\mathrm{d}y = G(r)$, the first term evaluates to $rG(r)$. This leads directly to our familiar result:

$$\beta^r(x) = \frac{rG(r) + \int_r^x yg(y)\mathrm{d}y}{G(x)}$$

- **First Order Approach**

  Suppose a bidder has a true valuation $x > r$, but considers reporting a valuation $z$ (and bidding $\beta(z)$). The bidder's expected utility is the probability of winning multiplied by the surplus:

$$U(z, x) = (x - \beta(z))G(z)$$

  To find the symmetric equilibrium, truth-telling must be optimal, meaning the first-order derivative with respect to $z$ evaluated at $z = x$ must be zero:

$$\left.\frac{\partial U(z, x)}{\partial z}\right|_{z=x} = -\beta'(x)G(x) + (x - \beta(x))g(x) = 0$$

  Rearranging terms yields a linear ordinary differential equation (ODE):

$$\beta'(x)G(x) + \beta(x)g(x) = xg(x) \implies \frac{d}{dx}[\beta(x)G(x)] = xg(x)$$

  Integrating both sides from the reserve price $r$ to $x$:

$$\beta(x)G(x) - \beta(r)G(r) = \int_r^x yg(y)\mathrm{d}y$$

  Consider the marginal buyer whose valuation is exactly $x = r$. To acquire the item without strictly losing money, this bidder must bid exactly the reserve price. Thus, $\beta(r) = r$. Substituting this boundary condition into the equation gives:

$$\beta(x)G(x) - rG(r) = \int_r^x yg(y)\mathrm{d}y \implies \beta^r(x) = \frac{rG(r) + \int_r^x yg(y)\mathrm{d}y}{G(x)}$$

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**Proposition 6.1: Optimal Reserve Price For a FPA**

The optimal reserve price $r^*$ for a revenue-maximizing seller in a First-Price Auction with $n$ bidders and valuations drawn from a distribution $F$ is given by:

$$r^* = \frac{1 - F(r^*)}{f(r^*)}.$$

***Proof for Proposition.***

For a FPA with a reserve price $r$, the symmetric bidding strategy is then

$$\beta^r(x) = \begin{cases} \frac{rG(r) + \int_r^x yg(y)\mathrm{d}y}{G(x)} & \text{if } x > r, \\ 0 & \text{if } x \leq r. \end{cases}$$

So the expected payment for a bidder with value $x > r$ is given by

$$m^r(x) = rG(r) + \int_r^x tg(t)\mathrm{d}t.$$

And the expected payment for a bidder with value $x \leq r$ is zero: $m^r(x) = 0$.
The *ex-ante* expected payment for a bidder with valuation drawn from $F$ is then

$$m^r = \int_0^1 m^r(x)f(x)dx = r\left[1 - F(r)\right]G(r) + \int_r^1 t\left[1 - F(t)\right]g(t)\mathrm{d}t.$$

Hence, the optimal reserve price satisfies

$$\frac{\mathrm{d}m^r}{\mathrm{d}r} = \left[1 - F(r)\right]G(r) - rG(r)f(r) + r\left[1 - F(r)\right]g(r) - r\left[1 - F(r)\right]g(r) = 0.$$

Therefore, the optimal reserve price $r^*$ is

$$r^* = \frac{1 - F(r^*)}{f(r^*)}.$$

Interestingly, the optimal reserve price does not depend on the number of bidders $n$.

## 6.4 Revenue Equivalence Theorem

We make a comparison of the expected payment in a FPA and a SPA with private value $x$.

In a SPA, you bid $x$, and you pay the second-highest bid, which is $Y_1$. Conditional on winning, your expected payment is $\mathbb{E}\left[Y_1 | Y_1 < x\right]$. Since you win the auction with probability $G(x)$, your unconditional expected payment (prior to knowing if you won) is $m(x) = G(x) \cdot \mathbb{E}\left[Y_1 | Y_1 < x\right]$.

In a FPA, you bid $\beta(x) = \mathbb{E}\left[Y_1 | Y_1 < x\right]$, and you pay your own bid if you win the auction. You win the auction if $Y_1 < x$, which happens with probability $G(x)$.

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The expected payment in both mechanisms for a bidder with valuation $x$ is:

$$m(x) = G(x) \cdot \mathbb{E}\left[Y_1 | Y_1 < x\right] = \int_0^x y g(y) \mathrm{d}y.$$

*Ex-ante*, the expected payment for each player is given by

$$m = \mathbb{E}_X\left[m(X)\right] = \int_0^1 m(x) f(x) \mathrm{d}x = \int_0^1 \left( \int_0^x y g(y) \mathrm{d}y \right) f(x) \mathrm{d}x.$$

Hence, the expected revenue (oftentimes called the expected selling price) is the same in a FPA and a SPA, which is just $n \times m$.

This result is more general than just comparing the FPA and SPA. It actually applies to any two auction mechanisms that satisfy certain conditions, as stated in the following theorems.

### Theorem 6.2: Revenue Equivalence Theorem (RET)

Assume there are $n$ risk-neutral bidders with independent private values drawn from a strictly increasing cumulative distribution function $F$ on an interval $[0, \bar{x}]$. Any auction mechanism that satisfies the following two conditions:

1. **Allocative Efficiency:** The object is always allocated to the bidder with the highest valuation.
2. **Zero Payment at the Bottom:** Any bidder with the lowest possible valuation ($x = 0$) has an expected payment of zero.

will yield the exact same expected payment $m(x)$ for every bidder type $x$, and consequently, generate the exact same expected revenue for the seller.

### Corollary 6.3

Standard auction formats such as the FPA and the SPA all satisfy these conditions and are therefore revenue equivalent.

**Remark (The "Pie-Splitting" Economic Intuition).**

- **Baking the Pie (Total Social Surplus):** The allocation rule decides who ultimately receives the object, which dictates the total value created by the mechanism (the size of the pie). Because standard auctions always allocate the object to the bidder with the highest valuation, they are all efficient and thus "bake" the exact same size of pie.
- **The Bidder's Slice (Information Rent):** Because valuations are private, the seller must leave some surplus to the bidders to induce truth-telling. By the Envelope Theorem, this information rent is dictated *entirely* by the allocation probability ($V(x) = \int_0^x G(y)\mathrm{d}y$). Since the allocation rules are identical across these standard auctions, the winning probabilities are identical, meaning the bidders carve out the exact same slice of the pie.

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- **The Seller's Slice (Expected Revenue):** The fundamental accounting identity of any mechanism is:

$$\text{Expected Revenue} = \text{Total Social Surplus} - \text{Information Rent}$$

If two auctions bake the exact same total pie, and the bidders are guaranteed to take the exact same size slice in both formats, then the remaining portion—the revenue left on the table for the seller—must logically be absolutely identical.

Having established the revenue equivalence of standard auctions, a natural question arises for a profit-maximizing seller: *Are standard auctions the best we can do?* To answer this, we must step out of the rigid, predefined rules of specific auction formats (like FPA or SPA) and adopt the "God's-eye view" of a mechanism designer. Instead of asking "what happens in an auction," we ask: "over the space of *all possible mechanisms*, which one maximizes the seller's expected revenue?" This requires a more fundamental, generalized mathematical law that governs *any* allocation rule, not just the efficient ones.

### Theorem 6.4: General RET (Payoff Equivalence)

Assume there are $n$ risk-neutral bidders with independent private values drawn from a strictly increasing cumulative distribution function $F$ on an interval $[0, \bar{x}]$. Consider any two incentive-compatible mechanisms. If both mechanisms satisfy the following two conditions:

1. **Identical Allocation Rule:** The interim probability of receiving the object for a bidder with valuation $x$, denoted by $q(x)$, is exactly the same across both mechanisms.
2. **Identical Payment at the Bottom:** The expected payment of a bidder with the lowest possible valuation ($x = 0$) is the same across both mechanisms (typically zero).

Then, both mechanisms yield the exact same expected payment $m(x)$ for every bidder type $x$, and consequently, they generate the exact same expected revenue for the seller.

**Remark (Why is this theorem more fundamental than the Standard RET?).**

The Standard RET rigorously requires **Allocative Efficiency**—meaning the object *must* always go to the bidder with the highest valuation. The General RET, however, removes this straitjacket. It mathematically proves that for *any* arbitrary, monotonic allocation rule $q(x)$ (even inefficient ones), the payment function is completely and uniquely pinned down by the Envelope Theorem:

$$m(x) = m(0) + xq(x) - \int_0^x q(t)\mathrm{d}t$$

This "Payoff Equivalence" is the ultimate theoretical weapon for the seller. It implies that the seller's revenue maximization problem is no longer a daunting search through

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> complex, multidimensional payment rules, but can be *entirely reduced to designing the optimal one-dimensional allocation rule* $q(x)$.

Armed with the General RET, the seller arrives at a somewhat counterintuitive economic truth: *To strictly maximize revenue, the seller must be willing to destroy social efficiency.* If the seller simply uses a standard efficient auction, they leave money on the table in the form of excessive information rents conceded to high-value bidders. To extract more surplus, for example, the seller can strategically introduce a *reserve price (r).* With a reserve price, if the highest bidder's valuation is positive but falls below $r$, the seller retains the object. This explicitly violates the "*Allocative Efficiency*" condition of the Standard RET (because mutually beneficial trade is artificially blocked). However, the *General RET still applies perfectly*! We merely apply the generalized math to a newly shaped allocation rule $q^r(x)$, where the probability of winning is squashed to zero for all types below $r$. This strategic manipulation of the allocation rule to squeeze information rents leads directly to the seminal framework of Optimal Mechanism Design.

## 6.5 Risk Attitude and the Breakdown of Revenue Equivalence

All the analysis before was built on the assumption that bidders are risk-neutral. What happens when we relax this assumption? Suppose now that the bidders are strictly *risk averse*, with a utility function $u(\cdot)$ such that $u' > 0$, $u'' < 0$, and $u(0) = 0$.

First, for the SPA, risk aversion *has no effect* on the equilibrium of an SPA. Truth-telling ($\beta(x) = x$) remains a weakly dominant strategy.

However, the equilibrium in a FPA changes dramatically. Suppose all other $n-1$ bidders follow a strictly increasing and differentiable symmetric strategy $\gamma(x)$, with $\gamma(0) = 0$. The expected utility for a bidder with true value $x$ who submits a bid $b$ is given by:

$$\begin{aligned} U(b,x) &= u(x-b) \cdot \Pr\left(b > \max_{j \neq i} \gamma(X_j)\right) \\ &= u(x-b) \cdot \Pr\left(\gamma^{-1}(b) > \max_{j \neq i} X_j\right) \\ &= u(x-b) \cdot G(\gamma^{-1}(b)), \end{aligned}$$

where $G(\cdot)$ is the distribution function of the highest opposing valuation.

To find the optimal bid, we take the FOC with respect to $b$:

$$\frac{\partial U(b,x)}{\partial b} = -u'(x-b) \cdot G(\gamma^{-1}(b)) + u(x-b) \cdot \frac{g(\gamma^{-1}(b))}{\gamma'(\gamma^{-1}(b))} = 0$$

In a symmetric equilibrium, the optimal bid must be $b = \gamma(x)$, which implies $\gamma^{-1}(b) = x$. Substituting this into the FOC gives:

$$-u'(x-\gamma(x)) \cdot G(x) + u(x-\gamma(x)) \cdot \frac{g(x)}{\gamma'(x)} = 0$$

Rearranging this equation yields the differential equation for the bidding strategy under risk

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aversion:

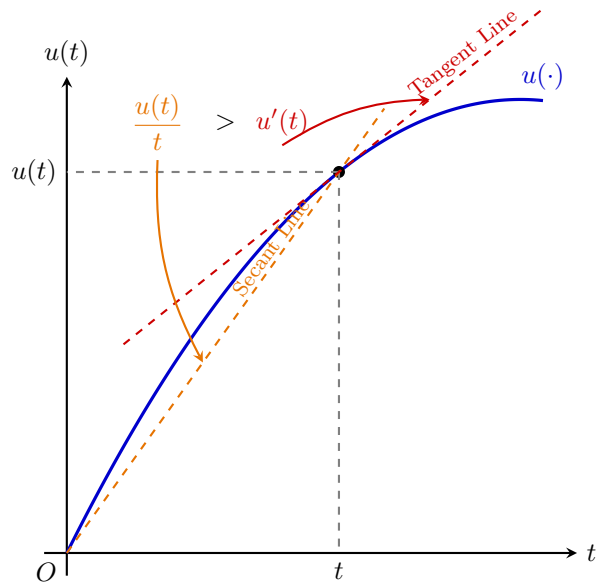
$$\gamma'(x) = \frac{u(x-\gamma(x))}{u'(x-\gamma(x))} \cdot \frac{g(x)}{G(x)}$$

To see how this compares to the risk-neutral case, recall that the risk-neutral bidding strategy $\beta(x)$ satisfies:

$$\beta'(x) = (x - \beta(x)) \cdot \frac{g(x)}{G(x)}$$

Note that because $u(\cdot)$ is a strictly concave function with $u(0) = 0$, its graph lies strictly *below* its tangent lines. Geometrically, for any point $t > 0$, the slope of the secant line connecting the origin to $(t, u(t))$ is strictly steeper than the slope of the tangent line at that exact point. Therefore, we directly have the slope inequality:

$$\frac{u(t)}{t} > u'(t) \implies \frac{u(t)}{u'(t)} > t$$




**Proposition 6.5: Overbidding under Risk Aversion**

For any strictly positive valuation $x > 0$, a risk-averse bidder bids strictly higher than a risk-neutral bidder:

$$\gamma(x) > \beta(x).$$

***Proof for Proposition.***

Note that a bidder with the lowest possible valuation ($x = 0$) bids zero regardless of their risk attitude, so we have the boundary condition $\gamma(0) = \beta(0) = 0$.

Suppose, for the sake of contradiction, that there exists some $x > 0$ where $\gamma(x) \leq \beta(x)$. Let $t = x - \gamma(x) > 0$. By the concavity property discussed above, $\frac{u(t)}{u'(t)} > t$, meaning:

$$\frac{u(x-\gamma(x))}{u'(x-\gamma(x))} > x - \gamma(x) \geq x - \beta(x).$$

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Substituting this strict inequality into our differential equation for $\gamma'(x)$:

$$\gamma'(x) = \frac{u(x-\gamma(x))}{u'(x-\gamma(x))} \cdot \frac{g(x)}{G(x)} > (x-\beta(x)) \cdot \frac{g(x)}{G(x)} = \beta'(x).$$

This implies that at any point where $\gamma(x)$ is equal to or below $\beta(x)$, the slope of $\gamma$ is strictly steeper than the slope of $\beta$. Since both functions start at the same origin ($\gamma(0) = \beta(0) = 0$), $\gamma$ must immediately rise above $\beta$. Furthermore, $\gamma$ can never cross $\beta$ from above, because at any intersection, $\gamma$ must be steeper. Thus, we conclude $\gamma(x) > \beta(x)$ for all $x > 0$.

**Remark (The "Insurance Premium" Intuition).**

Why does a risk-averse bidder bid higher? In a First-Price Auction, a bidder faces a trade-off: bidding lower increases the profit *if* they win, but increases the probability of losing and getting exactly zero. A risk-averse individual deeply dislikes the downside risk of losing the item. To protect themselves against this zero-payoff outcome, they are willing to sacrifice some potential profit by bidding more aggressively. In essence, the difference $\gamma(x) - \beta(x)$ is an **insurance premium** the bidder pays to increase their probability of winning.

Because risk aversion causes bidders to bid more aggressively in the FPA, while leaving the dominant strategy in the SPA unchanged, the expected payments in the FPA are strictly higher. Therefore, under risk aversion, the First-Price Auction generates strictly higher expected revenue for the seller than the Second-Price Auction.

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# Chapter 7

# The Revelation Principle and Mechanism Design

## 7.1 Direct Mechanism and Revelation Principle

In exploring the optimal way to sell an object to $n$ buyers, we establish a rigorous mathematical framework. Suppose the seller's reservation value for the object is $x_0$, and the buyers' true valuations $X_i$ are drawn independently from a distribution $F_i$.

Any complex selling procedure in reality can be abstracted as a *general mechanism*. This mechanism consists of three mathematical components, denoted as $(\mathcal{B}, \pi, \mu)$:

- **Message Space ($\mathcal{B}$):** The set of messages (or "bids") that buyers can submit. For $n$ buyers, the global message space is $\mathcal{B} = \mathcal{B}_1 \times \mathcal{B}_2 \times \cdots \times \mathcal{B}_n$.
- **Allocation Rule ($\pi(b)$):** Given a profile of submitted messages $b$, $\pi_i(b)$ determines the probability that buyer $i$ receives the object.
- **Payment Rule ($\mu(b)$):** Given a profile of submitted messages $b$, $\mu_i(b)$ determines the amount that buyer $i$ must pay to the seller.

Under this mechanism, suppose the buyers play a game and reach an equilibrium. Each buyer employs an *equilibrium strategy* $\beta_i(x_i)$, which maps their true valuation $x_i$ to an optimal message $b_i$ that maximizes their expected utility.

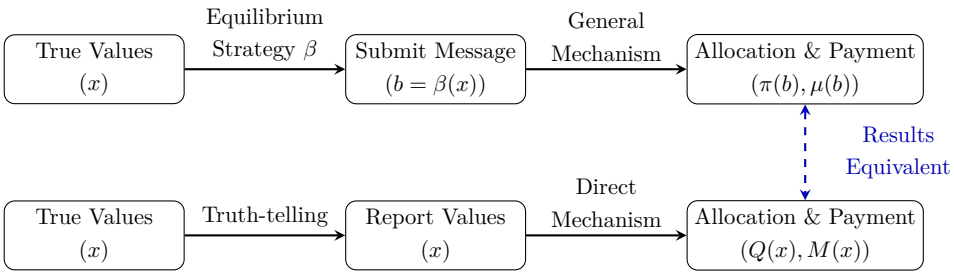
Analyzing general mechanisms can be mathematically daunting due to the potentially infinite complexity of the message space. To drastically simplify this problem, we introduce the concept of a *direct mechanism*.

In a direct mechanism $([0,1]^n, Q, M)$, the seller simply asks the buyers to report their true valuations directly. Thus, the message space is restricted to the valuation space itself (i.e., $\mathcal{B}_i = [0,1]$). The mechanism's allocation and payment rules are implemented by applying the equilibrium strategies $\beta(x)$ from the original mechanism "behind the scenes":

- **Direct Allocation Rule:** $Q_i(x) = \pi_i(\beta(x))$
- **Direct Payment Rule:** $M_i(x) = \mu_i(\beta(x))$

*The Revelation Principle* states a profound result: If there exists an equilibrium outcome in some complex general mechanism, we can always construct a corresponding direct mechanism where truth-telling (reporting true valuations honestly) constitutes an equilibrium for all buyers. Furthermore, this truth-telling equilibrium yields the exact same allocation probabilities and expected payments as the original mechanism.

This principle is the cornerstone of mechanism design. It greatly simplifies the search for an "optimal auction": instead of exhaustively searching through all possible auction formats and bidding rules, the mechanism designer only needs to optimize over the set of direct mechanisms subject to the constraint that buyers are willing to report truthfully (the *incentive compatibility constraint*).




## Theorem 7.1: Revelation Principle

If $\beta$ is an equilibrium of a general mechanism $(\mathcal{B}, \pi, \mu)$, then truth-telling (reporting $x_i$ honestly) is an equilibrium of the corresponding direct mechanism $([0,1]^n, Q, M)$.

***Proof for Theorem***

Intuitively, if a buyer found it optimal to play the strategy $\beta(x_i)$ in the original complex mechanism, they will find it optimal to simply report their true type $x_i$ in the direct mechanism, because the direct mechanism automatically plays their optimal strategy $\beta(x_i)$ on their behalf. Any deviation from truth-telling would result in the mechanism executing a sub-optimal strategy for that buyer.

## Corollary 7.2: Expected Payment Identity

In any incentive-compatible mechanism with a reserve price $r$, the expected payment $m(x)$ of a bidder with valuation $x \geq r$ is uniquely determined by the allocation rule $G(\cdot)$ and the expected payment of the marginal type $m(r)$. Specifically, it is given by:

$$m(x) = m(r) + \int_r^x y g(y) \mathrm{d}y,$$

where $g(y) = G'(y)$ is the density function associated with the allocation rule.

***Proof for Corollary.***

Let $U(z|x) = xG(z) - m(z)$ denote the expected utility of a bidder with true valuation $x$ who reports $z$. By the Revelation Principle, we focus on the truth-telling equilibrium

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where the value function is:

$$V(x) = U(x|x) = \max_z (xG(z) - m(z)).$$

By the Envelope Theorem, the derivative of the value function with respect to $x$ is simply $V'(x) = G(x)$. Integrating this from the reserve price $r$ to $x$ yields:

$$V(x) = V(r) + \int_r^x G(y)\mathrm{d}y.$$

Using the definition $V(x) = xG(x) - m(x)$, we can isolate the expected payment:

$$m(x) = xG(x) - V(r) - \int_r^x G(y)\mathrm{d}y.$$

Applying integration by parts to the term $\int_r^x yg(y)\mathrm{d}y = xG(x) - rG(r) - \int_r^x G(y)\mathrm{d}y$, we can rewrite the payment equation as:

$$m(x) = rG(r) + \int_r^x tg(y)\mathrm{d}y - V(r).$$

Recall that the equilibrium utility for the marginal bidder is $V(r) = rG(r) - m(r)$. Substituting this definition into the equation above immediately yields the desired result:

$$m(x) = m(r) + \int_r^x yg(y)\mathrm{d}y.$$

## 7.2 Optimal Mechanism Design (Myerson, 1981)

### 7.2.1 Motivating Example: Monopoly Pricing

What is the best way to sell an object to $n$ buyers? Let us start with the simplest case: $n = 1$.

Suppose the single buyer's valuation $X$ is drawn from a distribution with cumulative distribution function $F(x)$ and probability density function $f(x)$. The seller wants to post a take-it-or-leave-it price $p$.

Importantly, let $x_0$ be the *seller's own valuation* (or reservation value) of the object. This represents the utility the seller retains if the transaction fails and they keep the item.

The seller's expected profit $\mathbb{E}[\pi(p)]$ from posting price $p$ consists of two mutually exclusive scenarios:

- **Transaction succeeds (Probability $1 - F(p)$):** The buyer's valuation $X \geq p$. The seller collects the payment $p$.
- **Transaction fails (Probability $F(p)$):** The buyer's valuation $X < p$. The seller keeps the object and retains a value of $x_0$.

Thus, the expected profit is given by:

$$\mathbb{E}[\pi(p)] = p \cdot \Pr(X \geq p) + x_0 \cdot \Pr(X < p) = p(1 - F(p)) + x_0 F(p)$$

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To find the revenue-maximizing optimal price $p^*$, we take the first derivative of the expected profit with respect to $p$ and set it to zero:

$$\frac{\mathrm{d}\mathbb{E}[\pi(p)]}{\mathrm{d}p} = (1 - F(p)) - pf(p) + x_0 f(p) = 0$$

By dividing by $f(p)$ and rearranging the terms, we arrive at a profound economic condition:

$$p^* - \frac{1 - F(p^*)}{f(p^*)} = x_0$$

The expression $p - \frac{1-F(p)}{f(p)}$ is a fundamental concept in mechanism design known as the *virtual value*. The term $\frac{f(p)}{1-F(p)}$ is defined as the *hazard rate* of the valuation distribution. This equation dictates that a revenue-maximizing seller should set the optimal posted price $p^*$ exactly at the point where the buyer's *virtual value* equals the seller's *true valuation* $x_0$.

*In many standard textbook problems, we assume the object has zero use value to the seller, i.e., $x_0 = 0$. In such cases, the FOC simplifies to $p^* - \frac{1-F(p^*)}{f(p^*)} = 0$.*

### 7.2.2 Framing Seller's Problem

By the Revelation Principle, instead of searching through an infinite space of possible game rules and message spaces, the seller *only needs to find the optimal direct mechanism* $(Q, M)$ subject to incentive compatibility (truth-telling) constraints. There is no loss of generality in restricting our attention to direct mechanisms where the message space is simply the type space, $\mathcal{B}_i = [0, 1]$.

The mechanism designer (the seller) optimizes over two functions: the *allocation rule* $Q$ and the *payment rule* $M$:

$$Q : [0, 1]^n \to \Delta, \quad M : [0, 1]^n \to \mathbb{R}^n,$$

where $\Delta$ is the probability simplex over the $n$ buyers (i.e., $Q_i(x)$ is the probability that buyer $i$ receives the object given the profile of reported valuations $x$).

**Definition 7.3: Interim Expected Probability of Winning & Interim Expected Payment**

For any bidder $i$, suppose they report their valuation as $z_i$ while all other $n - 1$ bidders report truthfully. The *interim expected probability of winning* $q_i(z_i)$ and the *interim expected payment* $m_i(z_i)$ are defined as:

$$q_i(z_i) = \int_{[0,1]^{n-1}} Q_i(z_i, x_{-i}) f_{-i}(x_{-i}) \mathrm{d}x_{-i}$$
$$m_i(z_i) = \int_{[0,1]^{n-1}} M_i(z_i, x_{-i}) f_{-i}(x_{-i}) \mathrm{d}x_{-i}$$

where $f_{-i}(x_{-i}) = \prod_{j \neq i} f_j(x_j)$ is the joint density of all other bidders' true valuations.

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The expected payoff for bidder $i$ with true valuation $x_i$ who reports $z_i$ is therefore:

$$\pi_i(x_i, z_i) = q_i(z_i) \cdot x_i - m_i(z_i)$$

To ensure that truth-telling is a Bayesian Nash Equilibrium,[1] the mechanism must satisfy the *incentive compatibility constraint*:

### Incentive Compatibility Constraint (IC)

A mechanism $(Q, M)$ is *incentive compatible* if the following holds:

$$\pi_i(x_i, z_i) \geq \pi_i(x_i, x_i), \quad \forall i, \quad \forall x_i \in [0,1], \quad \forall z_i \in [0,1].$$

If the IC constraint is satisfied, every bidder will tell the truth. In this case, we can express the equilibrium payoff of bidder $i$ as a function of their true type $x_i$:

$$u_i(x_i) \equiv \pi_i(x_i, x_i) = \max_{z_i} \{q_i(z_i) \cdot x_i - m_i(z_i)\}$$

### Claim: Properties of the IC Mechanism

From the definition of the equilibrium payoff function $u_i(x_i)$,

$$u_i(x_i) \equiv \pi_i(x_i, x_i) = \max_{z_i} \{q_i(z_i) \cdot x_i - m_i(z_i)\},$$

we immediately obtain three fundamental results:

1. $u_i(x_i)$ is a convex function of $x_i$.
2. $u_i'(x_i) = q_i(x_i)$ almost everywhere.
3. $q_i(x_i)$ is non-decreasing in $x_i$.

***Proof for Claim.***

- **Convexity:** The value function $u_i(x_i)$ is defined as the upper envelope (the pointwise maximum) of a family of functions in $x_i$. Since $q_i(z_i)x_i - m_i(z_i)$ is an affine (linear) function of $x_i$, their upper envelope is globally convex.[a]
- **Envelope Theorem:** By the Envelope Theorem, the total derivative of the maximized value function with respect to the parameter $x_i$ equals the partial derivative of the objective function evaluated at the optimal choice ($z_i = x_i$). Thus, $u_i'(x_i) = q_i(x_i)$.
- **Monotonicity:** Because $q_i(x_i) = u_i(x_i)$ and thus $q_i'(x_i) = u_i'(x_i)$, and $u_i$ is convex ($u_i''(x_i) \geq 0$), $q_i(x_i)$ is naturally non-decreasing in $x_i$.

[1] **Nash Equilibrium (NE) vs. Bayesian Nash Equilibrium (BNE):** In a traditional NE, players have complete information about the game, including the exact payoffs of all other players. In a mechanism design setting, information is incomplete: a bidder knows their own valuation but only knows the *distribution* of others' valuations. A BNE is the extension of NE to games of incomplete information, where each player's strategy maximizes their *expected* payoff, taking expectations over the possible types of other players based on a common prior distribution.

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[a] **Linear vs. Affine:** Strictly speaking, a *linear* function must pass through the origin ($f(x) = ax$), whereas an *affine* function is a linear function plus a constant translation ($f(x) = ax + b$). Sometimes economists sometimes use the terms interchangeably.

### Proposition 7.4: Allocation Rule Uniquely Determines Payment Rule

In any incentive-compatible (IC) and individually rational (IR) mechanism designed by a profit-maximizing seller, the equilibrium expected payoff $u_i(x_i)$ and the expected payment $m_i(x_i)$ for any bidder $i$ are completely and uniquely determined by the allocation rule $Q$. Specifically, the lowest type extracts zero surplus ($u_i(0) = 0$), yielding the exact payoff function:

$$u_i(x_i) = \int_0^{x_i} q_i(t)\mathrm{d}t$$

***Proof for Proposition.***

Since $u_i'(x_i) = q_i(x_i)$, we can express the payoff function as an integral of the interim winning probability:

$$u_i(x_i) = u_i(0) + \int_0^{x_i} q_i(t)\mathrm{d}t$$

This implies that once the allocation rule $Q$ (and thus the interim probability $q_i$) is fixed, the equilibrium payoff function $u_i$ is entirely pinned down up to a constant $u_i(0)$. Loosely speaking, the IC constraint determines the *shape* of the payoff function. Consequently, the shape of the expected payment $m_i(x_i) = q_i(x_i)x_i - u_i(x_i)$ is also uniquely determined by $Q$.

Furthermore, to maximize revenue, the seller should extract as much surplus as possible without violating the *individual rationality constraint*:

### Individual Rationality Constraint (IR)

A mechanism $(Q, M)$ is *individually rational* if the following holds:

$$u_i(x_i) \geq 0, \quad \forall i.$$

Since $u_i(x_i)$ is non-decreasing, the IR constraint is satisfied everywhere if and only if $u_i(0) \geq 0$. Note that by definition, $u_i(0) = q_i(0) \cdot 0 - m_i(0) = -m_i(0)$, so:

$$u_i(0) \geq 0 \implies m_i(0) \leq 0.$$

Because the seller is a profit-maximizer, $m_i(0) \leq 0$ immediately implies they will set $m_i(0) = 0$, which means $u_i(0) = 0$. Thus:

$$u_i(x_i) = \int_0^{x_i} q_i(t)\mathrm{d}t.$$

That is, the IR constraint further pins down the constant term in the payoff function,

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leaving *zero degrees of freedom* for the payoff function (and thus the payment rule) once the allocation rule $Q$ is fixed.

The seller's ultimate problem is to maximize total expected revenue:

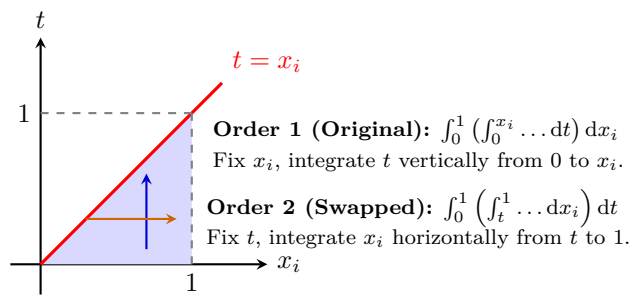
$$\max_{Q,M} \sum_{i=1}^{n} \int_0^1 m_i(x_i) f_i(x_i) \mathrm{d}x_i$$

Substitute the expected payment identity $m_i(x_i) = q_i(x_i)x_i - \int_0^{x_i} q_i(t)\mathrm{d}t$ into the integral for bidder $i$:

$$\int_0^1 m_i(x_i) f_i(x_i) \mathrm{d}x_i = \int_0^1 q_i(x_i) x_i f_i(x_i) \mathrm{d}x_i - \int_0^1 \left( \int_0^{x_i} q_i(t)\mathrm{d}t \right) f_i(x_i) \mathrm{d}x_i$$

We swap the order of integration for the second term (visual proof below):

$$\int_0^1 \int_0^{x_i} q_i(t)\mathrm{d}t f_i(x_i)\mathrm{d}x_i = \int_0^1 \int_t^1 f_i(x_i)\mathrm{d}x_i \, q_i(t)\mathrm{d}t = \int_0^1 (1 - F_i(x_i)) q_i(x_i) \mathrm{d}x_i$$




Substituting this back into the revenue equation gives us the expected revenue purely as a function of the allocation rule:

$$\int_0^1 m_i(x_i) f_i(x_i) \mathrm{d}x_i = \int_0^1 \left[ x_i - \frac{1 - F_i(x_i)}{f_i(x_i)} \right] q_i(x_i) f_i(x_i) \mathrm{d}x_i$$

We define the bracketed term as the *virtual value*:

**Definition 7.5: Virtual Value**

The *virtual value* of a bidder with true type $x_i$, denoted by $\phi_i(x_i)$, is defined as

$$\phi_i(x_i) = x_i - \frac{1 - F_i(x_i)}{f_i(x_i)}.$$

**Why?** To understand the logical leap to "money", we must look at the mechanics of the Incentive Compatibility (IC) constraint: $u_i(x_i) = \int_0^{x_i} q_i(t)\mathrm{d}t$.

Suppose the seller decides to increase the allocation probability $q_i(x)$ for a specific type $x$. Because the expected payoff $u_i$ is the integral of $q_i$ from 0 up to the true type, increasing $q_i(x)$ does not just increase the payoff for type $x$—it automatically increases the payoff for *every single type strictly greater than* $x$!

If the seller makes it attractive for type $x$ to win the object, mimicking type $x$ suddenly

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becomes very tempting for everyone with a higher valuation. To prevent all higher types from lying and claiming to be type $x$ to get a cheaper price, the seller is forced to "bribe" them by leaving more surplus (money) on the table.

How many higher types are there? The probability mass of bidders with a valuation $> x$ is exactly $1-F_i(x)$. Therefore, $1-F_i(x)$ represents the total volume of "bribes" (information rent) the seller must pay out to the upper tail of the distribution. Dividing this by the local density $f_i(x)$ simply normalizes this total cost into a *marginal* cost evaluated at type $x$.

Thus, $\frac{1-F_i(x_i)}{f_i(x_i)}$ is exactly the expected dollar amount of surplus the mechanism must concede *locally* at type $x_i$: allocating to $x_i$ forces the seller to surrender rent to the $1-F_i(x_i)$ mass of people above them!

**Remark (Understanding the Information Rent).**

The virtual value $\phi_i(x_i)$ is strictly less than the true valuation $x_i$. The subtracted term, $\frac{1-F_i(x_i)}{f_i(x_i)}$ (the inverse hazard rate), represents the *information rent*. This is because of the IC constraint: If the seller allocates the object to a bidder with valuation $x_i$, they cannot simply charge them $x_i$. To prevent all types strictly higher than $x_i$ from lying and mimicking type $x_i$ to get a cheaper price, the seller must leave some surplus (rent) on the table for those higher types.

By the Envelope Theorem, a bidder of type $x_i$ earns a payoff of $u_i(x_i) = \int_0^{x_i} q_i(t)\mathrm{d}t$. When we take the expectation of this payoff across all possible types, the mathematical result is $\mathbb{E}[u_i] = \mathbb{E}\left[q_i(x_i)\frac{1-F_i(x_i)}{f_i(x_i)}\right]$. Thus, $\frac{1-F_i(x_i)}{f_i(x_i)}$ is exactly the expected dollar amount of surplus (information rent) the mechanism must concede *locally* at type $x_i$ to ensure all higher types truthfully reveal themselves.

Therefore, $\phi_i(x_i)$ is the *actual marginal revenue* the seller extracts when allocating the object to a bidder with value $x_i$.

Recall that by definition, the interim allocation probability is the expected value of the ex-post allocation rule over all other bidders' types:

$$q_i(x_i) = \int_{[0,1]^{n-1}} Q_i(x_i, x_{-i}) f_{-i}(x_{-i})\mathrm{d}x_{-i}$$

By expanding $q_i(x_i)$ back into the joint integral over all $n$ bidders, the seller's optimization problem simplifies remarkably to maximizing the expected virtual surplus:

$$\max_Q \int_{[0,1]^n} \left[\sum_{i=1}^n \phi_i(x_i) Q_i(x)\right] f(x)\mathrm{d}x$$

subject to the physical constraints that $Q_i(x) \in [0,1]$, $\sum Q_i(x) \leq 1$, and the IC monotonicity constraint that $q_i(\cdot)$ is non-decreasing.

**Assumption 7.6: Regularity Condition on $\phi_i$**

The virtual value $\phi_i(x_i)$ is strictly increasing in $x_i$.

**Remark (Why Regularity Guarantees Monotonicity).**

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To understand why this condition is critical, recall that the seller's objective is to maximize the expected virtual surplus. For any given profile of reported types $x$, the seller wants to choose an allocation rule $q(x)$ to maximize:

$$\max_q \sum_{i=1}^{n} q_i(x)\phi_i(x_i)$$

subject to the feasibility constraint $\sum q_i(x) \leq 1$.

Ignoring the IC constraint for a moment, the unconstrained optimal way to maximize this sum is simply to allocate the object to the bidder with the highest strictly positive virtual value. That is, $q_i(x) = 1$ if $\phi_i(x_i) > \max_{j \neq i} \phi_j(x_j)$ and $\phi_i(x_i) > 0$.

Recall from the Envelope Theorem that for a mechanism to be Incentive Compatible (IC), the interim allocation probability $q_i(x_i)$ must be non-decreasing in $x_i$.

If the Regularity Condition holds (i.e., $\phi_i$ is strictly increasing), then a higher true type $x_i$ directly translates to a strictly higher virtual value $\phi_i(x_i)$. A higher virtual value makes it strictly more likely that bidder $i$ outbids all competitors and clears the seller's reserve price. Therefore, the allocation probability $q_i(x_i)$ naturally goes up as $x_i$ goes up.

In short, the regularity condition ensures that the greedy, unconstrained pointwise maximization algorithm *automatically* satisfies the global IC monotonicity constraint! (If regularity fails and $\phi_i$ decreases in some regions, higher types might be assigned lower probabilities, violating IC.)

This regularity condition eensures that pointwise maximization naturally satisfies the monotonicity constraint of $q_i$.

### Proposition 7.7: Optimal Allocation Rule and Reserve Prices

Under the regularity assumption, we can solve the integral via *pointwise maximization*. For any realized profile of valuations $x = (x_1, \ldots, x_n)$, the seller should simply examine the virtual values and assign the object to the bidder with the highest virtual value, provided it is non-negative.
Mathematically, the optimal allocation rule is:

$$Q_i^*(x) = \begin{cases} 1 & \text{if } \phi_i(x_i) = \max_j \phi_j(x_j) \text{ and } \phi_i(x_i) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

From this, the seller can define an optimal reserve price for each bidder as follows:

### Definition 7.8: Individual Reserve Price

The optimal reserve price $r_i^*$ for bidder $i$ is exactly the root of their virtual value function:

$$\phi_i(r_i^*) = 0 \implies r_i^* - \frac{1 - F_i(r_i^*)}{f_i(r_i^*)} = 0$$

Notably, to maximize revenue, the seller does *not* necessarily allocate the object to

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the person with the highest true value, but to the person with the highest *virtual value*. Furthermore, if no bidder's virtual value exceeds zero (i.e., everyone's valuation is below their respective $r_i^*$), the seller optimally retains the object to avoid paying excessive information rents. From this perspective, the optimal mechanism is to allocate the object to the buyer with the highest valuation *strictly above their respective reserve price*.

### 7.2.3 The Monopoly Pricing Isomorphism: What is Virtual Value?

While the derivation of the optimal mechanism using the Envelope Theorem and integration by parts is mathematically rigorous, the economic intuition behind the *virtual valuation* can be beautifully illuminated by a simple parallel: *Standard Monopoly Pricing*.

Imagine a monopolist selling a single object to a market of buyers whose valuations are distributed according to $F(\cdot)$ with density $f(\cdot)$.

**Claim: Virtual Valuation is Marginal Revenue**

If we view the probability of sale as the "quantity" demanded, the virtual valuation $\psi(p)$ is exactly the monopolist's marginal revenue evaluated at price $p$.

***Proof for Claim.***

Let $p$ be the price set by the monopolist. A buyer will purchase the object if their valuation is greater than or equal to $p$. Thus, the quantity demanded $q$ at price $p$ is exactly the survival function:

$$q(p) = 1 - F(p)$$

To find the marginal revenue, we first need the inverse demand curve $P(q)$. By inverting the demand function, we get:

$$P(q) = F^{-1}(1-q)$$

The monopolist's total revenue $R(q)$ as a function of quantity is price times quantity:

$$R(q) = q \cdot P(q) = q \cdot F^{-1}(1-q)$$

Now, we differentiate $R(q)$ with respect to $q$ to find the $MR$:

$$\begin{aligned} MR(q) &= \frac{\mathrm{d}R(q)}{\mathrm{d}q} \\ &= F^{-1}(1-q) + q\left[\frac{1}{f(F^{-1}(1-q))} \cdot (-1)\right] \\ &= F^{-1}(1-q) - \frac{q}{f(F^{-1}(1-q))} \end{aligned}$$

Finally, we substitute the original price variable $p = F^{-1}(1-q)$ and quantity $q = 1 - F(p)$ back into the $MR$ equation to express marginal revenue as a function of price:

$$MR(p) = p - \frac{1-F(p)}{f(p)}$$

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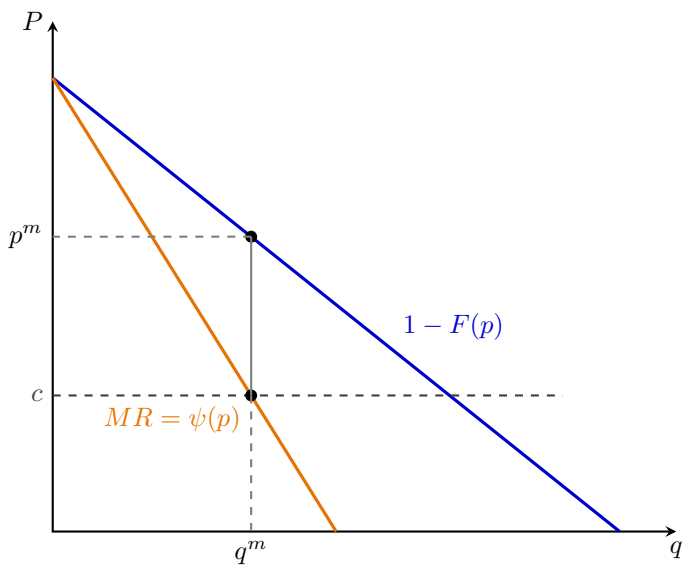
This expression is exactly the definition of the *virtual valuation*

This profound mathematical equivalence means we can analyze optimal auction mechanisms using the familiar geometric tools of intermediate microeconomics. The optimal mechanism simply allocates the good to the buyer with the highest $MR$, provided $MR \geq MC$ (where $MC$ could be the seller's own value $c$).




**Remark (Information Rent is the Inframarginal Loss).**

Why does the inverse hazard rate $\frac{1-F(p)}{f(p)}$ appear in both mechanism design and standard monopoly pricing?

In intermediate microeconomics, to sell one additional unit (i.e., to induce a buyer with a slightly lower valuation to buy), the monopolist must lower the price. However, because price discrimination is impossible, the monopolist must lower the price not just for this marginal buyer, but for *all inframarginal buyers* who would have been willing to pay a higher price. This loss in revenue on the inframarginal units drives a wedge between the price $p$ and the marginal revenue $MR$.

In mechanism design, this exact same dynamic is what we call the **Information Rent**. To convince a high-type buyer to reveal their true valuation rather than mimicking a lower type, the mechanism designer must leave them some surplus. The "loss of revenue on inframarginal buyers" in the monopoly model is mathematically identical to the "information rent paid to higher types" to satisfy the Incentive Compatibility (IC) constraint!

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# Chapter 8

# The Vickrey-Clarke-Groves (VCG) Mechanism

### Motivating Example: Public Good Provision

Suppose there are $N$ people in a society. Each person has a private value $x_i \in [\alpha, \omega]$ for a new public project (e.g., building a bridge), where $\alpha$ is the minimum possible value (might be negative) and $\omega$ is the maximum. The cost of building the bridge is $c$.

From a social planner's perspective, the efficient decision is simple:

- Build the bridge if $\sum_{i=1}^{N} x_i \geq c$.
- Do not build it if $\sum_{i=1}^{N} x_i < c$.

However, the planner does not know the true $x_i$. If the planner just asks people for their values, they have an incentive to exaggerate (if they want the bridge but don't pay) or understate (if they want to freeride). We need a mechanism to elicit the truth.

## 8.1 The General VCG Framework

Consider a general direct mechanism $(Q, M)$.

**Definition 8.1: Efficient Allocation Rule**

An allocation rule $Q^*$ is *efficient* if it maximizes the total reported social surplus:

$$Q^*(x) \in \arg\max_{Q} \sum_{j=1}^{N} Q_j(x) \cdot x_j,$$

subject to feasibility constraints.

- If the object is a private good, feasibility requires $\sum_{j=1}^{N} Q_j(x) \leq 1$.
- If it is a pure public good, then everyone consumes it equally:

$$Q_1(x) = \cdots = Q_N(x) \in \{0, 1\}.$$

### Definition 8.2: Social Surplus and VCG Payment Rule

Suppose $Q^*$ is an efficient allocation rule. We define two surplus functions based on the reported types $x$:

- **Total Maximized Surplus:**

$$W(x) = \sum_{j=1}^{N} Q_j^*(x) \cdot x_j.$$

- **Surplus of Players Other than $i$:**

$$W_{-i}(x) = \sum_{j \neq i} Q_j^*(x) \cdot x_j.$$

The **VCG Mechanism** pairs the efficient allocation $Q^*$ with the following payment rule $M_i^*$:

$$M_i^*(x) = W(\alpha_i, x_{-i}) - W_{-i}(x)$$

where $\alpha_i$ is the lowest possible type.

**Remark.**

- **Intuition: Pricing the Externality**
  The VCG payment is exactly the **social externality** that player $i$ imposes on the rest of the society.
  - $W(\alpha_i, x_{-i})$ is the maximum surplus the *other* players could have achieved if player $i$ simply did not exist (or reported the lowest type $\alpha_i$).
  - $W_{-i}(x)$ is the surplus the *other* players actually end up getting because player $i$ is present and changed the allocation to $Q^*(x)$.

  The difference between what others *could have had* and what they *actually have* is the harm (externality) $i$ causes. VCG forces player $i$ to internalize this exact harm by making them pay it out of pocket.

- **Connection to SPA:**
  The Second-Price Auction (SPA) is just a special case of VCG. If you win an SPA, your presence took the item away from the second-highest bidder. The value they lost is exactly the second-highest bid. Therefore, your externality on society is the second-highest bid, which is exactly what VCG/SPA makes you pay.

  **Example.**

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Suppose there are three bidders for a single object with true valuations $x = (5, 7, 4)$. Let's calculate the VCG payment for the winning bidder (Bidder 2, with value 7):

- If Bidder 2 is absent ($\alpha_2 = 0$), the object goes to Bidder 1 (value 5). The maximum surplus of the others is $W(\alpha_2, x_{-2}) = 5$.
- When Bidder 2 is present, the efficient rule allocates the object to Bidder 2. The surplus realized by the *other* players (Bidder 1 and 3) under this allocation is exactly zero: $W_{-2}(x) = 0$.
- Bidder 2's payment is $M_2^*(x) = W(\alpha_2, x_{-2}) - W_{-2}(x) = 5 - 0 = 5$.

Bidder 2 pays exactly 5, which is the externality they imposed on Bidder 1 by taking the object away.

- **What role does the first term $W(\alpha_i, x_{-i})$ play?**
  In the general VCG framework, the first term can be *any* arbitrary function $h_i(x_{-i})$ without violating IC. Because the first term depends ONLY on the reports of others ($x_{-i}$) and a constant ($\alpha_i$), it acts as a constant shift in player $i$'s optimization problem. When player $i$ takes the derivative to maximize their payoff, this term vanishes!

  However, specifically choosing $h_i(x_{-i}) = W(\alpha_i, x_{-i})$, the maximum social surplus achievable *without* player $i$, is known as the *Clarke Pivot Rule*. This specific choice analytically guarantees two crucial properties:

  - **No Deficit ($M_i \geq 0$):** No one is paid just to participate.

    *Proof:* $M_i^*(x) = W(\alpha_i, x_{-i}) - \sum_{j \neq i} Q_j^*(x) x_j$. The first term is the theoretical maximum surplus the others could achieve on their own. The second term is what they *actually* achieve when $i$ is present. Since the theoretical unconstrained maximum must be greater than or equal to any realized value under constraints, $M_i^* \geq 0$.

  - **Individual Rationality ($U_i \geq 0$):** No one gets a negative net utility from participating.

    *Proof:* Player $i$'s net utility is $U_i(x) = Q_i^*(x) x_i - M_i^*(x)$. Substituting the VCG payment rule, we can rewrite this as:

    $$U_i(x) = W(x) - W(\alpha_i, x_{-i})$$

    Here, $W(x)$ is the maximum social pie *with* player $i$, and $W(\alpha_i, x_{-i})$ is the maximum pie *without* player $i$. Because the social planner could always choose to ignore player $i$ and replicate the "without $i$" outcome, adding a player can never shrink the maximum possible social pie. Thus, $W(x) \geq W(\alpha_i, x_{-i})$, which guarantees $U_i(x) \geq 0$.

    In short, you only pay if your presence "pivots" the final allocation, but your payment will never exceed the value you personally gain!

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## 8.2 Truth-telling in VCG

**Proposition 8.3: Truth-telling is a Weakly Dominant Strategy**

In the VCG mechanism, reporting the true valuation $x_i$ is a weakly dominant strategy for every player $i$.

***Proof for Proposition.***

Let $x_i$ be player $i$'s true valuation. Suppose player $i$ reports $z_i$, while the other players report some arbitrary profile $x_{-i}$.

*Notice that we do NOT assume $x_{-i}$ are truthful. They can be any arbitrary lies. We will show $z_i = x_i$ is optimal regardless of what $x_{-i}$ is. This is the definition of a dominant strategy, which is strictly stronger than a Bayesian Nash Equilibrium.*

Player $i$'s true payoff given these reports is:

$$\begin{aligned}\pi_i(z_i|x_i, x_{-i}) &= Q_i^*(z_i, x_{-i})x_i - M_i^*(z_i, x_{-i}) \\ &= Q_i^*(z_i, x_{-i})x_i + W_{-i}(z_i, x_{-i}) - W(\alpha_i, x_{-i}) \\ &= \left[Q_i^*(z_i, x_{-i})x_i + \sum_{j\neq i} Q_j^*(z_i, x_{-i})x_j\right] - W(\alpha_i, x_{-i})\end{aligned}$$

The term in the brackets is exactly the total social surplus evaluated using player $i$'s *true* type $x_i$ and the others' *reported* types $x_{-i}$, under the allocation rule $Q^*(z_i, x_{-i})$.

By definition, the efficient allocation rule $Q^*$ is the mathematical operator that maximizes the sum of the inputs it is given. Therefore, to make the bracketed term as large as mathematically possible, player $i$ should hand $Q^*$ their true type $x_i$.

Formally, for any lie $z_i$:

$$\left[Q_i^*(z_i, x_{-i})x_i + \sum_{j\neq i} Q_j^*(z_i, x_{-i})x_j\right] \leq \max_Q \left[Q_i(x_i, x_{-i})x_i + \sum_{j\neq i} Q_j(x_i, x_{-i})x_j\right]$$

The right-hand side is achieved perfectly by reporting $z_i = x_i$. The final term $W(\alpha_i, x_{-i})$ is completely independent of $z_i$, so it cannot be manipulated. Thus, truth-telling maximizes $i$'s payoff for *any* $x_{-i}$.

**Corollary 8.4**

$(Q^*, M^*)$ is incentive compatible.

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## 8.3 VCG Maximizes Revenue Among Efficient Mechanisms

Let the interim expected payoff (indirect utility) of player $i$ in the VCG mechanism be:

$$U_i^*(x_i) = \int \left[Q_i^*(x_i, x_{-i})x_i - M_i^*(x_i, x_{-i})\right] f_{-i}(x_{-i})\mathrm{d}x_{-i} \equiv q_i^*(x_i)x_i - m_i^*(x_i).$$

Recall that IC implies $U_i^*(x_i)$ is convex and strictly increasing.

Suppose $(Q^*, \bar{M})$ is *another* efficient mechanism that is IC. Let $\bar{U}_i$ be its indirect utility function. Because the allocation rule $Q^*$ is identical, the derivative $\bar{U}_i'(x_i) = q_i^*(x_i)$ must also be identical. Therefore, $\bar{U}_i(x_i)$ must have the *exact same shape* as $U_i^*(x_i)$, differing at most by a constant.

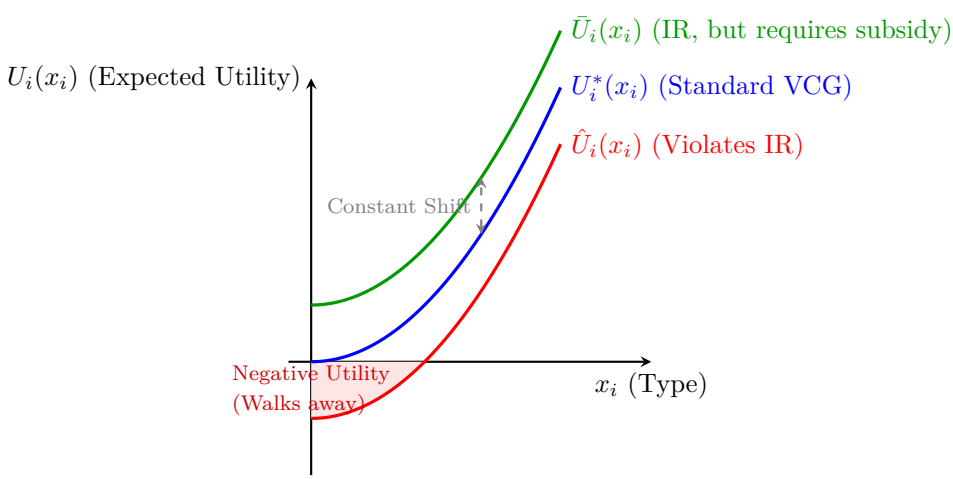
Note that in the definition of the Clarke Pivot Rule payment, we carefully included the constant term $W_{-i}(\alpha_i, x_{-i})$ to guarantee that the *lowest type gets exactly zero surplus*: $U_i^*(\alpha_i) = 0$.

**Claim: The Individual Rationality (IR) Boundary**

For any IC and efficient mechanism to satisfy the IR constraint for all possible types, its utility curve must lie weakly above the standard VCG utility curve.

We have established that any IC and efficient mechanism yields a utility curve with the exact same shape as the VCG utility curve. By design (the Clarke Pivot Rule), the standard VCG utility curve passes through the origin ($U_i^*(\alpha_i) = 0$).

Any alternative mechanism's utility curve $\bar{U}_i$ that is shifted downwards (lying strictly below $U_i^*$) would yield negative utility for the lowest types, directly violating the IR constraint. Conversely, any utility curve shifted upwards (lying strictly above $U_i^*$) perfectly satisfies IR, but implicitly requires the mechanism designer to pay an upfront lump-sum subsidy to the participants.




**Remark (Why do we ask for IR?).**

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Could there be a mechanism that is not IR for some regions, but is still reasonable? Yes! It entirely depends on the socio-economic context. Note that in standard mechanism design, we typically assume "interim" decision-making, where a bidder evaluates IR *after* knowing their own true type $x_i$, but before knowing others'.

- **Voluntary Participation (Default setting: Auctions, Private Markets):** Here, IR is strictly required for *all* possible types. If $U_i(x_i) < 0$, a bidder of type $x_i$ will simply refuse to participate. Because the designer must ensure the mechanism works regardless of the realized types, the mechanism must universally guarantee $U_i(x_i) \geq 0$.
- **Mandatory Participation (Taxes, Public Goods):** If a government is building a public good or regulating a congestion problem, it has coercive taxing power. You cannot "opt out" of a society just because the mechanism yields a negative payoff for your specific type. In such compulsory environments, IR is *not* a strict constraint.

**Proposition 8.5: VCG Maximizes Revenue Among Efficient Mechanisms**

Among all efficient, IC, and IR mechanisms, the standard VCG mechanism maximizes the seller's total revenue $\sum_{i=1}^{n} m_i$.

***Proof for Proposition.***

From the previous claim, any alternative efficient, IC, and IR mechanism must provide a utility curve that lies weakly above the standard VCG curve. Providing strictly higher utility to the buyers mathematically necessitates extracting strictly less payment from them. Thus, any deviation from the standard VCG payment rule (such as adding a lump-sum subsidy) strictly decreases the seller's expected revenue. ∎

## 8.4 Budget Balance and the AGV Mechanism

We formally define two expected payment concepts:

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### Definition 8.6: Interim and Ex-ante Expected Payments

Consider a direct mechanism $(Q, M)$ in an environment with $n$ agents, where $M_i(x)$ is the *ex-post* payment exacted from agent $i$ when the reported type profile is $x = (x_i, x_{-i})$. Assume that each agent's type $x_j$ is drawn independently from a type space $X_j$ according to a density function $f_j$. Let $f_{-i}(x_{-i}) = \prod_{j \neq i} f_j(x_j)$ denote the joint density of the other agents' types.
Define two expected payment concepts corresponding to different information stages:

- **Interim Expected Payment:** The expected payment of agent $i$ evaluated at the interim stage—conditional on knowing their own realized type $x_i$, but prior to observing the types of other agents. It is defined as:

$$m_i(x_i) = \mathbb{E}_{x_{-i}}[M_i(x_i, x_{-i})] = \int_{X_{-i}} M_i(x_i, x_{-i}) f_{-i}(x_{-i}) \mathrm{d}x_{-i}$$

- **Ex-ante Expected Payment:** The expected payment of agent $i$ evaluated at the ex-ante stage—before any agent's type is realized. It is the unconditional expectation of the payment rule over the entire type profile:

$$m_i = \mathbb{E}_{x_i}[m_i(x_i)] = \int_{X_i} m_i(x_i) f_i(x_i) \mathrm{d}x_i = \mathbb{E}_x[M_i(x)]$$

While VCG is elegantly efficient and IC, it is notoriously bad at balancing the budget. It often runs at a deficit (e.g., the mechanism has to inject outside money to pay for positive externalities).

**Example (VCG Deficit in the Bridge Problem).**

Consider the 3-person bridge problem with cost $c = 10$ and values $x = (5, 6, 2)$. The sum of values is $13 \geq 10$, so the bridge is built. Let's calculate the VCG payments:

- Player 1 ($x_1 = 5$): Without P1, others value it at $8 < 10$. Not built. P1's payment is $M_1^* = 0 - (6 + 2 - 10) = 2$.
- Player 2 ($x_2 = 6$): Without P2, others value it at $7 < 10$. Not built. P2's payment is $M_2^* = 0 - (5 + 2 - 10) = 3$.
- Player 3 ($x_3 = 2$): Without P3, others value it at $11 > 10$. Built anyway. P3's payment is $M_3^* = (5 + 6 - 10) - (5 + 6 - 10) = 0$.

The total payment collected from all players is $2 + 3 + 0 = 5$. However, the physical cost to build the bridge is 10. The central planner (or the mechanism) must absorb a massive deficit of 5 to execute this socially optimal decision!

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### Definition 8.7: Budget Balanced

A mechanism is *Budget Balanced (BB)* if, for *all* realized profiles $x$:

$$\sum_{i=1}^{n} M_i(x) = 0.$$

This is a very strong *ex-post* condition: the agents simply transfer money among themselves under the mechanism.

Naturally, a subsequent question is: **Does there exist a mechanism that is efficient ($Q^*$), IC, IR, and BB?**

### Theorem 8.8: Arrow-d'Aspremont-Gerard-Varet (AGV)

There exists an efficient, IC, IR, and BB mechanism if and only if the standard VCG mechanism runs at an ex-ante surplus. That is:

$$\sum_{i=1}^{n} m_i^* \geq 0.$$

***Proof for Theorem***

- $\Longrightarrow$:
  It is straightforward that if VCG runs at an ex-ante deficit ($\sum m_i^* < 0$), no other mechanism can balance the budget. As discussed, any other IC and IR mechanism must provide a utility curve equal to or above the VCG's curve. Giving everyone more utility means the mechanism must extract even less payment. Thus, the deficit would only widen.

- $\Longleftarrow$:
  We prove the other direction constructively. This is the celebrated AGV mechanism. Define a new ex-post payment rule $\bar{M}_i(x)$ as:

$$\bar{M}_i(x) = \underbrace{\left[ m_i^*(x_i) - \frac{1}{n-1} \sum_{j \neq i} m_j^*(x_j) \right]}_{\text{Zero-sum transfers}} + \underbrace{\left[ \frac{1}{n-1} \sum_{j \neq i} m_j^* - \frac{1}{n} \sum_{j=1}^{n} m_j^* \right]}_{\text{Constant adjustment}}$$

  Summing the payments across all $n$ players:

$$\begin{aligned} \sum_{i=1}^{n} \bar{M}_i(x) = & \left\{ \sum_{i=1}^{n} m_i^*(x_i) - \sum_{i=1}^{n} \left( \frac{1}{n-1} \sum_{j \neq i} m_j^*(x_j) \right) \right\} \\ & + \left\{ \sum_{i=1}^{n} \left( \frac{1}{n-1} \sum_{j \neq i} m_j^* \right) - \sum_{i=1}^{n} \left( \frac{1}{n} \sum_{j=1}^{n} m_j^* \right) \right\} \end{aligned}$$

  Notice that $\sum_{i=1}^{n} \sum_{j \neq i} m_j^*(x_j)$ is simply summing every player's interim payment exactly $n-1$ times. Divided by $n-1$, it precisely cancels out the first term $\sum m_i^*(x_i)$.

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The first curly brace is strictly 0. The same combinatorial logic applies to the constants in the second curly brace. Thus, $\sum \bar{M}_i(x) = 0$. The budget balances ex-post.

To see if this new payment rule alters bidding incentives, we must find the new interim expected payment $\bar{m}_i(x_i)$. We take the expectation of $\bar{M}_i(x)$ over all other players' types $x_{-i}$:

$$\bar{m}_i(x_i) = \mathbb{E}_{x_{-i}}[\bar{M}_i(x)]$$

Let's pass the expectation operator through the terms linearly:

1. $\mathbb{E}_{x_{-i}}[m_i^*(x_i)] = m_i^*(x_i)$ (since it is already an interim function depending only on $x_i$).
2. $\mathbb{E}_{x_{-i}}\left[\frac{1}{n-1}\sum_{j\neq i} m_j^*(x_j)\right] = \frac{1}{n-1}\sum_{j\neq i}\mathbb{E}_{x_j}[m_j^*(x_j)] = \frac{1}{n-1}\sum_{j\neq i} m_j^*$.
3. The terms in the second bracket are already absolute constants, so their expectation is simply themselves.

Substituting these back in:

$$\begin{aligned}\bar{m}_i(x_i) &= m_i^*(x_i) - \frac{1}{n-1}\sum_{j\neq i} m_j^* + \frac{1}{n-1}\sum_{j\neq i} m_j^* - \frac{1}{n}\sum_{j=1}^{n} m_j^* \\ &= m_i^*(x_i) - \frac{1}{n}\sum_{j=1}^{n} m_j^*\end{aligned}$$

Note that $\bar{m}_i(x_i)$ only deviates from the standard VCG interim payment $m_i^*(x_i)$ by an absolute constant $\left(\frac{1}{n}\sum m_j^*\right)$. Because the slope of the payment function is unchanged, $(Q^*, \bar{M})$ is perfectly IC.

Furthermore, since the theorem condition explicitly requires the ex-ante surplus to be non-negative ($\sum_{j=1}^{n} m_j^* \geq 0$), the constant term we are subtracting is positive. This means the overall payment is reduced (or at worst, unchanged) for every player: $\bar{m}_i(x_i) \leq m_i^*(x_i)$. Consequently, players are receiving weakly more utility than in the VCG mechanism, trivially preserving IR.

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# Chapter 9

# Matching Theory

## 9.1 The Housing Market: Allocation with Endowments

We now move from settings where monetary transfers are available (auctions, mechanisms with payments) to *allocation problems without money*. No taxes, no subsidies, no side payments—agents simply trade indivisible objects among themselves. This is the classic *housing market* model introduced by Shapley and Scarf (1974).

### 9.1.1 Housing Market and the Core

**Definition 9.1: Housing Market**

A *housing market* is a triple $(\mathcal{P}, H, \succ)$ where:

- $\mathcal{P} = \{P_1, P_2, \ldots, P_n\}$ is a finite set of *agents* ("people");
- $H = \{h_1, h_2, \ldots, h_n\}$ is a finite set of *houses* ("offices"), with $|H| = |\mathcal{P}|$;
- Each agent $P_i$ is initially endowed with house $h_i$ and has a strict preference ordering $\succ_i$ over $H$.

An *allocation* is a bijection $Q : \mathcal{P} \to H$. That is, $Q$ assigns exactly one house to each agent.

The key question is: can we find an allocation that is *stable* in the sense that no group of agents can profitably rearrange houses among themselves?

**Definition 9.2: Core**

An allocation $Q^*$ is in the core if there is no coalition $S$ and no allocation $Q'$ of $\{h_i : P_i \in S\}$ to $S$ such that $Q'(P_i) \succsim_i Q^*(P_i)$ for all $P_i \in S$ and $Q'(P_j) \succ_j Q^*(P_j)$ for some $P_j \in S$.

Intuitively, $Q^*$ is in the *core* if there is no subset $S \subseteq \mathcal{P}$ (a *blocking coalition*) such that the members of $S$ can reallocate their initial endowments $\{h_i : P_i \in S\}$ among themselves

in a way that makes every member of $S$ weakly better off, and at least one member strictly better off.

**Remark (Why "Core" and Not Just Pareto Efficiency?).**

> Pareto efficiency only rules out improvements where *everyone* can be made better off (the grand coalition $S = \mathcal{P}$). The core is a much stronger requirement: it rules out deviations by *any* subset of agents, including small coalitions. In particular, individual rationality (no agent envies their own initial endowment) is a special case of the core condition with $|S| = 1$.

### 9.1.2 Top Trading Cycle (TTC)

The *Top Trading Cycle* algorithm, attributed to David Gale by Shapley and Scarf (1974), provides an elegant procedure to find a core allocation.

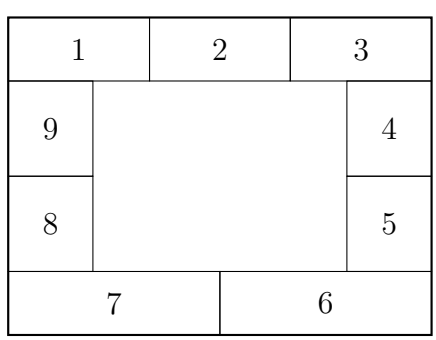
**Top Trading Cycle Algorithm**

Given a housing market $(\mathcal{P}, H, \succ)$ with initial endowment $P_i \mapsto h_i$:

1. **Point:** Each remaining agent points to the agent who currently owns their most preferred house (among remaining houses).
2. **Trade:** Since the number of agents is finite and each agent points to exactly one other agent, at least one cycle must exist. Identify all cycles. Each agent in a cycle receives the house they pointed to (i.e., the house owned by the agent they pointed to).
3. **Remove:** Remove all agents and houses involved in cycles from the market.
4. **Repeat:** Return to Step 1 with the reduced market. Terminate when all agents have been assigned a house.

**Example (The Office Problem).**

> Consider 9 people $\mathcal{P} = \{P_1, \ldots, P_9\}$ and 9 offices $H = \{1, \ldots, 9\}$, arranged around a corridor as follows:




> Person $P_i$ initially occupies office $i$. The agents' strict preference orderings (listed from most preferred to least preferred) are:

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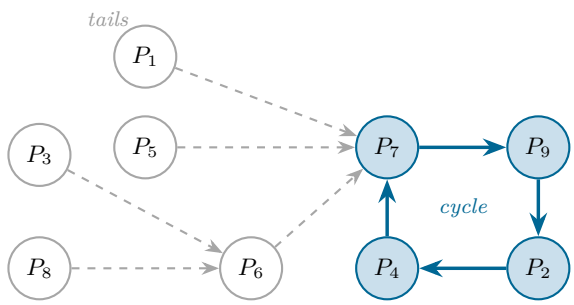
| Rank | $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_5$ | $P_6$ | $P_7$ | $P_8$ | $P_9$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 7 | 4 | 6 | 7 | 7 | 7 | 9 | 6 | 2 |
| 2 | 6 | 9 | 2 | 1 | 1 | 1 | 3 | 2 | 5 |
| 3 | 9 | 6 | 8 | 5 | 8 | 2 | 6 | 4 | 4 |
| 4 | 4 | 2 | 1 | 8 | 4 | 4 | 2 | 3 | 8 |
| 5 | 8 | 5 | 7 | 9 | 3 | 5 | 5 | 7 | 3 |
| 6 | 5 | 3 | 4 | 3 | 5 | 3 | 4 | 5 | 1 |
| 7 | 1 | 7 | 5 | 2 | 6 | 8 | 8 | 1 | 9 |
| 8 | 2 | 8 | 9 | 6 | 9 | 9 | 7 | 8 | 6 |
| 9 | 3 | 1 | 3 | 4 | 2 | 6 | 1 | 9 | 7 |

We apply TTC to the office problem defined above. The logic of each round is as follows:

- Every remaining agent *points to* the current owner of their most-preferred available office, forming a directed graph in which each node has out-degree exactly one.
- Following the arrows traces directed **chains**. Because each node has exactly one outgoing edge, every chain must eventually revisit a node—so every chain either *closes into a cycle* or *feeds into* a cycle found by another chain.
- **Cycle members** trade immediately (each receives the office of the agent they pointed to). **Tail agents** (those whose chains lead into the cycle without closing) are held over for the next round.

**Round 1.** All nine agents with offices $\{1, \ldots, 9\}$.




*Chains and cycle detection.*

- Start from $P_1$: $P_1 \to P_7 \to P_9 \to P_2 \to P_4 \to P_7$ — the chain revisits $P_7$. **Cycle found:** $P_7 \to P_9 \to P_2 \to P_4 \to P_7$.
- $P_1$ (arriving at $P_7$), as well as $P_5$ (pointing directly to $P_7$), $P_6$ (pointing to $P_7$), $P_3$ and $P_8$ (pointing to $P_6$) are all tails. They do not close any new cycle.

*Cycle trade:*

$$P_7 \leftarrow \text{office } 9, \quad P_9 \leftarrow \text{office } 2, \quad P_2 \leftarrow \text{office } 4, \quad P_4 \leftarrow \text{office } 7.$$

Remove $\{P_2, P_4, P_7, P_9\}$ and offices $\{2, 4, 7, 9\}$. Tails $\{P_1, P_3, P_5, P_6, P_8\}$ proceed.

**Round 2.** Remaining agents: $\{P_1, P_3, P_5, P_6, P_8\}$; available offices: $\{1, 3, 5, 6, 8\}$.

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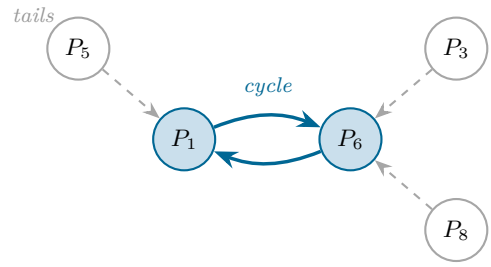
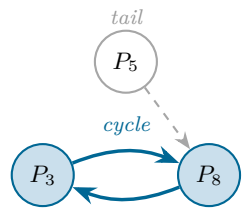



*Chains and cycle detection.*

- Start from $P_5$: $P_5 \to P_1 \to P_6 \to P_1$ — revisits $P_1$. **Cycle found:** $P_1 \leftrightarrow P_6$.
- $P_3$ and $P_8$ both point to $P_6$ (into the cycle), and $P_5$ points to $P_1$ (into the cycle). All three are tails.

*Cycle trade:* $P_1 \leftarrow$ office 6, $\quad P_6 \leftarrow$ office 1.
Remove $\{P_1, P_6\}$ and offices $\{1, 6\}$. Tails $\{P_3, P_5, P_8\}$ proceed.

**Round 3.** Remaining: $\{P_3, P_5, P_8\}$; available offices: $\{3, 5, 8\}$.




*Chains and cycle detection.*

- $P_3 \to P_8 \to P_3$ — revisits $P_3$. **Cycle found:** $P_3 \leftrightarrow P_8$.
- $P_5 \to P_8$ feeds into the cycle; $P_5$ is a tail.

*Cycle trade:* $P_3 \leftarrow$ office 8, $\quad P_8 \leftarrow$ office 3.
Remove $\{P_3, P_8\}$ and offices $\{3, 8\}$.

**Round 4.** Only $P_5$ remains with office 5. With a single agent, the "chain" is a trivial self-loop: $P_5 \to P_5$. The self-loop is itself a cycle, so $P_5$ keeps office 5.

The final TTC allocation is:

| Agent | $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_5$ | $P_6$ | $P_7$ | $P_8$ | $P_9$ |
|---|---|---|---|---|---|---|---|---|---|
| Office | 6 | 4 | 8 | 7 | 5 | 1 | 9 | 3 | 2 |

**Remark (Cycle Structure in TTC).**

In each round, the pointing graph is a *functional graph*: every node has out-degree exactly one. A fundamental property of functional graphs is that each weakly connected component contains *exactly one* directed cycle, with all remaining nodes forming tails that feed into it. Consequently:

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- **Existence:** At least one cycle is guaranteed to exist in every round, since the finite graph must eventually revisit a node along any directed path.
- **Multiplicity:** The number of cycles per round equals the number of weakly connected components—it may be one or many, depending on the preference profile. Uniqueness of cycles *per round* does not hold in general.
- **Uniqueness within a chain:** Following the arrows from any fixed starting node traces a single directed path, which must eventually close into *exactly one* cycle. This per-chain uniqueness is what makes TTC well-defined: each agent belongs to at most one cycle and receives a unique assignment in the round that cycle is identified.

### 9.1.3 Properties of TTC

**TTC Implements the Core**

**Theorem 9.3: TTC Allocation is in the Core**

The allocation produced by the TTC algorithm is in the core of the housing market. Moreover, when preferences are strict, the core contains a unique allocation, and TTC finds it.

***Proof for Theorem***

*The intuition is as follows. Agents removed in Round 1 each obtain their overall top choice—no coalition including any of them can make them strictly better off, since they already have their first-best house. Once Round 1 agents are settled, Round 2 agents obtain their best feasible house given Round 1 assignments, and so on. Any blocking coalition would need to offer some agent a house better than what TTC gave them, but that house has already been claimed by someone removed in an earlier round who would never agree to give it up.*

We proceed by induction on the round of removal. For each round $r$, let $C_r$ denote the set of agents removed in round $r$ (i.e., the union of all cycles identified and executed in that round), and let $H_r = \{\mu(i) : i \in C_r\}$ denote the corresponding set of houses allocated in round $r$.

**Base case (Round 1).** Every agent removed in Round 1 belongs to a cycle in which each member points to the owner of their globally top-ranked house. Each such agent therefore receives their first-best house. No coalition $S$ can block this allocation: to improve upon the outcome for any Round-1 agent $i \in S$, the coalition would need to reassign $i$'s own top choice to $i$—but $i$ already has it.

**Inductive step.** Suppose that for all rounds $r' < r$, the agents removed in rounds $1, \ldots, r' - 1$ receive their best house within the set of houses held by agents in those same rounds, and no coalition can improve upon their assignment using only houses from $\bigcup_{r'' \leq r'} H_{r''}$. Consider agents removed in round $r$. Each such agent receives their most preferred house among all houses still available—i.e., not yet claimed by any agent removed in rounds $1, \ldots, r - 1$.

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Suppose for contradiction that some blocking coalition $S$ exists. Let $i \in S$ be an agent who is made strictly better off under the proposed reassignment, and let $h'$ be the house $S$ proposes to give $i$, where $h' \succ_i \mu(i)$. Since $\mu(i)$ is already $i$'s best available house in round $r$, the house $h'$ must have been removed in some earlier round $r' < r$—it belongs to some agent $j \in C_{r'}$ who exited in round $r'$ with $\mu(j) = h'$. For $S$ to reassign $h'$ to $i$, it must include $j$. But by the inductive hypothesis, $\mu(j) = h'$ is $j$'s best house within $H_{r'}$; giving up $h'$ makes $j$ strictly worse off, so $j$ would not participate in $S$. Contradiction.

Hence the TTC allocation is in the core. For uniqueness under strict preferences: any core allocation must give Round-1 agents their top choice (otherwise the coalition $C_1$ itself is a blocking coalition), and by induction any core allocation must coincide with TTC round by round. Thus the core is a singleton, and TTC finds it.

### Corollary 9.4: TTC is Individually Rational

The allocation produced by TTC is individually rational: every agent weakly prefers their assigned house to their own initial endowment. That is, $\mu(i) \succsim_i h_i$ for all $i \in \mathcal{P}$.

***Proof for Corollary.***

This follows immediately from the core property. Suppose for contradiction that some agent $i$ strictly prefers their endowment to their TTC allocation: $h_i \succ_i \mu(i)$. Then the singleton coalition $S = \{i\}$, using the trivial allocation $Q'(i) = h_i$ (agent $i$ keeps their own house), satisfies $Q'(i) \succ_i \mu(i)$. This means $S$ blocks $\mu$, contradicting the fact that $\mu$ is in the core. Hence no such $i$ can exist.

**TTC is Order Independent**

### Theorem 9.5: TTC is Order Independent (Strict Preferences)

Suppose all agents have strict preferences over objects. When there are multiple cycles in a given round, the TTC outcome does not depend on the order in which cycles are identified and removed. That is, under strict preferences, the final allocation is unique regardless of which cycle is processed first.

***Proof for Theorem***

The proof reduces to one key structural observation about functional graphs.

#### Claim

Under strict preferences, in any round of TTC, all cycles in the pointing graph are *vertex-disjoint*.

***Proof for Claim.***

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Under strict preferences, each agent has a unique top-ranked available object, so the pointing graph is a functional graph: every node has out-degree exactly one. Suppose for contradiction that two distinct cycles $\mathcal{C}_1$ and $\mathcal{C}_2$ share a node $v$. Then $v$ would need two distinct outgoing edges—one continuing around $\mathcal{C}_1$ and one around $\mathcal{C}_2$—contradicting out-degree one. Hence all cycles are vertex-disjoint.

Since any two coexisting cycles $\mathcal{C}_1$ and $\mathcal{C}_2$ are vertex-disjoint, the agents and houses in $\mathcal{C}_1$ are entirely distinct from those in $\mathcal{C}_2$. Removing $\mathcal{C}_1$ first therefore does not affect the membership or the pointing structure of $\mathcal{C}_2$, and vice versa. More precisely:

- The agents in $\mathcal{C}_2$ are still present after $\mathcal{C}_1$ is removed, and their preferences over the remaining houses are unchanged.
- The agents in $\mathcal{C}_2$ still point to the same targets they did before (none of their targets were in $\mathcal{C}_1$), so $\mathcal{C}_2$ remains a valid cycle in the reduced graph.

By symmetry, removing $\mathcal{C}_2$ first leaves $\mathcal{C}_1$ intact. In either order, both cycles are eventually removed and their members receive exactly the same houses. Since the argument applies to every pair of coexisting cycles in every round, the final allocation is independent of the order in which cycles are processed.

**Remark (Order Dependence Under Weak Preferences).**

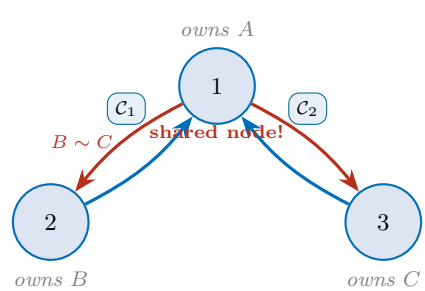
The strictness assumption embedded in our definition of a housing market is not merely a technical convenience—it is essential for TTC to be well-defined. To see what goes wrong if we relax it, suppose we allow agents to be *indifferent* between objects (i.e., $\succ_i$ is replaced by a weak order $\succsim_i$). Then an agent with multiple top-ranked objects has out-degree greater than one in the pointing graph, vertex-disjointness of cycles can fail, and different cycle-processing orders may yield different allocations.

**Counterexample.** Consider three agents with three houses:

$$\text{agent 1 owns } A, \quad \text{agent 2 owns } B, \quad \text{agent 3 owns } C,$$

with weak preferences $1 : B \sim C \succ A, \quad 2 : A \succ B \succ C, \quad 3 : A \succ C \succ B.$

*Pointing graph.* Agent 1 is indifferent between $B$ and $C$ and simultaneously points to both agents 2 and 3. Agents 2 and 3 each point to agent 1. The resulting graph has out-degree 2 at node 1, and contains two cycles sharing that node:




The two cycles $\mathcal{C}_1 = (1 \to 2 \to 1)$ and $\mathcal{C}_2 = (1 \to 3 \to 1)$ are *not* vertex-disjoint. Processing

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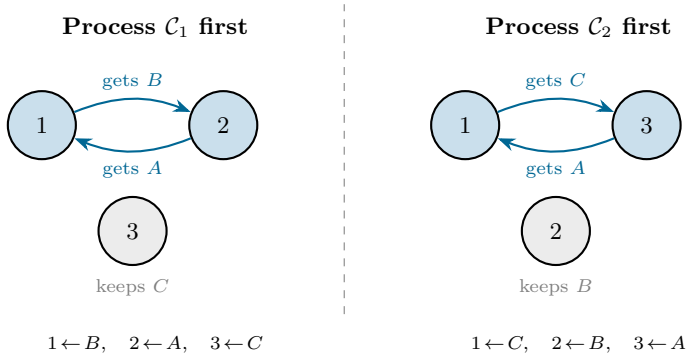
them in different orders gives different outcomes:




The two orders produce distinct allocations, confirming that TTC is not well-defined under weak preferences without an additional tie-breaking rule.

The standard remedy is to impose an exogenous **tie-breaking rule** (e.g., a lottery over objects) before running TTC, converting weak preferences into strict ones. This restores out-degree one, vertex-disjoint cycles, and a well-defined outcome. However, the final allocation then depends on the tie-breaking rule chosen, so TTC under weak preferences is best understood as a *family* of mechanisms indexed by tie-breaking rules, rather than a single deterministic procedure.

### TTC is Strategy-Proof

**Theorem 9.6: TTC is Individual Strategy-Proof**

Under the TTC mechanism, truthful reporting of preferences is a dominant strategy for every agent. That is, regardless of what other agents report, no agent can obtain a strictly better outcome by misreporting their preferences.

***Proof for Theorem***

Fix any agent $i$ and any reports $P_{-i}$ by all other agents. Let $\mu$ denote the TTC outcome under truthful reporting, with $\mu(i) = o^*$, and let $r$ be the round in which $i$ exits under truthful TTC. We show that no misreport $P_i'$ yields $i$ a strictly better object.

**Key observation.** An agent's report determines only *whom they point to*; it does not affect whom others point to. Since all $j \neq i$ report truthfully, the pointing behavior of every $j \neq i$ depends only on $j$'s own preferences and the current available set—both of which are unaffected by $i$'s report.

We present two proofs, each illuminating a different aspect of the mechanism.

**Proof 1 (by contradiction on the object obtained).** Suppose $i$ misreports and obtains $o' \succ_i o^*$. We derive a contradiction in two cases.

- **Case 1: $o'$ was taken in round $r' < r$ under truth-telling.** For $i$ to obtain $o'$, $i$ must be part of the cycle containing $o'$ in round $r'$ under the misreport. This requires some agent $j \neq i$ to point to $i$ in round $r'$, closing the cycle through $i$. But

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$j$'s pointing depends only on $j$'s preferences and the available set at round $r'$. As long as $i$ is still present at round $r'$, the agents and objects removed in rounds $1, \ldots, r'-1$ are exactly those from cycles not involving $i$—the same as under truth-telling (since those cycles form independently of $i$'s report). Hence the available set at round $r'$ is identical under both runs, and so is $j$'s pointing. Since no $j$ points to $i$ in round $r'$ under truth-telling (no cycle involving $i$ forms before round $r$), no $j$ does so under misreport either. The cycle closes without $i$. Contradiction.

- **Case 2: $o'$ is available in round $r$ under truth-telling.** Then $o' \in A_r$, the available set when $i$'s truthful cycle forms. But under truth-telling $i$ points to the owner of their top choice in $A_r$, which would be $o'$ if $o' \succ_i o^*$—contradicting $\mu(i) = o^*$.

**Proof 2 (by analyzing the exit round).** Suppose $i$ misreports and exits in round $r' \neq r$. We show neither case improves $i$'s outcome.

- **Case 1: $r' > r$ (i exits later).** In TTC, each agent exits in the round their cycle forms, receiving their most preferred object among those still available *at that round*. By construction, $A_{r'} \subseteq A_r$: the longer $i$ stays, the more objects get claimed by earlier cycles. Hence $i$'s best available object weakly worsens as the round increases, and exiting later cannot improve $i$'s outcome.

- **Case 2: $r' < r$ (i exits earlier).** For $i$ to exit in round $r' < r$, $i$ must join a cycle in that round. This requires some $j \neq i$ to point toward $i$ (to close the cycle through $i$). Under truth-telling, $j$ points to $i$ in round $r'$ if and only if $h_i$ is $j$'s top available object at round $r'$. Since the available set at round $r'$ is the same under both runs (cycles before round $r'$ do not involve $i$ and form identically), $j$ already points to $i$ in round $r'$ under truth-telling. But then $j$ never exits before round $r'$, so $j$ is still present in round $r$ under truth-telling. This means $h_j \in A_r$, the available set when $i$ exits truthfully. Since $\mu(i) = o^*$ is $i$'s top choice from $A_r$ and $h_j \in A_r$, we have $o^* \succsim_i h_j$. Joining $j$'s cycle early to obtain $h_j$ cannot improve upon $o^*$.

In all cases, no misreport yields $i$ a strictly better object than $o^*$. Since $i$ and $P_{-i}$ were arbitrary, truth-telling is a dominant strategy for every agent. ∎

The two proofs above capture complementary intuitions about why manipulation fails in TTC.

- **Proof 1** asks: *can $i$ obtain a specific better object $o'$?* It shows the answer is no by ruling out each possible location of $o'$: either $o'$ was already taken (and $i$ cannot intercept its cycle, since no one points back to $i$), or $o'$ was available all along (and $i$ would have gotten it honestly).

- **Proof 2** asks: *can $i$ benefit by changing which round it exits?* It shows the answer is no in both directions. Exiting later only shrinks the available set. Exiting earlier forces $i$ to join a chain that was already pointing toward $i$ under truth-telling—meaning the object at the end of that chain was already in $i$'s truthful available set $A_r$, and therefore no better than $o^*$.

At the deepest level, both proofs rest on the same asymmetry: *$i$ controls only whom it*

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*points to, not who points back.* The objects reachable by $i$ are ultimately determined by who is willing to trade with $i$—and that depends on others' preferences, which no misreport can change.

A natural follow-up question is whether TTC is also *group strategy-proof*: can a coalition of agents coordinate their misreports to make every member of the coalition strictly better off? The answer is **NO**—TTC is group strategy-proof. No coalition can jointly misreport preferences and make all coalition members strictly better off. This is a stronger result that follows from the fact that TTC implements the unique strict core allocation, combined with the structure of the trading cycle procedure.

### Theorem 9.7: TTC is Strictly Group Strategy-Proof

TTC is group strategy-proof: there is no coalition $S \subseteq \mathcal{P}$ and no coordinated misreport $P'_S$ such that every member of $S$ obtains a strictly better object than under truthful reporting.

***Proof for Theorem***

*The proof has two parts that mirror—and extend—the individual strategy-proofness argument. Individual SP shows that no single agent can infiltrate a better cycle, because they cannot control who points back to them. Group SP faces a richer challenge: even if coalition members coordinate their pointing, they still cannot control the pointing of agents outside $S$, and—crucially—the core uniqueness of TTC prevents them from profitably rearranging objects among themselves either.*

Suppose for contradiction that $S$ misreports $P'_S$ and obtains allocation $\mu'$ with $\mu'(i) \succ_i \mu(i)$ for all $i \in S$, where $\mu$ is the truthful TTC outcome. Partition $S$'s gains by the origin of the improved objects: outside $S$, and inside $S$ (redistribution).

### Claim

No $S$-member can gain an object initially owned by an agent outside $S$.

***Proof for Claim.***

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Suppose $i \in S$ gains object $o' = \mu'(i) \succ_i \mu(i)$, where $o'$ is initially owned by some $j \notin S$. Fix the reports of all other agents at $(P'_{S\setminus\{i\}}, P_{-S})$ and apply the individual strategy-proofness argument to $i$ alone. Regardless of what others report, $i$ cannot benefit from misreporting; in particular, the same two-case argument applies verbatim:

- If $o'$ was removed in an earlier round under this fixed profile, $j \notin S$ reports truthfully, so $j$'s pointing behavior depends only on $j$'s own preferences and the available set—neither of which $i$'s report can affect. No agent in the cycle containing $o'$ points back to $i$, so $i$ cannot join that cycle regardless of $i$'s misreport. Contradiction.

- If $o'$ is still available in $i$'s exit round, then $o' \succ_i \mu(i)$ implies $i$ would have pointed to $j$ (owner of $o'$) under truthful reporting and obtained $o'$ honestly— contradicting $\mu(i) \neq o'$.

Hence every $i \in S$ can only gain objects initially endowed to other $S$-members. That is, $\{\mu'(i) : i \in S\} \subseteq \{h_i : i \in S\}$.

**Claim**

$S$ cannot profitably redistribute its own endowments.

***Proof for Claim.***

By the previous claim, the objects received by $S$-members under $\mu'$ are drawn entirely from $\{h_i : i \in S\}$. Define an allocation $\nu$ for $S$ by $\nu(i) = \mu'(i)$ for all $i \in S$. Then $\nu$ is a redistribution of $\{h_i : i \in S\}$ among $S$ such that $\nu(i) \succ_i \mu(i)$ for all $i \in S$. This means $S$ constitutes a *blocking coalition* for $\mu$ under true preferences, contradicting the fact that $\mu$ is in the strict core.

Both parts yield contradictions, so no coalition $S$ can jointly misreport to make all its members strictly better off.

**Theorem 9.8: TTC is Weakly Group Strategy-Proof**

The TTC mechanism is weakly group strategy-proof: there is no coalition $S \subseteq \mathcal{P}$ and joint misreport $P'_S$ such that the resulting outcome $\mu'$ satisfies

$$\mu'(i) \succsim_i \mu(i) \text{ for all } i \in S, \quad \text{with} \quad \mu'(i^*) \succ_{i^*} \mu(i^*) \text{ for some } i^* \in S.$$

***Proof for Theorem***

Suppose for contradiction that such $S$ and $P'_S$ exist. Since preferences are strict, every $i \in S$ either keeps exactly the same object ($\mu'(i) = \mu(i)$) or gets a strictly better one. Partition $S$ accordingly:

$$S^+ = \{i \in S : \mu'(i) \succ_i \mu(i)\} \neq \emptyset, \qquad S^0 = \{i \in S : \mu'(i) = \mu(i)\}.$$

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Run TTC under the misreport profile $(P'_S, P_{-S})$. Among all $S^+$-members, let $r^*$ be the earliest round in which any $S^+$-member exits, and let $i^{**} \in S^+$ be such a member. Let $\mathcal{C}$ be the TTC cycle that $i^{**}$ belongs to in round $r^*$.

In cycle $\mathcal{C}$, agents trade their initial endowments cyclically: writing $\mathcal{C} = (a_1 \to a_2 \to \cdots \to a_k \to a_1)$, each $a_l$ gets $h_{a_{l+1}}$ (indices mod $k$). In particular, $\mu'(a_l) = h_{a_{l+1}}$ for each $l$.

**Case 1: $\mathcal{C} \subseteq S$.**

All members of $\mathcal{C}$ belong to $S$, so by the weak improvement assumption,

$$\mu'(a_l) = h_{a_{l+1}} \succsim_{a_l} \mu(a_l) \quad \text{for all } a_l \in \mathcal{C},$$

with $h_{a_{l^*+1}} \succ_{i^{**}} \mu(i^{**})$ for $i^{**} \in \mathcal{C}$. The coalition $\mathcal{C}$ can execute exactly this trade under their *true* preferences: each $a_l$ contributes endowment $h_{a_l}$ and receives $h_{a_{l+1}}$, a redistribution of $\{h_{a_l} : a_l \in \mathcal{C}\}$. Every member weakly benefits, and $i^{**}$ strictly benefits. This means $\mathcal{C}$ is a blocking coalition for $\mu$ under true preferences, contradicting the fact that $\mu$ is in the strict core.

**Case 2: $\mathcal{C} \not\subseteq S$.**

$\mathcal{C}$ contains $i^{**} \in S^+$ and at least one agent $j_0 \notin S$. We consider two sub-cases based on whether the non-$S$ members of $\mathcal{C}$ are harmed by the trade.

**Sub-case 2a:** $\mu'(j) = h_{n(j)} \succsim_j \mu(j)$ for every $j \in \mathcal{C} \setminus S$.

Then every agent in $\mathcal{C}$ weakly benefits from the cycle trade (under true preferences), and $i^{**}$ strictly benefits. Again, $\mathcal{C}$ is a blocking coalition for $\mu$, contradicting the strict core.

**Sub-case 2b:** Some $j_0 \in \mathcal{C} \setminus S$ is strictly harmed: $\mu'(j_0) \prec_{j_0} \mu(j_0)$.

Let $\mu(j_0) = h_l$ (the object $j_0$ receives under truthful TTC, which is $l$'s endowment). Since $j_0$ reports truthfully in the misreport run and always picks their top available object, the fact that $j_0$ gets $\mu'(j_0) \prec_{j_0} h_l$ at round $r^*$ means $h_l$ is *not available* at the start of round $r^*$ in the misreport run. So $h_l$ was taken in some round $r' < r^*$ by some agent $m \neq j_0$.

We now derive a contradiction by tracing who took $h_l$:

- $m \notin S^+$: by choice of $r^*$, no $S^+$-member exits before round $r^*$. So $m \notin S^+$.
- $m \notin S^0$: if $m \in S^0$, then $\mu'(m) = \mu(m)$. In round $r'$, $m$ exits and gets $h_l$, so $\mu'(m) = h_l$. But then $\mu(m) = h_l = \mu(j_0)$, implying $m = j_0$ (since $\mu$ is a bijection). This contradicts $m \in S^0 \subseteq S$ and $j_0 \notin S$.
- Therefore $m \notin S$, and $m$ reports truthfully. In the truthful run, $m$'s exit cycle forms at round $r_m$ (the round $m$ exits under truth-telling). Under truthful TTC, $m$ gets $\mu(m) \neq h_l$ (since $\mu^{-1}(h_l) = j_0 \neq m$ and $\mu$ is a bijection). So $m$ is *not* in the cycle containing $l$ in the truthful run: under truthful TTC, $m$ does not point to $l$ at the round where $h_l$ changes hands.

So in the misreport run, the truthful agent $m \notin S$ ends up in a different cycle than in the truthful run, specifically one that includes $l$ and yields $h_l$ to $m$. Since $m$

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always reports truthfully, $m$'s pointing changes only if the available set at $m$'s exit round changes. This change in the available set must itself be caused by some $S$-member's misreport altering an earlier cycle. Tracing this chain: let $\mathcal{C}'$ be the TTC cycle in round $r'$ that contains $m$ (and $l$) in the misreport run. By the same case analysis applied to $\mathcal{C}'$:

- If $\mathcal{C}' \subseteq S$: a sub-coalition of $S$ is trading endowments in $\mathcal{C}'$. Since $m \notin S$ is in $\mathcal{C}'$, this is impossible.
- If $\mathcal{C}' \not\subseteq S$: $\mathcal{C}'$ contains some $S$-member. If all agents in $\mathcal{C}'$ weakly benefit (sub-case 2a for $\mathcal{C}'$), then $\mathcal{C}'$ is a blocking coalition—contradiction. If some agent in $\mathcal{C}' \setminus S$ is strictly harmed, the same tracing argument applies recursively to a strictly earlier round.

Since the rounds are finite and strictly decreasing at each recursive step, this process must terminate. The terminal case is either sub-case 2a (blocking coalition, contradiction with core) or Case 1 (coalition entirely within $S$, contradiction with core). Either way, we reach a contradiction.

Since all cases yield contradictions, no coalition $S$ can achieve a weak Pareto improvement via coordinated misreporting. TTC is weakly group strategy-proof. ■

**Remark (Why Weak GSP Is Harder Than Strong GSP).**

The additional difficulty in the weak GSP proof relative to strong GSP comes entirely from $S^0$-members. In strong GSP, every coalition member is assumed strictly better off, so individual strategy-proofness can be applied to each member to show they can only receive objects from within $S$ (Claim 1 of the strong GSP proof). This argument fails for $S^0$-members: they are indifferent between their truthful and misreport outcomes, so ISP provides no constraint on where their object originates.

$S^0$-members may misreport not to benefit themselves, but to act as "facilitators"—adjusting the cycle structure in early rounds to create a path that allows $S^+$-members to enter better cycles. Sub-case 2b captures exactly this scenario and requires the recursive tracing argument above: each step of the chain that $S^0$-members create must eventually close into a cycle, and that cycle is forced to be a blocking coalition for $\mu$ under true preferences—which is impossible since $\mu$ is the unique strict core.

## 9.2 The Assignment Problem: Allocation without Endowments

The housing market model rests critically on the assumption that each agent holds an initial endowment: this is what gives IR its bite, and what makes the core a meaningful solution concept. Once we drop this assumption—handing all objects to a central planner and asking how to distribute them from scratch—both IR and the core lose their reference point. We enter the *assignment problem*, where no agent has a prior claim to any object and the relevant normative goals reduce to *Pareto efficiency* and *strategy-proofness*.

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### 9.2.1 Serial Dictatorship (SD)

The serial dictatorship (SD) mechanism addresses this setting directly. Agents are given a predetermined priority ordering and pick their favorite available object in turn. Compared to TTC, SD trades the structural richness of cycle-based trading for sheer simplicity: there are no endowments to track, no cycles to identify, and no core to verify—just a queue and a sequence of greedy choices.

**Serial Dictatorship**

Fix an ordering $\sigma = (\sigma(1), \sigma(2), \ldots, \sigma(n))$ of the agents. The mechanism proceeds as follows:

1. Agent $\sigma(1)$ (the "dictator") picks their most preferred house from $H$.
2. Agent $\sigma(2)$ picks their most preferred house from the remaining houses.
3. Continue until all agents are assigned.

**Example (Serial Dictatorship on the Office Problem).**

We run SD on the office problem with the natural ordering $\sigma = (P_1, P_2, \ldots, P_9)$. At each step, the active agent scans their preference list top-to-bottom and picks the highest-ranked office that is still available. Taken offices are shown in gray.

| Step | Available offices | Agent picks | Gets |
|---|---|---|---|
| 1 | $\{1,2,3,4,5,6,7,8,9\}$ | $P_1$: top choice is 7 ✓ | 7 |
| 2 | $\{1,2,3,4,5,6,8,9\}$ | $P_2$: top choice is 4 ✓ | 4 |
| 3 | $\{1,2,3,5,6,8,9\}$ | $P_3$: top choice is 6 ✓ | 6 |
| 4 | $\{1,2,3,5,8,9\}$ | $P_4$: 7 taken; next available: 1 ✓ | 1 |
| 5 | $\{2,3,5,8,9\}$ | $P_5$: $7,1$ taken; next available: 8 ✓ | 8 |
| 6 | $\{2,3,5,9\}$ | $P_6$: $7,1$ taken; next available: 2 ✓ | 2 |
| 7 | $\{3,5,9\}$ | $P_7$: top choice is 9 ✓ | 9 |
| 8 | $\{3,5\}$ | $P_8$: $6,2,4$ taken; next available: 3 ✓ | 3 |
| 9 | $\{5\}$ | $P_9$: 2 taken; next available: 5 ✓ | 5 |

The SD allocation under ordering $\sigma = (P_1, \ldots, P_9)$ is:

| Agent | $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_5$ | $P_6$ | $P_7$ | $P_8$ | $P_9$ |
|---|---|---|---|---|---|---|---|---|---|
| Office | 7 | 4 | 6 | 1 | 8 | 2 | 9 | 3 | 5 |

Compare this with the TTC allocation on the same problem:

| Agent | $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_5$ | $P_6$ | $P_7$ | $P_8$ | $P_9$ |
|---|---|---|---|---|---|---|---|---|---|
| Office | 6 | 4 | 8 | 7 | 5 | 1 | 9 | 3 | 2 |

Several agents differ: $P_1$ gets office 7 under SD (their top choice) versus 6 under TTC;

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$P_4$ gets 1 under SD versus 7 under TTC. This illustrates a key tradeoff: the agent who goes first under SD always receives their top choice, but later agents may fare worse than under TTC, where the cycle structure can give non-first agents access to offices they could not reach by waiting their turn.

### 9.2.2 Properties of SD

**Theorem 9.9: Serial Dictatorship is Pareto Efficient**

The allocation produced by SD is Pareto efficient: there is no other allocation that makes every agent weakly better off and some agent strictly better off.

***Proof for Theorem***

*The argument is essentially one of optimality at the margin. Consider the first agent $i$ who would need to be reassigned to achieve the supposed Pareto improvement. Agent $i$ already holds their top choice among everything not taken by higher-priority agents—there is simply nothing better left for them. Any Pareto improvement for $i$ is therefore impossible, and the supposed dominating allocation cannot exist.*

Suppose for contradiction that some allocation $\mu'$ Pareto dominates the SD allocation $\mu$. Let $i = \sigma(k)$ be the *first* agent in the priority ordering such that $\mu'(i) \neq \mu(i)$. Since $\mu'$ Pareto dominates $\mu$, we have $\mu'(i) \succ_i \mu(i)$.

But $\mu(i)$ is $i$'s most preferred object among all objects available when it was $i$'s turn—that is, among all objects not already taken by $\sigma(1), \ldots, \sigma(k-1)$. Since $i$ is the first agent to differ between $\mu$ and $\mu'$, every agent $\sigma(1), \ldots, \sigma(k-1)$ receives the same object under both $\mu$ and $\mu'$. Therefore the set of objects available to $i$ at step $k$ is identical under both allocations, and $\mu(i)$ is $i$'s top choice from that set. The existence of $\mu'(i) \succ_i \mu(i)$ with $\mu'(i)$ in the same available set is a contradiction.

*Notice that this argument does **not** require every agent to get their globally top-ranked object. It only requires that each agent gets their top choice **given the residual supply**, which is precisely what the greedy sequential structure of SD guarantees.* ■

**Theorem 9.10: Serial Dictatorship is Strategy-Proof**

Under SD, truthful reporting of preferences is a dominant strategy for every agent. No agent can obtain a strictly better object by misreporting their preferences, regardless of what other agents report.

***Proof for Theorem***

*In SD, your report only affects* what you pick*—it has no effect whatsoever on what is available to you, since the available set is determined entirely by agents ahead of you in the queue. You always face the same menu $A_k$, and truth-telling simply ensures you pick your genuine favorite from it. Lying can only cause you to pick something you like less.*

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Fix any agent $i = \sigma(k)$ and any reports by all other agents. Suppose $i$ misreports, yielding object $o'$ under the misreport, compared to $o^* = \mu(i)$ under truthful reporting. We show $o' \not\succ_i o^*$.

The key observation is that, the set of objects available to $i$ at step $k$ is determined entirely by the choices of $\sigma(1), \ldots, \sigma(k-1)$, each of whom picks their own top available object according to their own report. Since $i = \sigma(k)$ has no influence over any earlier agent's report or choice, the available set $A_k$ at step $k$ is **identical** whether $i$ reports truthfully or not.

Under truthful reporting, $i$ picks $o^* = \mu(i)$, their most preferred object in $A_k$ (the available set at step $k$). Under misreport, $i$ picks according to a false preference list, and receives some $o' \in A_k$. Since $o^*$ is $i$'s true top choice from $A_k$ and $o' \in A_k$, we have $o^* \succsim_i o'$. In particular, $o' \not\succ_i o^*$. Since the reports of other agents and the choice of $i = \sigma(k)$ were arbitrary, truth-telling is a dominant strategy. ∎

### 9.2.3 Equivalence of Random SD and TTC with Random Endowments

The separation between the housing market (with endowments, TTC) and the assignment problem (without endowments, SD) may seem sharp. Yet there is a striking bridge between the two frameworks: once we introduce randomness into either model in the natural way, the two mechanisms become outcome-equivalent. This result, due to Abdulkadiroğlu and Sönmez (1998), reveals that the endowment structure and the priority structure are, at a deeper level, two sides of the same coin.

The key observation is that both frameworks have one remaining degree of freedom that the model leaves unspecified:

- In the *assignment problem*, objects have no owner, so the mechanism designer must choose a **priority ordering** over agents. If this choice seems arbitrary, a natural response is to randomize: draw a uniformly random ordering.
- In the *housing market*, agents must have initial endowments, but if no agent has a natural prior claim to any particular object, a natural response is again to randomize: draw a uniformly random endowment assignment, then run TTC.

Randomizing the one free parameter in each model yields two randomized mechanisms. The theorem below says they are the same mechanism.

**Definition 9.11: Randomized Mechanisms**

- **Random Serial Dictatorship (RSD):** Draw a uniformly random ordering $\sigma$ of agents (each of the $n!$ orderings equally likely), then run SD under $\sigma$.
- **Core from Random Endowments (CFRE):** Draw a uniformly random assignment of objects to agents as initial endowments (each of the $n!$ bijections equally likely), then run TTC to find the core allocation.

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**Theorem 9.12: RSD = CFRE**

RSD and CFRE induce the *same* probability distribution over allocations: for every deterministic allocation $\mu$,

$$\Pr_\sigma\big[\mu_\sigma^{\mathrm{SD}} = \mu\big] \;=\; \Pr_e\big[\mu_e^{\mathrm{TTC}} = \mu\big],$$

where $\sigma$ is a uniformly random priority ordering and $e$ is a uniformly random endowment assignment.

***Proof for Theorem***

**Lemma 9.13: Within-Cycle Commutativity**

Fix any endowment $e$ and let $\mu = \mu_e^{\mathrm{TTC}}$, with TTC producing cycles $C_1, C_2, \ldots, C_m$ in successive rounds. Let $\Sigma(\mu, e)$ denote the set of all priority orderings that:

1. place all agents in $C_k$ before all agents in $C_{k+1}$ (for each $k$), and
2. within each $C_k$, use any permutation of its members.

Then SD under every $\sigma \in \Sigma(\mu, e)$ produces $\mu$.

***Proof for Lemma***

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We establish one fact about the TTC cycle structure.

**Fact 9.14: Within-Cycle Self-Preference**

For any cycle $C_k$ and any two distinct members $a, b \in C_k$: $\mu(a) \succ_a \mu(b)$.

Take any $\sigma \in \Sigma(\mu, e)$ and any agent $a \in C_k$. When $a$'s turn comes in SD, the available objects are a subset of those available in TTC round $k$, for the following reason: the objects removed before round $k$ in TTC are exactly $\bigcup_{l<k}\{\mu(x) : x \in C_l\}$, and by definition of $\Sigma(\mu, e)$, all agents in $C_1, \ldots, C_{k-1}$ pick before $a$. By the Fact above and induction, each such agent picks their own $\mu$-assignment, so the objects removed before $a$'s turn in SD are exactly $\bigcup_{l<k}\{\mu(x) : x \in C_l\}$ plus the $\mu$-assignments of those $C_k$-members who pick before $a$.

We verify that $\mu(a)$ is $a$'s top available object when $a$ picks:

- **Objects from earlier cycles are unavailable.** All $\mu(i)$ for $i \in C_l$, $l < k$ have been claimed. In TTC, $\mu(a)$ is $a$'s top choice among all objects available in round $k$—this already accounts for the absence of earlier objects. So no unavailable object is preferred over $\mu(a)$.
- $\mu(a)$ **has not been taken by earlier** $C_k$ **members.** By the Fact, every other $b \in C_k$ prefers $\mu(b)$ over $\mu(a)$, so no earlier $C_k$-member claims $\mu(a)$.
- **Objects from later cycles are weakly worse.** All objects $\mu(x)$ for $x \in C_l$, $l > k$ were available in TTC round $k$, and $a$ chose $\mu(a)$ over all of them.

Therefore $a$ claims $\mu(a)$ in SD. Since $a$ was arbitrary, SD under $\sigma$ produces $\mu$. ∎

It remains to show that the randomization in CFRE and RSD induce the same distribution. We do this by mapping both the valid orderings $\sigma$ and the valid endowments $e$ to a canonical block partition uniquely defined by $\mu$ and the agents' preferences.

Fix an allocation $\mu$. Define an ordered partition $\mathcal{B} = (B_1, B_2, \ldots, B_m)$ of the agents recursively as follows: Let $B_1$ be the minimal nonempty set of agents whose top choices among all objects are exactly $\mu(B_1)$. Let $B_2$ be the minimal nonempty set of agents whose top choices among the remaining objects $N \setminus \mu(B_1)$ are exactly $\mu(B_2)$, and so on. Notice that $\mathcal{B}$ is uniquely determined independent of $\sigma$ or $e$.

**Counting Orderings for RSD:**
An ordering $\sigma$ yields $\mu$ under SD if and only if it respects this dependency structure. Specifically, agents in $B_k$ must appear after all agents in $B_1, \ldots, B_{k-1}$ to ensure their preferred objects in those blocks are already taken. Because $B_k$ is defined as the minimal closed set under top available choices, the agents within $B_k$ can be processed in *any* internal order without disrupting the assignment $\mu$. Thus, the total number of orderings yielding $\mu$ is exactly the number of ways to permute agents within each block:

$$\left|\{\sigma : \mu_\sigma^{\mathrm{SD}} = \mu\}\right| = \prod_{k=1}^{m} |B_k|!$$

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**Counting Endowments for CFRE:**
An endowment $e$ yields $\mu$ under TTC if and only if for every block $k$, $e$ endows the objects $\mu(B_k)$ strictly to the agents in $B_k$ (i.e., $e(B_k) = \mu(B_k)$). To see why, consider the TTC pointer function $p(i) = e^{-1}(\mu(i))$. If $e(B_k) = \mu(B_k)$, then $p$ forms a bijection from $B_k$ to itself. Consequently, $p$ perfectly decomposes into a set of disjoint cycles covering all of $B_k$. Since their top available choices are in $\mu(B_k)$, every agent $i \in B_k$ participates in a cycle and receives exactly $\mu(i)$.

The number of such valid endowments is simply the number of possible bijections from $B_k$ to $\mu(B_k)$ for each block $k$. Therefore:

$$\left|\{e : \mu_e^{\mathrm{TTC}} = \mu\}\right| = \prod_{k=1}^{m} |B_k|!$$

Since there are exactly $n!$ total orderings and $n!$ total endowments, and the sizes of the pre-images yielding $\mu$ are strictly equal, we conclude:

$$\Pr_\sigma\left[\mu_\sigma^{\mathrm{SD}} = \mu\right] = \frac{\prod_{k=1}^{m} |B_k|!}{n!} = \Pr_e\left[\mu_e^{\mathrm{TTC}} = \mu\right]$$

This holds for any deterministic allocation $\mu$, completing the proof. ■

**Remark (Proof Intuition).**

The bijection has a clean interpretation. In SD, the priority ordering determines who gets served first. In TTC, the endowment determines who holds what. The bijection $\sigma \mapsto e(\sigma)$ encodes the SD priority order *into* the endowment structure by a cyclic shift: agent $\sigma_k$ is endowed with the object that $\sigma_{k-1}$ ends up with under SD. This creates a pointing chain in TTC that exactly mirrors the SD queue—agent $\sigma_k$ points to $\sigma_{k+1}$ because $\sigma_{k+1}$ holds what $\sigma_k$ ultimately wants. The TTC cycles that form then execute the same trades as the SD steps, in the same order.

At a higher level, both mechanisms are doing the same thing: translating an underlying permutation (of agents, or of objects) into an allocation by a greedy sequential procedure. Randomizing the permutation uniformly in either case produces the same lottery, because the two "greedy procedures" are isomorphic via the cyclic shift bijection.

The equivalence has several consequences worth noting.

- **Mechanism design flexibility.** A planner who knows only agents' preferences—but has no natural endowment structure or priority structure—can implement the same outcome-distribution using whichever model is more convenient for the context.
- **Properties transfer.** Since RSD inherits strategy-proofness and Pareto efficiency from SD, and CFRE inherits the core property and IR from TTC, the equivalence implies that both randomized mechanisms simultaneously satisfy all four properties in expectation.
- **The endowment is not deep.** The result suggests that the distinction between "having endowments" and "not having endowments" is less fundamental than it appears—at least

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once both are randomized. What matters is the underlying combinatorial structure of sequential greedy assignment, not the narrative framing of endowments versus priorities.

## 9.3 Two-Sided Matching

We now transition from one-sided allocation problems (housing markets) to *two-sided matching* problems, where agents on both sides of the market have preferences over agents on the other side. The canonical example is the *college admissions* problem introduced by Gale and Shapley (1962).

### 9.3.1 Two-Sided Matching and Pairwise Stability

**Definition 9.15: Two-Sided Matching Market**

A **two-sided matching market** (or *marriage model*) consists of:

- Two disjoint, finite sets of agents: $\mathcal{S} = \{s_1, \ldots, s_n\}$ (students) and $\mathcal{C} = \{c_1, \ldots, c_m\}$ (colleges).
- Each student $s \in \mathcal{S}$ has a strict preference ordering $\succ_s$ over $\mathcal{C}$.
- Each college $c \in \mathcal{C}$ has a strict preference ordering $\succ_c$ over $\mathcal{S}$.
- Each college has a **quota** $q_c \geq 1$ specifying the number of seats available. In the basic model, $q_c = 1$ for all $c$, and $m = n$.

**Definition 9.16: Matching**

A **matching** $\mu$ is a bijection $\mu : \mathcal{C} \to \mathcal{S}$ (when $|\mathcal{C}| = |\mathcal{S}|$ and all quotas are 1). That is, $\mu(c)$ denotes the student assigned to college $c$, and $\mu^{-1}(s)$ denotes the college assigned to student $s$.

In the housing market (Shapley–Scarf), only one side—the agents—had preferences; houses were passive objects. Here, *both* sides have preferences, which fundamentally changes the problem. The notion of "core" from the housing market generalizes to the concept of *stability*, but the structure of stable allocations is much richer.

The central solution concept in two-sided matching is *pairwise stability*. Unlike the core (which considers arbitrary coalitions), pairwise stability focuses on deviations by pairs $(c, s)$.

**Definition 9.17: Blocking Pair and Pairwise Stability**

A pair $(c, s) \in \mathcal{C} \times \mathcal{S}$ is a **blocking pair** for matching $\mu$ if:

1. College $c$ prefers student $s$ to its current match: $s \succ_c \mu(c)$;
2. Student $s$ prefers college $c$ to their current match: $c \succ_s \mu^{-1}(s)$.

A matching $\mu$ is **pairwise stable** if there exists no blocking pair.

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**Remark.**

Pairwise stability captures a minimal notion of "**no justified envy**": there is no student-college pair who would both prefer to be matched with each other over their current assignments. If such a pair existed, the college would have an incentive to admit the student, and the student would have an incentive to accept—destabilizing the matching.

### 9.3.2 Deferred Acceptance (DA)

Gale and Shapley (1962) introduced the *Deferred Acceptance* (DA) algorithm, which always produces a pairwise stable matching.

**Student-Proposing Deferred Acceptance (DA)**

1. **Propose:** Each unmatched student proposes to the highest-ranked college on their list to which they have not yet proposed.
2. **Hold/Reject:** Each college that receives proposals *tentatively* holds the most preferred applicant (according to its own ranking) among the new proposals and its currently held student (if any), and rejects all others.
3. **Repeat:** Rejected students propose to their next choice. Continue until no rejections occur (every student is either tentatively held or has been rejected by all colleges).
4. **Finalize:** All tentative matches become permanent.

**Example (Student-Proposing DA on the Matching Problem).**

Consider 10 students $S = \{s_1, \ldots, s_{10}\}$ and 10 colleges $C = \{c_1, \ldots, c_{10}\}$, each with quota $q = 1$.

**College preferences over students (Final matches highlighted):**

| Rank | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $s_1$ | $s_1$ | $s_{10}$ | $s_6$ | $s_7$ | $\boldsymbol{s_4}$ | $s_1$ | $s_6$ | $s_8$ | $s_6$ |
| 2 | $s_2$ | $s_6$ | $s_2$ | $s_1$ | $s_2$ | $s_8$ | $s_4$ | $s_4$ | $\boldsymbol{s_{10}}$ | $s_5$ |
| 3 | $\boldsymbol{s_3}$ | $s_9$ | $s_4$ | $s_5$ | $s_8$ | $s_2$ | $s_3$ | $s_2$ | $s_6$ | $\boldsymbol{s_1}$ |
| 4 | $s_4$ | $s_2$ | $\boldsymbol{s_6}$ | $\boldsymbol{s_2}$ | $s_4$ | $s_6$ | $s_8$ | $s_{10}$ | $s_9$ | $s_4$ |
| 5 | $s_5$ | $s_5$ | $s_8$ | $s_7$ | $s_{10}$ | $s_{10}$ | $\boldsymbol{s_5}$ | $s_8$ | $s_5$ | $s_{10}$ |
| 6 | $s_6$ | $\boldsymbol{s_8}$ | $s_3$ | $s_3$ | $s_5$ | $s_3$ | $s_2$ | $s_3$ | $s_7$ | $s_8$ |
| 7 | $s_7$ | $s_3$ | $s_5$ | $s_8$ | $\boldsymbol{s_9}$ | $s_7$ | $s_9$ | $s_1$ | $s_4$ | $s_9$ |
| 8 | $s_8$ | $s_4$ | $s_7$ | $s_4$ | $s_6$ | $s_1$ | $s_6$ | $s_9$ | $s_3$ | $s_2$ |
| 9 | $s_9$ | $s_7$ | $s_9$ | $s_{10}$ | $s_3$ | $s_5$ | $s_7$ | $\boldsymbol{s_7}$ | $s_1$ | $s_3$ |
| 10 | $s_{10}$ | $s_{10}$ | $s_1$ | $s_9$ | $s_1$ | $s_9$ | $s_{10}$ | $s_5$ | $s_2$ | $s_7$ |

**Student preferences over colleges (Final matches highlighted):**

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| Rank | $s_1$ | $s_2$ | $s_3$ | $s_4$ | $s_5$ | $s_6$ | $s_7$ | $s_8$ | $s_9$ | $s_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $c_5$ | $\boldsymbol{c_4}$ | $c_5$ | $\boldsymbol{c_6}$ | $c_3$ | $\boldsymbol{c_3}$ | $\boldsymbol{c_8}$ | $c_1$ | $c_1$ | $c_2$ |
| 2 | $\boldsymbol{c_{10}}$ | $c_9$ | $c_6$ | $c_2$ | $\boldsymbol{c_7}$ | $c_4$ | $c_6$ | $c_4$ | $\boldsymbol{c_5}$ | $c_{10}$ |
| 3 | $c_6$ | $c_{10}$ | $\boldsymbol{c_1}$ | $c_5$ | $c_9$ | $c_5$ | $c_4$ | $\boldsymbol{c_2}$ | $c_2$ | $\boldsymbol{c_9}$ |
| 4 | $c_4$ | $c_6$ | $c_7$ | $c_8$ | $c_{10}$ | $c_1$ | $c_2$ | $c_6$ | $c_{10}$ | $c_4$ |
| 5 | $c_7$ | $c_3$ | $c_4$ | $c_1$ | $c_1$ | $c_9$ | $c_9$ | $c_3$ | $c_3$ | $c_1$ |
| 6 | $c_2$ | $c_7$ | $c_8$ | $c_7$ | $c_5$ | $c_7$ | $c_7$ | $c_7$ | $c_8$ | $c_7$ |
| 7 | $c_8$ | $c_1$ | $c_9$ | $c_3$ | $c_4$ | $c_{10}$ | $c_5$ | $c_5$ | $c_6$ | $c_5$ |
| 8 | $c_9$ | $c_5$ | $c_{10}$ | $c_9$ | $c_2$ | $c_8$ | $c_{10}$ | $c_{10}$ | $c_7$ | $c_6$ |
| 9 | $c_1$ | $c_2$ | $c_2$ | $c_{10}$ | $c_6$ | $c_2$ | $c_1$ | $c_8$ | $c_4$ | $c_3$ |
| 10 | $c_3$ | $c_8$ | $c_3$ | $c_4$ | $c_8$ | $c_6$ | $c_3$ | $c_9$ | $c_9$ | $c_8$ |

We execute the student-proposing DA algorithm. We track the process using a chalkboard-style simulation: the colleges are listed across the top, and proposing students are written below them. In each round, if a college receives multiple proposals (or a new proposal challenges a held student), the college's less preferred student is simply crossed out (~~struck through~~).

**Round 1.**
All 10 students propose to their first-choice colleges. Colleges $c_1$, $c_3$, and $c_5$ receive multiple proposals and must reject the less preferred students.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_8$ | $s_{10}$ | $s_6$ | $s_2$ | $s_3$ | $s_4$ | | $s_7$ | | |
| $\cancel{s_9}$ | | $\cancel{s_5}$ | | $\cancel{s_1}$ | | | | | |

*Rejected:* $s_9$ (next $c_5$), $s_5$ (next $c_7$), $s_1$ (next $c_{10}$).

**Round 2.**
The three rejected students propose to their next choices. $s_9$ challenges $s_3$ at $c_5$. College $c_5$ prefers $s_9$ (#7) over $s_3$ (#9).

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_8$ | $s_{10}$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | | $s_1$ |
| | | | | $\cancel{s_3}$ | | | | | |

*Rejected:* $s_3$ (next $c_6$).

**Round 3.**
$s_3$ proposes to $c_6$. College $c_6$ evaluates the held $s_4$ and the new $s_3$. $s_4$ is $c_6$'s top pick and is retained.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_8$ | $s_{10}$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | | $s_1$ |
| | | | | | $\cancel{s_3}$ | | | | |

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*Rejected:* $s_3$ (next $c_1$).

**Round 4.**
$s_3$ proposes to $c_1$. College $c_1$ prefers $s_3$ (#3) over the currently held $s_8$ (#8).

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_3$ | $s_{10}$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | | $s_1$ |
| $\cancel{s_8}$ | | | | | | | | | |

*Rejected:* $s_8$ (next $c_4$).

**Round 5.**
$s_8$ proposes to $c_4$. College $c_4$ prefers the held $s_2$ (#4) over $s_8$ (#7).

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_3$ | $s_{10}$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | | $s_1$ |
| | | | $\cancel{s_8}$ | | | | | | |

*Rejected:* $s_8$ (next $c_2$).

**Round 6.**
$s_8$ proposes to $c_2$. College $c_2$ prefers $s_8$ (#6) over the held $s_{10}$ (#10).

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_3$ | $s_8$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | | $s_1$ |
| | $\cancel{s_{10}}$ | | | | | | | | |

*Rejected:* $s_{10}$ (next $c_{10}$).

**Round 7.**
$s_{10}$ proposes to $c_{10}$. College $c_{10}$ prefers the held $s_1$ (#3) over $s_{10}$ (#5).

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_3$ | $s_8$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | | $s_1$ |
| | | | | | | | | | $\cancel{s_{10}}$ |

*Rejected:* $s_{10}$ (next $c_9$).

**Round 8.**
$s_{10}$ proposes to $c_9$. College $c_9$ is empty and accepts immediately.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_3$ | $s_8$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | $s_{10}$ | $s_1$ |

*No further rejections. The algorithm terminates.*

The **student-optimal stable matching** $\mu_S$ is directly read from the final chalkboard state:

| College | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Student | $s_3$ | $s_8$ | $s_6$ | $s_2$ | $s_9$ | $s_4$ | $s_5$ | $s_7$ | $s_{10}$ | $s_1$ |

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### 9.3.3 Properties of DA

**Theorem 9.18: DA is Pairwise Stable**

The matching produced by the student-proposing DA algorithm is pairwise stable.

***Proof for Theorem***

The argument proceeds as follows. Suppose, for contradiction, that $(c, s)$ is a blocking pair under the DA outcome $\mu$. Then:

1. Student $s$ prefers $c$ to $\mu^{-1}(s)$, meaning $c \succ_s \mu^{-1}(s)$. Since DA has students propose in order of preference, $s$ must have proposed to $c$ at some earlier round before being eventually matched with $\mu^{-1}(s)$.

2. But $s$ was rejected by $c$ at that round, which means $c$ was holding (tentatively matched with) a student it prefers to $s$. Since colleges only ever improve their tentative match over the course of the algorithm, $c$'s final match $\mu(c)$ is at least as preferred as the student it held when it rejected $s$. Hence $\mu(c) \succ_c s$.

This contradicts the requirement that $c$ prefers $s$ to $\mu(c)$. Therefore no blocking pair can exist.

**Theorem 9.19: Strategy-Proofness for the Proposing Side**

In the student-proposing DA, truthful reporting of preferences is a dominant strategy for every **student**. That is, no student can obtain a strictly better outcome by misreporting their preferences, regardless of what other students or colleges report.

***Proof for Theorem***

We present the proof for the general many-to-one matching market where each college $c$ has a quota $q_c \geq 1$.

We proceed by contradiction. Suppose there exists a true preference profile $P$, but some student $s^*$ submits a fake preference list $P'_{s^*}$. Let $\mu$ be the truthful DA outcome, and let $\mu'$ be the DA outcome under the misreported profile $P' = (P'_{s^*}, P_{-s^*})$.

Suppose the misreport is strictly profitable for $s^*$, meaning $\mu'(s^*) \succ_{s^*} \mu(s^*)$.

**Step 1: Identify the "Improving" Coalition.**
Let $S'$ be the set of all students who are strictly better off under $\mu'$ than under $\mu$ according to their true preferences:

$$S' = \{s \in \mathcal{S} \mid \mu'(s) \succ_s \mu(s)\}$$

By our assumption, $s^* \in S'$. Let $C' = \mu'(S')$ be the set of colleges these students are matched to in $\mu'$.

**Step 2: The Rejection Implication.**
For every student $s \in S'$, because $\mu'(s) \succ_s \mu(s)$, $s$ must have proposed to $\mu'(s)$ and been rejected during the *truthful* DA. Consequently, every college $c \in C'$ must have rejected at least one student from $S'$ in the truthful DA.

Under the rules of DA with quotas, a college only rejects an acceptable student if

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its quota is completely full. This implies that in the truthful outcome $\mu$, every college $c \in C'$ exactly filled its capacity ($|\mu(c)| = q_c$), and it strictly prefers **every single student** currently in $\mu(c)$ to the $S'$ student it rejected.

**Step 3: All Displaced Students must belong to $S'$.**
Take any college $c \in C'$. Since $c \in C'$, there is some student $s \in S'$ such that $s \in \mu'(c)$. Let $k \in \mu(c)$ be **any** student assigned to $c$ in the truthful DA. We claim $k$ must also belong to $S'$.

Suppose for contradiction that $k \notin S'$. This means $k$ does not strictly prefer $\mu'$ over $\mu$. Since $c$ must be filled with different students or drop some students to accommodate $s$ in $\mu'$, if $k$ is displaced from $c$, $k$ strictly prefers $\mu(k) = c$ over $\mu'(k)$. Since $k \notin S'$ but $s^* \in S'$, $k \neq s^*$. Thus, $k$ reported truthfully.

Now consider the matching $\mu'$ under the reported profile $P'$. College $c$ admits $s \in S'$ in $\mu'$. But from Step 2, $c$ strictly prefers $k$ over $s$ ($k \succ_c s$). At the same time, $k$ strictly prefers $c$ over $\mu'(k)$. Because $k$'s reported preference is their true preference, $(c, k)$ forms a **blocking pair** for $\mu'$ under $P'$! This contradicts the fact that $\mu'$ is the stable DA outcome under $P'$.

Hence, our claim holds: **every** student $k \in \mu(c)$ sitting in college $c$ during the truthful DA must belong to $S'$.

**Step 4: The Counting Contradiction.**
Let us count the seats in $C'$. The total capacity of all colleges in $C'$ is $\sum_{c \in C'} q_c$.

By Step 2, every college in $C'$ is completely full under $\mu$. By Step 3, every single student occupying these seats in $\mu$ belongs to $S'$. Therefore, $S'$ contains **at least** $\sum_{c \in C'} q_c$ students.

Now look at $\mu'$. By the definition of $C'$, every student in $S'$ is assigned to a college in $C'$. Since the maximum number of students $C'$ can hold is $\sum_{c \in C'} q_c$, $S'$ can contain **at most** $\sum_{c \in C'} q_c$ students.

Thus, $|S'| = \sum_{c \in C'} q_c$. The students assigned to $C'$ in the truthful matching $\mu$ are *exactly* the students in $S'$, occupying all available seats.

In $\mu'$, this exact same group of students $S'$ is completely reallocated among the exact same seats in $C'$. But by the definition of $S'$, **every single student in $S'$ strictly prefers their new seat in $\mu'$ to their old seat in $\mu$.**

This implies $\mu'$ **strictly Pareto dominates** $\mu$ for this subset of students, without affecting anyone else. However, it is a fundamental property of the student-proposing DA that its outcome ($\mu$) is the **student-optimal stable matching** (which is Pareto efficient for the students). A strict Pareto improvement via a pure internal reallocation is mathematically impossible.

This contradiction proves that no such profitable misreport $P'_{s^*}$ can exist. ■

**Remark.**

The intuition is clean: under DA, a student who misreports can only cause themselves to be rejected by colleges they would have been admitted to, or cause themselves to propose to less preferred colleges earlier. Misreporting can never lead to admission at a college that would have rejected the student under truthful reporting.

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**Theorem 9.20: Impossibility of Two-Sided Strategy-Proofness**

No stable matching mechanism is strategy-proof for **both** sides simultaneously. In particular, under the student-proposing DA, colleges may have incentives to misreport their preferences.

**Remark (How Can the Accepting Side Manipulate?).**

The accepting side (colleges in student-proposing DA) can potentially manipulate by *truncating* their preference lists or *reordering* them. For instance, a college might strategically rank a less-preferred student higher to trigger a chain of rejections that ultimately results in a more favorable match. This impossibility result (Roth, 1982) highlights a fundamental tension: **any stable matching mechanism must sacrifice strategy-proofness on at least one side**.

### 9.3.4 Many-to-One Matching (Quotas)

So far, our two-sided matching model assumed $q_c = 1$ for all colleges, mathematically mirroring a strict one-to-one "marriage" market. In reality, the classic Gale-Shapley model was built for the *college admissions* problem, where institutions admit entire incoming classes. Generalizing from one-to-one to many-to-one matching introduces capacity constraints, but the core theoretical machinery translates beautifully.

**Definition 9.21: Many-to-One Matching Market**

Each college $c \in \mathcal{C}$ now has a maximum capacity or **quota** $q_c \geq 1$. A **matching** $\mu$ is a mapping such that:

- For each student $s$, $\mu(s) \in \mathcal{C} \cup \{\emptyset\}$ (a student attends at most one college or remains unmatched).
- For each college $c$, $\mu(c) \subseteq \mathcal{S}$ with $|\mu(c)| \leq q_c$ (a college admits a *set* of students up to its quota).
- $\mu(s) = c \iff s \in \mu(c)$.

In a many-to-one matching market, a college $c$ with quota $q_c$ ultimately admits a *set* of students. While the model specifies the college's strict ranking over individual students (denoted by $\succ_c$), we must define how it ranks different incoming classes (subsets of students). To keep the model tractable and rule out complex complementarities (e.g., "I only want student A if I also get student B"), we impose a standard behavioral assumption on how colleges evaluate groups.

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### Assumption 9.22: Responsive Preferences

Let $\succ_c^*$ denote college $c$'s preference relation over *sets* of students. We assume $\succ_c^*$ is **responsive** to the college's preferences over individual students $\succ_c$. Specifically, for any set of students $S$ strictly below capacity ($|S| < q_c$), and any two students $s, s' \notin S$:

1. **Substitution:** Replacing a student with a strictly more preferred individual always makes the college strictly better off:

$$S \cup \{s\} \succ_c^* S \cup \{s'\} \iff s \succ_c s'$$

2. **Addition:** Adding an acceptable student (one who is preferred to leaving a seat empty, $s \succ_c \emptyset$) always makes the college strictly better off:

$$S \cup \{s\} \succ_c^* S \iff s \succ_c \emptyset$$

**Remark (Why do we need this assumption?).**

Without responsiveness, evaluating stability becomes a combinatorial nightmare. A college might reject a globally top-ranked student simply because they do not "fit" well with the rest of the admitted cohort.

By assuming responsive preferences, we guarantee that a college's evaluation of a group is strictly and additively tied to the independent quality of its individual members. *Crucially, this mathematically justifies why we only need to check for pairwise blocking pairs.* If a college ever wants to deviate and alter its admitted class, responsiveness ensures that this desire will manifest as a simple pairwise blocking condition with a single better student, relieving us from checking complex multi-student coordinated deviations.

### Definition 9.23: Pairwise Stability with Quotas

A matching $\mu$ is pairwise stable if it is individually rational and contains no **blocking pair**. A pair $(c, s)$ blocks $\mu$ if:

1. Student $s$ prefers $c$ to their current match: $c \succ_s \mu(s)$;
2. **Either** $c$ has an empty seat ($|\mu(c)| < q_c$),
   **Or** $c$ is full ($|\mu(c)| = q_c$) but prefers $s$ to at least one of its currently admitted students ($s \succ_c s'$ for some $s' \in \mu(c)$).

### Student-Proposing DA (with Quotas)

The DA algorithm adapts to quotas with a single, straightforward modification:

1. **Propose:** Each unassigned student proposes to their most preferred acceptable college that has not yet rejected them.

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2. **Hold/Reject:** Each college $c$ considers its new applicants alongside the students it tentatively held from the previous round. It ranks this combined pool, **tentatively holds up to $q_c$ of its most preferred students**, and rejects the rest.

3. **Repeat** until no student is rejected.

**Example (DA with Quotas).**

Consider 6 students $\{s_1, \ldots, s_6\}$ and 2 colleges $\{c_1, c_2\}$. College $c_1$ has a quota $q_1 = 2$, and college $c_2$ has a quota $q_2 = 3$. Notice that total demand (6 students) exceeds total supply (5 seats), meaning at least one student will be left unmatched.

**College preferences (Quotas in parentheses, Final matches highlighted):**

| Rank | $c_1$ $(q = 2)$ | $c_2$ $(q = 3)$ |
|---|---|---|
| 1 | $\boldsymbol{s_1}$ | $\boldsymbol{s_6}$ |
| 2 | $\boldsymbol{s_2}$ | $\boldsymbol{s_5}$ |
| 3 | $s_3$ | $\boldsymbol{s_4}$ |
| 4 | $s_4$ | $s_3$ |
| 5 | $s_5$ | $s_2$ |
| 6 | $s_6$ | $s_1$ |

**Student preferences (Final matches highlighted):**

| Rank | $s_1$ | $s_2$ | $s_3$ | $s_4$ | $s_5$ | $s_6$ |
|---|---|---|---|---|---|---|
| 1 | $c_2$ | $\boldsymbol{c_1}$ | $c_1$ | $c_1$ | $\boldsymbol{c_2}$ | $c_1$ |
| 2 | $\boldsymbol{c_1}$ | $c_2$ | $c_2$ | $\boldsymbol{c_2}$ | $c_1$ | $\boldsymbol{c_2}$ |

We run the student-proposing DA algorithm. Remember that colleges rank their combined pool of applicants and simply draw a "cutoff line" at their quota.

**Round 1.**
Students propose to their top choices: $s_1, s_5 \to c_2$, while $s_2, s_3, s_4, s_6 \to c_1$.

| $c_1$ $(q = 2)$ | $c_2$ $(q = 3)$ |
|---|---|
| $s_2$ | $s_1$ |
| $s_3$ | $s_5$ |
| $\cancel{s_4}$ | |
| $\cancel{s_6}$ | |

*Explanation:* $c_1$ receives 4 applicants for 2 seats. Looking at $c_1$'s preference list, $s_2 \succ s_3 \succ s_4 \succ s_6$. It holds the top two $(s_2, s_3)$ and rejects the rest. $c_2$ only has 2 applicants for 3 seats, so it holds both.
*Rejected:* $s_4$ (next $c_2$), $s_6$ (next $c_2$).

**Round 2.**
The rejected students $s_4$ and $s_6$ propose to $c_2$. $c_2$'s new pool is $\{s_1, s_5\}$ (held) $\cup$ $\{s_4, s_6\}$ (new) $= \{s_1, s_4, s_5, s_6\}$.

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| $c_1\,(q=2)$ | $c_2\,(q=3)$ |
|---|---|
| $s_2$ | $s_6$ |
| $s_3$ | $s_5$ |
| | $s_4$ |
| | $\cancel{s_1}$ |

*Explanation:* $c_2$ ranks its pool: $s_6 \succ s_5 \succ s_4 \succ s_1$. It keeps the top 3. Notice that $s_1$, who was safely held in Round 1, is now bumped out by stronger applicants!
*Rejected:* $s_1$ (next $c_1$).

**Round 3.**
$s_1$ proposes to $c_1$. $c_1$'s pool is now $\{s_2, s_3\}$ (held) $\cup$ $\{s_1\}$ (new).

| $c_1\,(q=2)$ | $c_2\,(q=3)$ |
|---|---|
| $s_1$ | $s_6$ |
| $s_2$ | $s_5$ |
| $\cancel{s_3}$ | $s_4$ |

*Explanation:* $c_1$ ranks its pool: $s_1 \succ s_2 \succ s_3$. It keeps the top 2. Another devastating bump occurs: $s_3$ is kicked out to make room for $s_1$.
*Rejected:* $s_3$ (next $c_2$).

**Round 4.**
$s_3$ proposes to $c_2$. $c_2$'s pool is $\{s_4, s_5, s_6\}$ (held) $\cup$ $\{s_3\}$ (new).

| $c_1\,(q=2)$ | $c_2\,(q=3)$ |
|---|---|
| $s_1$ | $s_6$ |
| $s_2$ | $s_5$ |
| | $s_4$ |
| | $\cancel{s_3}$ |

*Explanation:* $c_2$ ranks its pool: $s_6 \succ s_5 \succ s_4 \succ s_3$. The cutoff is 3, so $s_3$ is rejected immediately.
*Rejected:* $s_3$ (no choices remaining).

**Round 5.**
Student $s_3$ has exhausted their preference list. No unassigned student has an acceptable college left to propose to. The algorithm terminates.

The final student-optimal stable matching $\mu_S$ with quotas is:

$$\mu_S(c_1) = \{s_1, s_2\}, \quad \mu_S(c_2) = \{s_4, s_5, s_6\}, \quad \mu_S(s_3) = \emptyset$$

### 9.3.5 Truncated Preferences (Unacceptable Matches)

In our previous examples, we implicitly assumed that every college was acceptable to every student, and every student was acceptable to every college. In reality, matching markets

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are characterized by **outside options**. A student might prefer to stay home or enter the labor market rather than attend a poorly ranked college. Similarly, a college might prefer to leave a seat empty rather than admit a student who does not meet its minimum academic standards.

We model this by introducing **truncated preferences**. An agent ranks a subset of the opposite side and deems the rest *unacceptable*.

### Definition 9.24: Unacceptable Matches, Matching with Truncations and Individual Rationality

We use the symbol $\emptyset$ to denote the outside option of remaining unmatched.

- A college $c$ is **acceptable** to student $s$ if $c \succ_s \emptyset$.
- A student $s$ is **acceptable** to college $c$ if $s \succ_c \emptyset$.

Any agent ranked below $\emptyset$ in a preference list is strictly worse than being unmatched.

### Definition 9.25: Matching with Truncations

Let $\mathcal{C} = \{c_1, \ldots, c_m\}$ (with quotas $q_j$) and $\mathcal{S} = \{s_1, \ldots, s_n\}$. Each student $s$ has a strict preference $\succ_s$ over $\mathcal{C}$, and each college $c$ has a strict preference $\succ_c$ over $\mathcal{S}$. Agents may rank only a subset of the other side as acceptable, leaving the ranking below $\emptyset$ as unacceptable. A **matching** is a function $\mu : \mathcal{S} \to \mathcal{C} \cup \{\emptyset\}$ (with the constraint that $|\{s : \mu(s) = c\}| \leq q_c$ for each $c$). Here $\mu(s) = \emptyset$ means student $s$ is unmatched.

### Definition 9.26: Individual Rationality

A matching $\mu$ is **Individually Rational (IR)** if no agent is matched with an unacceptable partner: $\mu(s) \succsim_s \emptyset$ and $\mu(c) \succsim_c \emptyset$ for all $s \in \mathcal{S}, c \in \mathcal{C}$.

### Definition 9.27: Stability with Truncations

A matching $\mu$ is **stable** if:

(i) **Individual rationality:** Every matched student prefers their match to being unmatched: if $\mu(s) \in \mathcal{C}$, then $\mu(s) \succ_s \emptyset$. Similarly, every college finds each of its matched students acceptable.

(ii) **No blocking pair:** There is no pair $(s, c)$ with $\mu(s) \neq c$ such that $c \succ_s \mu(s)$ and either $c$ has an unfilled seat or $s \succ_c s'$ for some $s' \in \mu(c)$.

The DA algorithm handles truncated preferences effortlessly by embedding the IR constraint directly into its rules:

1. Students never propose to unacceptable colleges (they drop out of the market once they reach $\emptyset$ on their lists).

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2. Colleges immediately reject any proposing student who is unacceptable to them, even if they have empty seats.

This slight modification is not just algorithmic; it provides a constructive proof for a fundamental existence theorem in matching theory.

**Theorem 9.28: Existence of Stable Matchings with Truncations**

For any two-sided matching market with strict, truncated preferences, there always exists at least one matching that is both pairwise stable and individually rational.

***Proof for Theorem***

*The proof is constructive. We show that the Deferred Acceptance (DA) algorithm, modified with the two IR rules above, always terminates and outputs a matching that satisfies both properties.*

**Part 1: Individual Rationality (IR).**
By the modified rules of the DA algorithm, no student ever proposes to an unacceptable college. Furthermore, no college ever tentatively holds (or permanently admits) an unacceptable student. Therefore, in the final matching $\mu$, every student either receives an acceptable college or remains unmatched ($\mu(s) \succsim_s \emptyset$), and every college only admits acceptable students. Thus, $\mu$ is individually rational.

**Part 2: Pairwise Stability.**
Suppose, for contradiction, that there exists a blocking pair $(c, s)$ in the final matching $\mu$. By definition of a blocking pair under quotas and truncations:

1. Student $s$ strictly prefers $c$ over their final match: $c \succ_s \mu(s) \succsim_s \emptyset$.
2. College $c$ strictly prefers $s$ over $\emptyset$ (meaning $s$ is acceptable to $c$), AND either $c$ has an empty seat ($|\mu(c)| < q_c$) or $c$ strictly prefers $s$ over some admitted student $s' \in \mu(c)$.

Since $c \succ_s \mu(s)$, student $s$ must have proposed to college $c$ at some round prior to proposing to $\mu(s)$ (or before dropping out, if $\mu(s) = \emptyset$).

Since $s$ is not matched with $c$ in the end, college $c$ must have rejected $s$ at some round. Under the DA rules, a college only rejects an *acceptable* student if it has already filled its entire quota $q_c$ with students it strictly prefers to the rejected student.

Moreover, as the DA algorithm progresses, a college's pool of tentatively held students can only improve (a college only drops a held student if a strictly better one proposes). Therefore, in the final matching $\mu$, college $c$ must be exactly at full capacity ($|\mu(c)| = q_c$), and it must strictly prefer every student in $\mu(c)$ to $s$.

This directly contradicts the second condition of the blocking pair (that $c$ has an empty seat or prefers $s$ to some admitted student $s'$).

Hence, no blocking pair can exist. The modified DA algorithm always terminates yielding an individually rational and pairwise stable matching, proving existence. ∎

**Example (DA with Truncated Preferences).**

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Consider our 10-student, 10-college market again, but now with severe truncations. The symbol $\emptyset$ marks the end of the acceptable options. (For columns that do not explicitly show $\emptyset$, we assume the list naturally ends there or the unlisted options are unacceptable).

**College preferences (Final matches highlighted):**

| Rank | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $s_1$ | $s_1$ | $s_{10}$ | $s_6$ | $s_7$ | $\boldsymbol{s_4}$ | $s_1$ | $s_6$ | $s_8$ | $s_6$ |
| 2 | $s_2$ | $s_6$ | $s_2$ | $s_1$ | $s_2$ | $s_8$ | $s_4$ | $s_4$ | $\boldsymbol{s_{10}}$ | $\boldsymbol{s_5}$ |
| 3 | $\boldsymbol{s_3}$ | $\boldsymbol{s_9}$ | $s_4$ | $s_5$ | $s_8$ | $s_2$ | $s_3$ | $s_2$ | $s_6$ | $s_1$ |
| 4 | $s_4$ | $s_2$ | $\boldsymbol{s_6}$ | $\boldsymbol{s_2}$ | $s_4$ | $s_6$ | $\boldsymbol{s_8}$ | $s_{10}$ | $s_9$ | $s_4$ |
| 5 | $s_5$ | $\emptyset$ | $s_8$ | $s_7$ | $\emptyset$ | $s_{10}$ | $s_5$ | $s_8$ | $s_5$ | $s_{10}$ |
| 6 | $s_6$ | | $s_3$ | $s_3$ | | $s_3$ | $s_2$ | $s_3$ | $s_7$ | $s_8$ |
| 7 | $\emptyset$ | | $s_5$ | $s_8$ | | $s_7$ | $s_9$ | $s_1$ | $s_4$ | $s_9$ |
| 8 | | | $s_7$ | $s_4$ | | $s_1$ | $s_6$ | $\emptyset$ | $s_3$ | $s_2$ |
| 9 | | | $s_9$ | $s_{10}$ | | $s_5$ | $s_7$ | | $s_1$ | $s_3$ |
| 10 | | | $s_1$ | $s_9$ | | $s_9$ | $s_{10}$ | | $s_2$ | $s_7$ |
| 11 | | | $\emptyset$ | $\emptyset$ | | | | | | |

**Student preferences (Final matches highlighted):**

| Rank | $s_1$ | $s_2$ | $s_3$ | $s_4$ | $s_5$ | $s_6$ | $s_7$ | $s_8$ | $s_9$ | $s_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $c_5$ | $\boldsymbol{c_4}$ | $c_5$ | $\boldsymbol{c_6}$ | $c_3$ | $\boldsymbol{c_3}$ | $c_8$ | $c_1$ | $c_1$ | $c_2$ |
| 2 | $c_{10}$ | $c_9$ | $c_6$ | $c_2$ | $c_7$ | $c_4$ | $c_6$ | $c_4$ | $c_5$ | $c_{10}$ |
| 3 | $c_6$ | $c_{10}$ | $\boldsymbol{c_1}$ | $c_5$ | $c_9$ | $c_5$ | $c_4$ | $c_2$ | $\boldsymbol{c_2}$ | $\boldsymbol{c_9}$ |
| 4 | $\emptyset$ | $c_6$ | $c_7$ | $c_8$ | $\boldsymbol{c_{10}}$ | $c_1$ | $c_2$ | $c_6$ | $c_{10}$ | $c_4$ |
| 5 | | $c_3$ | $c_4$ | $c_1$ | $c_1$ | $c_9$ | $c_9$ | $c_3$ | $\emptyset$ | $c_1$ |
| 6 | | $c_7$ | $c_8$ | $c_7$ | $c_5$ | $c_7$ | $\emptyset$ | $\boldsymbol{c_7}$ | | $c_7$ |
| 7 | | $c_8$ | $c_3$ | $c_3$ | $c_4$ | $c_{10}$ | | $c_5$ | | $c_5$ |
| 8 | | $c_1$ | $c_9$ | $c_9$ | $c_2$ | $c_8$ | | $c_{10}$ | | $c_6$ |
| 9 | | $c_5$ | $c_{10}$ | $c_{10}$ | $c_6$ | $c_2$ | | $c_8$ | | $c_3$ |
| 10 | | $c_2$ | $c_2$ | | $c_8$ | | | $c_9$ | | $c_8$ |

We execute the student-proposing DA. Notice how truncations cause severe "immediate rejections" even when colleges are empty.

**Round 1.**

Students propose to their top choices.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $\cancel{s_8}$ | $\cancel{s_{10}}$ | $s_6$ | $s_2$ | $\cancel{s_1}$ | $s_4$ | | $\cancel{s_7}$ | | |
| $\cancel{s_9}$ | | $\cancel{s_5}$ | | $\cancel{s_3}$ | | | | | |

*Explanation:* Look closely at the rejections. College $c_5$ rejects *both* $s_1$ and $s_3$ immediately, because neither student is on $c_5$'s acceptable list $(s_7, s_2, s_8, s_4, \emptyset)$. Similarly,

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$c_1, c_2, c_8$ reject their applicants purely because they fail to meet the individual rationality threshold.
*Rejected:* $s_1, s_3, s_5, s_7, s_8, s_9, s_{10}$.

**Round 2.**
The mass of rejected students propose to their second choices.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| | | $s_6$ | $s_2$ | $\cancel{s_9}$ | $s_4$ | $s_5$ | | | $s_1$ |
| | | | $\cancel{s_8}$ | | $\cancel{s_3}$ | | | | $\cancel{s_{10}}$ |
| | | | | | $\cancel{s_7}$ | | | | |

*Explanation:* $c_{10}$ receives $s_1$ and $s_{10}$; it prefers $s_1$. $c_6$ holds $s_4$ and rejects both $s_3$ and $s_7$. $c_4$ rejects $s_8$. $c_5$ continues to be empty because $s_9$ is unacceptable.
*Rejected:* $s_3, s_7, s_8, s_9, s_{10}$.

**Round 3.**

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_3$ | $s_9$ | $s_6$ | $s_2$ | | $s_4$ | $s_5$ | | $s_{10}$ | $s_1$ |
| | $\cancel{s_8}$ | | $\cancel{s_7}$ | | | | | | |

*Explanation:* $c_1$ accepts $s_3$. $c_9$ accepts $s_{10}$. $c_2$ receives $s_9$ and $s_8$; $s_9$ is acceptable, but $s_8$ is unacceptable! Thus $c_2$ holds $s_9$ and rejects $s_8$.
*Rejected:* $s_7, s_8$.

**Rounds 4 & 5.**
$s_7 \to c_2$ (unacceptable, rejected). $s_8 \to c_6$ (rejected, $s_4 \succ s_8$).
Next: $s_7 \to c_9$ (rejected, $s_{10} \succ s_7$). $s_8 \to c_3$ (rejected, $s_6 \succ s_8$).
*Rejected:* $s_7, s_8$.

**Round 6 (The Dropouts Begin).**
Student $s_7$ has exhausted their list (reaching $\emptyset$ after $c_9$). $s_7$ **drops out of the market and remains permanently unmatched.**
Meanwhile, $s_8$ proposes to $c_7$.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $s_3$ | $s_9$ | $s_6$ | $s_2$ | | $s_4$ | $s_8$ | | $s_{10}$ | $s_1$ |
| | | | | | | $\cancel{s_5}$ | | | |

*Explanation:* $c_7$ prefers the new $s_8$ over the held $s_5$.
*Rejected:* $s_5$ (next $c_9$).

**Rounds 7, 8, & 9.**
$s_5 \to c_9$ (rejected, $s_{10} \succ s_5$).
$s_5 \to c_{10}$ (held, $s_5 \succ s_1$). $s_1$ is bumped!
$s_1 \to c_6$ (rejected, $s_4 \succ s_1$).

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**Round 10.**
Student $s_1$ has exhausted their truncated list (reaching $\emptyset$ after $c_6$). $s_1$ **drops out of the market.** No further proposals can be made. The algorithm terminates.

The student-optimal stable matching $\mu_S$ under truncated preferences is:

| College | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Student | $s_3$ | $s_9$ | $s_6$ | $s_2$ | $\emptyset$ | $s_4$ | $s_8$ | $\emptyset$ | $s_{10}$ | $s_5$ |

Notice that students $s_1$ and $s_7$, as well as colleges $c_5$ and $c_8$, remain completely unmatched. This is the profound impact of the Individual Rationality constraint: stability now requires that some participants go home empty-handed rather than accept sub-par matches.

The introduction of truncated preferences is not just a way to model outside options (Individual Rationality); it also reveals a powerful method of strategic manipulation in matching markets.

If the mechanism asks participants to submit a rank-order list, can an agent benefit by *pretending* that certain acceptable partners are actually unacceptable? In other words, can an agent game the system by submitting an artificially shortened list? This question was famously answered by Alvin Roth.

### Theorem 9.29: Strategic Truncation (Roth, 1982)

- **The Proposing Side:** In the college-proposing DA, no college can ever strictly improve its match by truncating its true preference list. (Symmetrically, no student can benefit from truncation in the student-proposing DA).
- **The Receiving Side:** In the student-proposing DA, a college **CAN** strictly improve its match by artificially truncating its preference list!

**Remark (The Intuition Behind Strategic Truncation).**

That the proposing side cannot benefit from truncation is a direct corollary of the Gale-Shapley Strategy-Proofness theorem: since truthful reporting is a dominant strategy for the proposers, no manipulation (including truncation) can help them.

But why does truncation work for the receiving side? Consider a college under the student-proposing DA. Suppose its true preference is $s_1 \succ_c s_2 \succ_c s_3$.

If the college reports truthfully, it might receive an application from $s_2$ early in the algorithm and tentatively hold them. Holding $s_2$ acts as a "safety net" for the college, but it also permanently removes $s_2$ from the broader applicant pool, effectively stabilizing the rest of the market.

If the college instead **truncates** its list to only include its top choice ($P'_c : s_1 \succ \emptyset$), it will ruthlessly reject $s_2$. This rejection forces $s_2$ to apply to their next-choice college, potentially triggering a massive **chain of rejections** across the market. If this chain reaction eventually bumps $s_1$ from whichever college was holding them, $s_1$ will fall down their own preference list and apply to $c$. By burning its safety net, the college forcefully

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destabilizes the market to shake loose its top choice!

Of course, truncation is a high-risk, high-reward strategy: if the rejection chain does not circle back to bump $s_1$, the college will end up completely empty-handed ($\emptyset$).

### 9.3.6 The Set of Stable Matchings

We now investigate the structure of the set of all stable matchings. The key results are: (i) the student-proposing DA produces the best stable matching for students, (ii) the interests of the two sides are diametrically opposed, and (iii) the set of stable matchings forms a lattice.

**Example (A 4×4 Matching Problem).**

Consider 4 students and 4 colleges with the following preferences:

| Rank | $c_1$ | $c_2$ | $c_3$ | $c_4$ |
|---|---|---|---|---|
| 1 | $s_1$ | $s_2$ | $s_3$ | $s_4$ |
| 2 | $s_2$ | $s_1$ | $s_4$ | $s_3$ |
| 3 | $s_3$ | $s_4$ | $s_1$ | $s_2$ |
| 4 | $s_4$ | $s_3$ | $s_2$ | $s_1$ |

| Rank | $s_1$ | $s_2$ | $s_3$ | $s_4$ |
|---|---|---|---|---|
| 1 | $c_4$ | $c_3$ | $c_2$ | $c_1$ |
| 2 | $c_3$ | $c_4$ | $c_1$ | $c_2$ |
| 3 | $c_2$ | $c_1$ | $c_4$ | $c_3$ |
| 4 | $c_1$ | $c_2$ | $c_3$ | $c_4$ |

Notice the "anti-diagonal" structure: colleges prefer students in order $s_1, s_2, s_3, s_4$, while students prefer colleges in the reverse order $c_4, c_3, c_2, c_1$.

**Student-Proposing DA.** Each student proposes to their top choice: $s_1 \to c_4$, $s_2 \to c_3$, $s_3 \to c_2$, $s_4 \to c_1$. Every college receives exactly one proposal. No rejections—the algorithm terminates immediately.

$$\mu_S : \quad (c_1, s_4),\ (c_2, s_3),\ (c_3, s_2),\ (c_4, s_1).$$

Every student gets their *first* choice. Every college gets their *last* choice.

**College-Proposing DA.** Each college proposes to their top choice: $c_1 \to s_1$, $c_2 \to s_2$, $c_3 \to s_3$, $c_4 \to s_4$. Every student receives exactly one proposal. No rejections.

$$\mu_C : \quad (c_1, s_1),\ (c_2, s_2),\ (c_3, s_3),\ (c_4, s_4).$$

Every college gets their *first* choice. Every student gets their *last* choice.

**A third stable matching.** Consider:

$$\bar{\mu} : \quad (c_1, s_3),\ (c_2, s_1),\ (c_3, s_4),\ (c_4, s_2).$$

One can verify that $\bar{\mu}$ has no blocking pair and is therefore stable, yet $\bar{\mu} \neq \mu_S$ and $\bar{\mu} \neq \mu_C$. In fact, this small market has **10 stable matchings** in total—the two DA solutions plus 8 others.

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**Student-Optimality of DA**

### Theorem 9.30: Student-Optimality of DA

The stable matching $\mu_S$ produced by the student-proposing DA is **weakly preferred by every student** to any other stable matching. That is, for every student $s$ and every stable matching $\mu$:

$$\mu_S^{-1}(s) \succsim_s \mu^{-1}(s).$$

By symmetry, the college-proposing DA produces the college-optimal stable matching $\mu_C$.

***Proof for Theorem***

We first introduce a key definition.

### Definition 9.31: Possible College

A college $c$ is **possible** for student $s$ if there exists some stable matching $\mu$ such that $\mu(s) = c$.

### Claim

In the student-proposing DA, no student $s$ is ever rejected by a college $c$ that is possible for $s$.

***Proof for Claim.***

By induction on the number of steps in the algorithm.

*Base case.* At step 1, every student proposes to their top choice. No rejections have occurred yet, so the claim holds vacuously.

*Inductive step.* Suppose that up to step $t-1$, no student has been rejected by a college that is possible for them. Suppose at step $t$, student $s$ is rejected by college $c$, because $c$ prefers another student $s'$ (whom it holds) to $s$:

$$s' \succ_c s.$$

We need to show that $c$ is not possible for $s$.

Since $s'$ is currently proposing to $c$ at step $t$, student $s'$ must have already been rejected by every college they prefer to $c$. By the induction hypothesis, all those colleges that rejected $s'$ were not possible for $s'$. Therefore, $c$ is the best possible college for $s'$: in any stable matching $\mu$, we have $c \succsim_{s'} \mu(s')$.

Now suppose, for contradiction, that $c$ is possible for $s$. Then there exists a stable matching $\mu$ with $\mu(s) = c$. In this matching $\mu$, student $s'$ is matched to $\mu(s') \neq c$ (since $s$ has $c$). But we just argued $c \succsim_{s'} \mu(s')$. Since $\mu(s') \neq c$ (because $s$ occupies $c$ in $\mu$) and preferences are strict, this implies $c \succ_{s'} \mu(s')$. Moreover, $c$ strictly prefers $s'$ to $s = \mu(c)$. This means $(c, s')$ blocks $\mu$—contradicting the stability of $\mu$. Therefore $c$ is not possible for $s$. ∎

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The theorem follows immediately. Since no student is ever rejected by a possible college during the algorithm, each student's final match under $\mu_S$ is the *best* college that is achievable in any stable matching.

### Opposition of Interests

The following proposition reveals that the two sides of the market have *diametrically opposed* interests over stable matchings.

**Proposition 9.32: Opposition of Interests**

If $\mu$ and $\mu'$ are both stable matchings and every student weakly prefers $\mu$ to $\mu'$, then every college weakly prefers $\mu'$ to $\mu$.

***Proof for Proposition.***

Suppose all students weakly prefer $\mu$ to $\mu'$. Suppose for contradiction that some college $c$ strictly prefers $\mu$ to $\mu'$, i.e., $\mu(c) \succ_c \mu'(c)$. Let $s = \mu(c)$. Then:

- College $c$ prefers $s$ to $\mu'(c)$: $s \succ_c \mu'(c)$.
- Student $s$ weakly prefers $\mu$ to $\mu'$, meaning $c = \mu^{-1}(s) \succsim_s \mu'^{-1}(s)$.

If $s$ strictly prefers $c$ to $\mu'^{-1}(s)$, then $(c, s)$ blocks $\mu'$, contradicting stability. If $s$ is indifferent, then since preferences are strict, $\mu^{-1}(s) = \mu'^{-1}(s) = c$, which implies $\mu(c) = \mu'(c) = s$, contradicting our assumption that $c$ strictly prefers $\mu$ to $\mu'$.

**Corollary 9.33: Pessimality**

The student-optimal stable matching $\mu_S$ is simultaneously the **college-pessimal** stable matching: every college gets its worst partner among all stable matchings. Symmetrically, $\mu_C$ is the student-pessimal stable matching.

### The Lattice of Stable Matchings

**Definition 9.34: Lattice of Stable Matchings**

Let $\mathcal{M}$ denote the set of all stable matchings. Define a partial order $\geq_S$ on $\mathcal{M}$ from the students' perspective: $\mu \geq_S \mu'$ if every student weakly prefers $\mu$ to $\mu'$. The set $(\mathcal{M}, \geq_S)$ forms a **distributive lattice**. This means:

- For any two stable matchings $\mu$ and $\mu'$, there exist stable matchings $\mu \vee \mu'$ (the *join*: student-best of the two) and $\mu \wedge \mu'$ (the *meet*: student-worst of the two).
- The lattice has a unique maximum element $\mu_S$ (student-optimal) and a unique minimum element $\mu_C$ (college-optimal, equivalently student-pessimal).

By the Opposition of Interests, the college ordering is exactly the reverse: what is the join for students is the meet for colleges, and vice versa.

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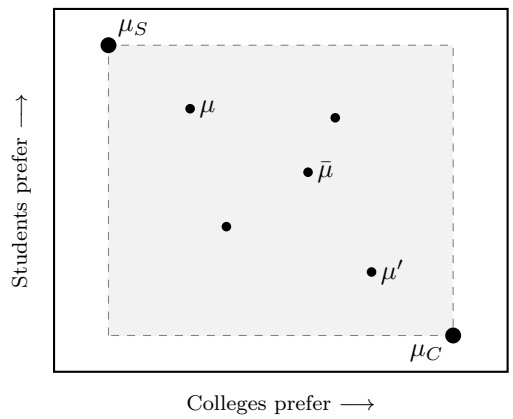
**Remark.**

In the 4×4 example above, $\mu_S$ sits at the top of the lattice (every student's favorite) and $\mu_C$ at the bottom. All 8 remaining stable matchings lie strictly between these two extremes. The following diagram illustrates the structure: the vertical axis represents student welfare (higher is better for students) and the horizontal axis represents college welfare (further right is better for colleges). By the Opposition of Interests, $\mu_S$ occupies the upper-left corner and $\mu_C$ the lower-right corner. All other stable matchings are confined to the rectangular region spanned by these two extremes.




Moving toward $\mu_S$ makes every student weakly better off and every college weakly worse off. Moving toward $\mu_C$ does the opposite. Any stable matching must lie weakly within the bounds set by these two extremes.

### The Rural Hospital Theorem

When we allow preferences to include truncations—i.e., agents may find some partners *unacceptable*, so that not every agent is necessarily matched—the following striking result holds.

**Theorem 9.35: Rural Hospital Theorem (Lone Wolf Theorem)**

If a student (or college) is unmatched in **some** stable matching, then that student (or college) is unmatched in **all** stable matchings.

***Proof for Theorem***

We prove the result for the case of one-to-one matching ($q_c = 1$ for all $c$).

Let $S_u \subseteq \mathcal{S}$ be the set of students unmatched in $\mu_S$ (the student-optimal stable matching), and let $C_u \subseteq \mathcal{C}$ be the set of colleges unmatched in $\mu_S$. Similarly, let $S'_u$ and $C'_u$ be the sets of students and colleges unmatched in $\mu_C$ (the college-optimal stable matching).

*Step 1:* $S_u \subseteq S'_u$. Since $\mu_S$ is the student-optimal stable matching, every student gets their best possible partner across all stable matchings. If a student $s$ is unmatched even in $\mu_S$, then $s$ cannot be matched in any stable matching. In particular, $s$ is unmatched in $\mu_C$. Hence $S_u \subseteq S'_u$.

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*Step 2:* $C'_u \subseteq C_u$. By symmetric reasoning, $\mu_C$ is the college-optimal stable matching, so every college gets its best possible partner. If a college $c$ is unmatched in $\mu_C$, then $c$ is unmatched in every stable matching, including $\mu_S$. Hence $C'_u \subseteq C_u$.

*Step 3: Counting argument.* In any one-to-one matching, the number of matched students equals the number of matched colleges. Therefore:

$$|\mathcal{S}| - |S_u| = |\mathcal{C}| - |C_u| \qquad \text{and} \qquad |\mathcal{S}| - |S'_u| = |\mathcal{C}| - |C'_u|.$$

From Step 1, $|S_u| \leq |S'_u|$, so the number of matched students in $\mu_S$ is at least the number in $\mu_C$. From Step 2, $|C'_u| \leq |C_u|$, so the number of matched colleges in $\mu_C$ is at least the number in $\mu_S$. But matched students = matched colleges in each matching, so:

$$\underbrace{|\mathcal{S}| - |S_u|}_{\text{matched in } \mu_S} \geq \underbrace{|\mathcal{S}| - |S'_u|}_{\text{matched in } \mu_C} = \underbrace{|\mathcal{C}| - |C'_u|}_{\text{matched in } \mu_C} \geq \underbrace{|\mathcal{C}| - |C_u|}_{\text{matched in } \mu_S}.$$

The leftmost and rightmost terms are equal (both count matched pairs in $\mu_S$), so all inequalities are equalities. This forces $|S_u| = |S'_u|$ and $|C_u| = |C'_u|$. Combined with $S_u \subseteq S'_u$ and $C'_u \subseteq C_u$, we conclude:

$$S_u = S'_u \qquad \text{and} \qquad C_u = C'_u.$$

Since every stable matching $\mu$ satisfies $S_u \subseteq S^\mu_u \subseteq S'_u$ (the first inclusion by student-optimality of $\mu_S$, the second by college-optimality of $\mu_C$), and $S_u = S'_u$, we have $S^\mu_u = S_u$ for all stable matchings $\mu$.

**Remark.**

The name "Rural Hospital" comes from the practical observation in medical residency matching: hospitals in rural areas that fail to fill all their positions under one stable matching will fail to fill them under *every* stable matching. Switching from the hospital-proposing to the resident-proposing DA (or any other stable mechanism) cannot help a rural hospital that is fundamentally undesirable to residents. The set of matched agents is invariant across stable matchings—only the *partners* can change, not whether one is matched at all.

## 9.4 The Boston Mechanism (Immediate Acceptance)

To truly appreciate the elegance and power of the Deferred Acceptance (DA) algorithm, it is instructive to study a mechanism that attempts to solve the same problem but fails spectacularly in its theoretical properties. The most famous example is the **Boston Mechanism**, historically used by the Boston Public School system (and many other school districts worldwide) before economists pointed out its severe flaws.

The Boston Mechanism is also known as the **Immediate Acceptance** mechanism. The critical difference from DA lies in the word "immediate": once a college accepts a student, that decision is permanent. Seats are locked in, and future applicants cannot displace already-admitted students, regardless of how highly the college ranks the new applicants.

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## Definition 9.36: The Boston Mechanism

1. **Round 1:** Each student proposes to their first-choice college. Each college looks at its proposals, accepts its most preferred students up to its quota, and rejects the rest. **These acceptances are final.** The college's quota is reduced accordingly.

2. **Round $k$ ($k \geq 2$):** Each student rejected in Round $k-1$ proposes to their $k$-th choice college. Each college looks at its *remaining* unfilled seats (if any). It accepts its most preferred students among the new applicants up to its remaining quota, and rejects the rest. **These acceptances are final.**

3. **Repeat** until all students are matched or all available seats are filled.

**Example (Boston Mechanism on the 10x10 Matching Problem).**

Let us apply the Boston Mechanism to the exact same 10-student, 10-college problem. All colleges have quota $q = 1$.

**College preferences (Boston Mechanism final matches highlighted):**

| Rank | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $s_1$ | $s_1$ | $s_{10}$ | $s_6$ | $s_7$ | $s_4$ | $s_1$ | $s_6$ | $s_8$ | $s_6$ |
| 2 | $s_2$ | $s_6$ | $s_2$ | $s_1$ | $s_2$ | $s_8$ | $s_4$ | $s_4$ | $s_{10}$ | $s_5$ |
| 3 | $s_3$ | $s_9$ | $s_4$ | $s_5$ | $s_8$ | $s_2$ | $s_3$ | $s_2$ | $s_6$ | $\boldsymbol{s_1}$ |
| 4 | $s_4$ | $s_2$ | $\boldsymbol{s_6}$ | $\boldsymbol{s_2}$ | $s_4$ | $s_6$ | $s_8$ | $s_{10}$ | $\boldsymbol{s_9}$ | $s_4$ |
| 5 | $s_5$ | $s_5$ | $s_8$ | $s_7$ | $s_{10}$ | $s_{10}$ | $\boldsymbol{s_5}$ | $s_8$ | $s_5$ | $s_{10}$ |
| 6 | $s_6$ | $s_8$ | $s_3$ | $s_3$ | $s_5$ | $s_3$ | $s_2$ | $s_3$ | $s_7$ | $s_8$ |
| 7 | $s_7$ | $s_3$ | $s_5$ | $s_8$ | $s_9$ | $s_7$ | $s_9$ | $s_1$ | $s_4$ | $s_9$ |
| 8 | $\boldsymbol{s_8}$ | $s_4$ | $s_7$ | $s_4$ | $s_6$ | $s_1$ | $s_6$ | $s_9$ | $s_3$ | $s_2$ |
| 9 | $s_9$ | $s_7$ | $s_9$ | $s_{10}$ | $\boldsymbol{s_3}$ | $s_5$ | $s_7$ | $\boldsymbol{s_7}$ | $s_1$ | $s_3$ |
| 10 | $s_{10}$ | $\boldsymbol{s_{10}}$ | $s_1$ | $s_9$ | $s_1$ | $s_9$ | $s_{10}$ | $s_5$ | $s_2$ | $s_7$ |

**Student preferences (Boston Mechanism final matches highlighted):**

| Rank | $s_1$ | $s_2$ | $s_3$ | $s_4$ | $s_5$ | $s_6$ | $s_7$ | $s_8$ | $s_9$ | $s_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $c_5$ | $\boldsymbol{c_4}$ | $\boldsymbol{c_5}$ | $\boldsymbol{c_6}$ | $c_3$ | $\boldsymbol{c_3}$ | $\boldsymbol{c_8}$ | $\boldsymbol{c_1}$ | $c_1$ | $\boldsymbol{c_2}$ |
| 2 | $\boldsymbol{c_{10}}$ | $c_9$ | $c_6$ | $c_2$ | $\boldsymbol{c_7}$ | $c_4$ | $c_6$ | $c_4$ | $c_5$ | $c_{10}$ |
| 3 | $c_6$ | $c_{10}$ | $c_1$ | $c_5$ | $c_9$ | $c_5$ | $c_4$ | $c_2$ | $c_2$ | $c_9$ |
| 4 | $c_4$ | $c_6$ | $c_7$ | $c_8$ | $c_{10}$ | $c_1$ | $c_2$ | $c_6$ | $c_{10}$ | $c_4$ |
| 5 | $c_7$ | $c_3$ | $c_4$ | $c_1$ | $c_1$ | $c_9$ | $c_9$ | $c_3$ | $c_3$ | $c_1$ |
| 6 | $c_2$ | $c_7$ | $c_8$ | $c_7$ | $c_5$ | $c_7$ | $c_7$ | $c_7$ | $c_8$ | $c_7$ |
| 7 | $c_8$ | $c_1$ | $c_9$ | $c_3$ | $c_4$ | $c_{10}$ | $c_5$ | $c_5$ | $c_6$ | $c_5$ |
| 8 | $c_9$ | $c_5$ | $c_{10}$ | $c_9$ | $c_2$ | $c_8$ | $c_{10}$ | $c_{10}$ | $c_7$ | $c_6$ |
| 9 | $c_1$ | $c_2$ | $c_2$ | $c_{10}$ | $c_6$ | $c_2$ | $c_1$ | $c_8$ | $c_4$ | $c_3$ |
| 10 | $c_3$ | $c_8$ | $c_3$ | $c_4$ | $c_8$ | $c_6$ | $c_3$ | $c_9$ | $\boldsymbol{c_9}$ | $c_8$ |

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We use the same chalkboard-style simulation. However, green text now represents a **permanent acceptance** (**admitted**), rather than a tentative hold. Once a column is filled with a green student, that college is closed forever.

**Round 1.**
All 10 students propose to their first-choice colleges.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $\mathbf{s_8}$ | $\mathbf{s_{10}}$ | $\mathbf{s_6}$ | $\mathbf{s_2}$ | $\mathbf{s_3}$ | $\mathbf{s_4}$ | | $\mathbf{s_7}$ | | |
| $\cancel{s_9}$ | | $\cancel{s_5}$ | | $\cancel{s_1}$ | | | | | |

*Explanation:* Colleges $c_2, c_4, c_6, c_8$ receive exactly one applicant and admit them immediately. Colleges $c_1, c_3, c_5$ face competition. They admit their preferred applicant ($s_8, s_6, s_3$ respectively) and reject the rest.
*Status:* Colleges $c_1, c_2, c_3, c_4, c_5, c_6, c_8$ are now **permanently closed**.
*Rejected:* $s_9$ (next $c_5$), $s_5$ (next $c_7$), $s_1$ (next $c_{10}$).

**Round 2.**
The three rejected students propose to their second-choice colleges.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| | | | | **FULL** | | $\mathbf{s_5}$ | | | $\mathbf{s_1}$ |
| | | | | $\cancel{s_9}$ | | | | | |

*Explanation:* Students $s_1$ and $s_5$ apply to $c_{10}$ and $c_7$, which are still empty. They are admitted permanently. Student $s_9$ applies to $c_5$. However, $c_5$ was already filled by $s_3$ in Round 1! Even though $c_5$ prefers $s_9$ (#7) over $s_3$ (#9), it is too late. The seat is gone. $s_9$ is automatically rejected.
*Status:* Colleges $c_7, c_{10}$ are now **permanently closed**. Only $c_9$ remains open.
*Rejected:* $s_9$ (next $c_2$).

**Rounds 3 through 9 (The Free-Fall of $s_9$).**
Student $s_9$ has suffered a "cascading failure" ( ). Because he missed his first choice, he is now applying to colleges that were filled in Round 1.

- **Round 3:** $s_9 \to c_2$. **FULL** (filled by $s_{10}$). Rejected.
- **Round 4:** $s_9 \to c_{10}$. **FULL** (filled by $s_1$). Rejected.
- **Round 5:** $s_9 \to c_3$. **FULL** (filled by $s_6$). Rejected.
- **Round 6:** $s_9 \to c_8$. **FULL** (filled by $s_7$). Rejected.
- **Round 7:** $s_9 \to c_6$. **FULL** (filled by $s_4$). Rejected.
- **Round 8:** $s_9 \to c_7$. **FULL** (filled by $s_5$). Rejected.
- **Round 9:** $s_9 \to c_4$. **FULL** (filled by $s_2$). Rejected.

**Round 10.**
Student $s_9$ finally reaches his 10th and absolute worst choice, $c_9$. Luckily, $c_9$ is still empty.

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| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | $\mathbf{s_9}$ | |

*The algorithm terminates.*

The **Boston Mechanism matching** $\mu_B$ is:

| College | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Student | $s_8$ | $s_{10}$ | $s_6$ | $s_2$ | $s_3$ | $s_4$ | $s_5$ | $s_7$ | $s_9$ | $s_1$ |

### 9.4.1 Properties of the Boston Mechanism

The simulation above provides everything we need to expose the theoretical failures of the Boston Mechanism. We examine the two most important properties: Pairwise Stability and Strategy-Proofness.

**Theorem 9.37: The Boston Mechanism is NOT Stable**

The allocation produced by the Boston Mechanism may contain blocking pairs, meaning it fails to guarantee a pairwise stable matching.

***Proof for Theorem***

We can prove this by simply identifying a blocking pair in the outcome $\mu_B$ of our example.

Under $\mu_B$, student $s_9$ is matched with $c_9$ (his 10th choice), and college $c_5$ is matched with $s_3$ (its 9th choice). Now look at their true preferences:

- Student $s_9$ ranks $c_5$ as his 2nd choice, meaning $c_5 \succ_{s_9} \mu_B(s_9) = c_9$.
- College $c_5$ ranks $s_9$ as its 7th choice, meaning $s_9 \succ_{c_5} \mu_B(c_5) = s_3$.

Both $s_9$ and $c_5$ would strictly prefer to be matched with each other rather than their current assignments. Thus, $(c_5, s_9)$ **forms a blocking pair**.

The instability arises precisely because of the "immediate acceptance" rule: $c_5$ was forced to permanently commit to $s_3$ in Round 1, preventing it from accepting the much stronger applicant $s_9$ who arrived in Round 2.

**Theorem 9.38: The Boston Mechanism is NOT Strategy-Proof**

Under the Boston Mechanism, truthful reporting is not a dominant strategy. Students can often obtain a strictly better outcome by misreporting their preferences.

***Proof for Theorem***

*Because missing out on your first choice is heavily penalized (the "free-fall" effect seen with $s_9$), the Boston Mechanism forces students to play a high-stakes psychological game. Instead of ranking their true favorite first, students are incentivized to rank a "safe" school first to secure a seat in Round 1.*

We demonstrate this with student $s_9$ from our example. Under truthful reporting,

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$s_9$ listed $c_1$ as his 1st choice and $c_5$ as his 2nd choice. Because he lost the tie-breaker at $c_1$ in Round 1, he free-fell all the way to $c_9$, his worst possible outcome.

Suppose $s_9$ anticipates this and decides to deviate. He submits a fake preference list $P'_{s_9}$ where he promotes his 2nd choice ($c_5$) to his 1st choice, abandoning the highly competitive $c_1$. The rest of the students report truthfully.

**Modified Student Preferences (with $s_9$'s misreport highlighted in blue):**

| Rank | $s_1$ | $s_2$ | $s_3$ | $s_4$ | $s_5$ | $s_6$ | $s_7$ | $s_8$ | $\mathbf{s'_9}$ | $s_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | $c_5$ | $c_4$ | $c_5$ | $c_6$ | $c_3$ | $c_3$ | $c_8$ | $c_1$ | $\mathbf{c_5}$ | $c_2$ |
| 2 | $c_{10}$ | $c_9$ | $c_6$ | $c_2$ | $c_7$ | $c_4$ | $c_6$ | $c_4$ | $\mathbf{c_1}$ | $c_{10}$ |
| 3 | $c_6$ | $c_{10}$ | $c_1$ | $c_5$ | $c_9$ | $c_5$ | $c_4$ | $c_2$ | $c_2$ | $c_9$ |
| 4 | $c_4$ | $c_6$ | $c_7$ | $c_8$ | $c_{10}$ | $c_1$ | $c_2$ | $c_6$ | $c_{10}$ | $c_4$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |

Let us trace **Round 1** of the Boston Mechanism under this new profile. All students apply to their reported 1st choices.

| $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | $c_6$ | $c_7$ | $c_8$ | $c_9$ | $c_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| $\mathbf{s_8}$ | $\mathbf{s_{10}}$ | $\mathbf{s_6}$ | $\mathbf{s_2}$ | $\mathbf{s_9}$ | $\mathbf{s_4}$ | | $\mathbf{s_7}$ | | |
| | | $\cancel{s_5}$ | | $\cancel{s_3}$ | | | | | |
| | | | | $\cancel{s_1}$ | | | | | |

*Explanation of $c_5$'s decision:* College $c_5$ now receives Round 1 proposals from $s_1$, $s_3$, and the strategically misreporting $s_9$. Looking at $c_5$'s true preference list: $s_9$ (#7) $\succ_{c_5}$ $s_3$ (#9) $\succ_{c_5}$ $s_1$ (#10). Because $c_5$ evaluates all three applicants simultaneously in Round 1, it permanently admits its favorite among them: $s_9$.

By lying and placing $c_5$ first, student $s_9$ completely avoided the free-fall and secured his true 2nd choice ($c_5$). This outcome is strictly better for him than the 10th choice ($c_9$) he received when telling the truth. Since a profitable deviation exists, the Boston mechanism is **not strategy-proof**.

**Remark (The Tragedy of the Boston Mechanism).**

The Boston Mechanism systematically rewards sophisticated, strategic families while punishing honest, naive ones. A naive student who truthfully applies to a competitive "reach" school wastes their critical Round 1 proposal. By the time they are rejected and move to their "match" schools in Round 2, those schools have already been filled by strategic students who ranked them first. This profound inequity is why many school districts have replaced it with the Strategy-Proof Deferred Acceptance algorithm.

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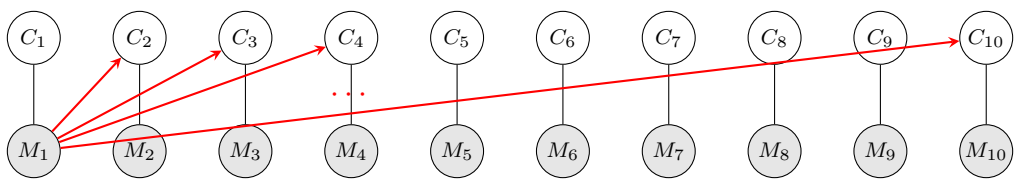
# Chapter 10

# Common Knowledge

## 10.1 Motivating Example: The Corrupted Industry

Consider an industry with 10 firms $C_1, C_2, \ldots, C_{10}$. Each firm $C_i$ has a manager $M_i$. The industry is thoroughly corrupted: every manager $M_i$ is secretly selling their own company's private information to all other companies. In addition, each company is buying private information from the managers of other companies.




Every $M_i$ sells to all $C_j$ ($j \neq i$); only $M_1$'s arrows shown.

There is one corporate governance rule: if a company discovers that its own manager is selling information to outsiders, it will fire the manager on the **Friday of that week**.

**What does each firm know?** Consider firm $C_1$. Since $C_1$ is buying information from the managers $M_2, M_3, \ldots, M_{10}$, it directly observes that those 9 managers are corrupt. However, $C_1$ **cannot observe whether its own manager $M_1$ is corrupt**—it does not know whether $M_1$ is selling to others. The same logic applies symmetrically to every firm. In summary:

- Each firm knows that *at least 9* managers are corrupt (the ones it buys from).
- No firm knows for certain whether its *own* manager is corrupt.

Now suppose that at the start of Week 1, a regulator makes a **public announcement**: *"At least one manager in this industry is corrupt."* This statement is heard by all 10 firms simultaneously, and everyone knows that everyone heard it.

**What happens?** Nothing happens in Week 1. Nothing happens in Week 2. In fact, nothing happens until **Week 10**, at which point all 10 managers are fired simultaneously.

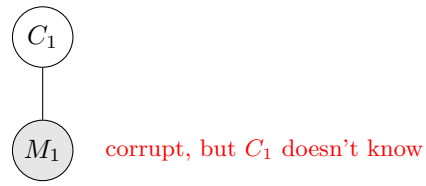
*At first glance, this is deeply puzzling. Every firm already knew that at least 9 managers were corrupt—far more than the "at least one" in the announcement. How can such a*

*seemingly vacuous statement trigger any action at all, let alone a cascade that takes 10 weeks to unfold?*

### 10.1.1 Iterated Reasoning

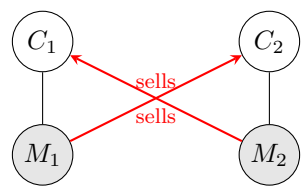
The answer lies in the chain of *iterated hypothetical reasoning* that the public announcement enables.

**The $n = 1$ Case.** Suppose there is only 1 firm with 1 corrupt manager.




Before the announcement, $C_1$ does not know its own manager is corrupt. After the announcement ("at least one is corrupt"), $C_1$ immediately deduces that $M_1$ must be the corrupt one, and fires him on Friday of **Week 1**.

**The $n = 2$ Case.** Suppose there are 2 firms, both with corrupt managers. Each firm observes that the other firm's manager is corrupt (by buying information from him), but does not know about its own.




Consider the reasoning from $C_1$'s perspective after the announcement. $C_1$ observes that $M_2$ is corrupt, but entertains two hypotheses about its own manager:

- **Hypothesis A:** $M_1$ is not corrupt. Then only $M_2$ is corrupt. Under this hypothesis, after the public announcement, $C_2$ would find itself in the $n = 1$ situation: $C_2$ would see no other corrupt manager, learn from the announcement that at least one exists, and deduce that $M_2$ is the one. So $C_2$ would fire $M_2$ in **Week 1**.
- **Hypothesis B:** $M_1$ is also corrupt.

$C_1$ waits and watches. At the end of Week 1, $M_2$ is *not* fired. This rules out Hypothesis A: if $M_1$ had been clean, $C_2$ would have acted already. Therefore $C_1$ concludes that $M_1$ must also be corrupt, and fires $M_1$ in **Week 2**.

By perfectly symmetric reasoning, $C_2$ also watches whether $C_1$ fires $M_1$ in Week 1. When it does not happen, $C_2$ likewise infers that $M_2$ is corrupt. Both managers are fired simultaneously in **Week 2**.

**The General Case: Induction.** By induction on the number of corrupt managers $n$. Each firm sees $n - 1$ corrupt managers and entertains the hypothesis that its own manager is clean (so that only $n - 1$ managers are corrupt). Under that hypothesis, by the inductive assumption, the other $n - 1$ firms would have fired their managers by Week $n - 1$. When

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this does not happen by the end of Week $n-1$, every firm deduces that the total must be $n$, meaning its own manager is also corrupt. All $n$ managers are fired simultaneously in **Week** $n$.

For our example with $n = 10$: nothing happens for 9 weeks, and then all 10 managers are fired in Week 10.

### 10.1.2 Why Does a "Vacuous" Announcement Matter?

The content of the announcement ("at least one is corrupt") was already known to every firm—each firm could see at least 9 corrupt managers. What was *not* in place before the announcement was the full tower of *hypothetical reasoning about hypothetical reasoning*.

To see why, trace the chain from $C_1$'s perspective. $C_1$ reasons: "I see 9 corrupt managers. But suppose $M_1$ is clean. Then $C_2$ would see only 8 corrupt managers. And $C_2$ might suppose $M_2$ is also clean. Then $C_3$ would see only 7…" Each level of nesting reduces the number of known-corrupt managers by one. After 9 levels of hypothetical reasoning, we reach a scenario where some firm sees *zero* other corrupt managers and does not know if any exist at all. Without the announcement, the induction has no base case in this deepest hypothetical—the chain of reasoning collapses before reaching its logical conclusion.

The public announcement plugs exactly this gap. By making "at least one is corrupt" publicly and simultaneously known to all, it provides the base case in every hypothetical scenario, no matter how deeply nested. This distinction—between knowledge that is shared but *finitely* nested, and knowledge that is shared *infinitely* deep—is precisely the distinction between *mutual knowledge* and *common knowledge*, which we now formalize.

## 10.2 Formal Definition

**Definition 10.1: Mutual Knowledge and Common Knowledge**

Let $E$ be an event and let there be $n$ agents.

- $E$ is **mutual knowledge of order 1** if every agent knows $E$.
- $E$ is **mutual knowledge of order** $k$ if every agent knows that $E$ is mutual knowledge of order $k-1$.
- $E$ is **common knowledge** if $E$ is mutual knowledge of order $k$ for *every* $k \in \mathbb{N}$.

In other words, common knowledge means: everyone knows $E$, everyone knows that everyone knows $E$, everyone knows that everyone knows that everyone knows $E$, and so on *ad infinitum*.

In the corrupted industry example, the event $E =$ "at least one manager is corrupt" was, before the announcement, mutual knowledge of high order but **not** common knowledge. Each firm's chain of hypotheticals ("suppose mine is clean, then you see one fewer, then suppose yours is clean too…") bottoms out after finitely many levels at a scenario in which one firm sees zero corrupt managers and cannot determine whether $E$ holds. Common knowledge requires that the chain never bottoms out—that at every depth of hypothetical

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reasoning, every agent in the hypothetical scenario still knows $E$.

The public announcement achieves exactly this. Because every firm heard the announcement, and every firm knows every other firm heard it, and so on without bound, $E$ becomes common knowledge. The base case of the induction ("a firm that sees zero corrupt managers and hears the announcement will fire its manager in Week 1") is now available at every depth, enabling the full chain of iterated reasoning to unfold.

## 10.3 The Coordinated Attack Problem

We now present a game-theoretic example that dramatically illustrates the difference between "almost common knowledge" and true common knowledge. This is based on Rubinstein's (1989) *Electronic Mail Game*.

### 10.3.1 Setup

Two divisions of an army (Player 1 and Player 2) must decide whether to **attack** ($A$) or **not attack** ($N$). The enemy is either **weak** (state $G$, the "good" state) or **strong** (state $B$, the "bad" state). The army wins if and only if *both* divisions attack *and* the enemy is weak. The prior probabilities are $\Pr(G) = \pi < \frac{1}{2}$ and $\Pr(B) = 1 - \pi$. **[TODO: learn how the command "Pr" is defined in the file commands.tex. and revise those for all the remaining Pr's. Also, if you find my current redefinition of Pr is not that appropriate, tell me and advise me how to revise it.]** The payoff matrices are:

**State** $G$ (prob $\pi$)

| | $A$ | $N$ |
|---|---|---|
| $A$ | $2,\ 2$ | $-3,\ 0$ |
| $N$ | $0,\ -3$ | $0,\ 0$ |

**State** $B$ (prob $1-\pi$)

| | $A$ | $N$ |
|---|---|---|
| $A$ | $-2,\ -2$ | $-2,\ 0$ |
| $N$ | $0,\ -2$ | $0,\ 0$ |

In state $B$, not attacking ($N$) is strictly dominant for both players. In state $G$, the game is a *coordination game*: $(A, A)$ is a Nash equilibrium with payoff $(2, 2)$, but attacking alone is severely punished ($-3$). Crucially, if the state $G$ were **common knowledge**, both $(A, A)$ and $(N, N)$ would be Nash equilibria—the game is a coordination game with two pure-strategy equilibria. The interesting question is whether the players can coordinate on the Pareto-superior equilibrium $(A, A)$.

### 10.3.2 Information Structure

The communication protocol is as follows:

1. If the state is $G$, Nature sends a signal to Player 1 with certainty (no error at this stage). If the state is $B$, no signal is sent.

2. Upon receiving a signal, Player 1's system *automatically* forwards a message to Player 2. This transmission fails (the message is lost) with probability $\varepsilon > 0$, independently.

3. If Player 2 receives the message, Player 2's system automatically sends a confirmation back to Player 1, again lost with probability $\varepsilon$.

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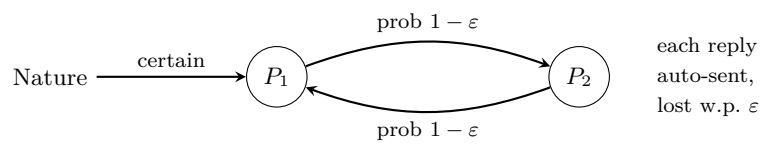
4. This back-and-forth continues indefinitely: each successfully received message triggers an automatic reply, each of which is independently lost with probability $\varepsilon$.

The message sending is *not* a strategic choice—it is automatic. The only strategic decision is whether to attack or not.

Let $t_i$ denote the **number of messages received** by Player $i$. In state $B$, no messages are sent, so $t_1 = t_2 = 0$. In state $G$, the message chain proceeds as:




The chain stops at the first lost message. The possible outcomes given state $G$ are:

| Chain stops at | $t_1$ | $t_2$ |
|---|---|---|
| $P_1 \to P_2$ lost | 1 | 0 |
| $P_2 \to P_1$ lost | 1 | 1 |
| $P_1 \to P_2$ lost | 2 | 1 |
| $P_2 \to P_1$ lost | 2 | 2 |
| $\vdots$ | $\vdots$ | $\vdots$ |

Observe that $t_1$ and $t_2$ always differ by at most 1: given $t_1 = T \geq 1$, the value of $t_2$ is either $T - 1$ or $T$.

**Remark ("Almost" Common Knowledge).**

When $\varepsilon$ is small, the message chain is very likely to be long. With high probability, both players receive many confirmations, so each player is "almost sure" the state is $G$, "almost sure" the other knows it, "almost sure" the other knows the other knows it, and so on. The state $G$ is "almost" common knowledge. One might expect that $(A, A)$ should be sustainable as an equilibrium. As we will see, this intuition is spectacularly wrong.

### 10.3.3 The Unique Equilibrium

A strategy for Player $i$ is a function $\sigma_i : \{0, 1, 2, \ldots\} \to \{A, N\}$ mapping the number of messages received to an action.

**Proposition 10.2: Unique Equilibrium of the Coordinated Attack Game**

There is a unique equilibrium: both players play $N$ regardless of the number of messages received. That is, $\sigma_i(t_i) = N$ for all $t_i \geq 0$ and $i \in \{1, 2\}$.

***Proof for Proposition.***

By induction on the number of messages received.

**Base case: $t_1 = 0$.** If Player 1 receives no messages, then state $B$ is certain (since Nature always signals Player 1 in state $G$). In state $B$, $N$ is strictly dominant. So

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$\sigma_1(0) = N$.

**Base case:** $t_2 = 0$**.** If Player 2 receives no messages, there are two possibilities:

(i) State is $B$ and no signal was ever sent (prior probability $1 - \pi$);

(ii) State is $G$, Player 1 received the signal, but the forward to Player 2 was lost (prior probability $\pi\varepsilon$).

By Bayes' rule:

$$\Pr(G \mid t_2 = 0) = \frac{\pi\varepsilon}{\pi\varepsilon + (1-\pi)}.$$

If Player 2 plays $N$, the payoff is 0. If Player 2 plays $A$, the payoff is at most (assuming Player 1 plays $A$ whenever state is $G$):

$$\Pr(G \mid t_2 = 0) \times 2 + \Pr(B \mid t_2 = 0) \times (-2) = \frac{2\pi\varepsilon - 2(1-\pi)}{\pi\varepsilon + (1-\pi)} < 0,$$

since $\pi\varepsilon \ll 1 - \pi$ (recall $\pi < \frac{1}{2}$ and $\varepsilon$ is small). So $\sigma_2(0) = N$.

**Inductive step.** Suppose that for all $t_1, t_2 < T$, both players play $N$: $\sigma_1(t_1) = N$ and $\sigma_2(t_2) = N$.

Consider Player 1 with $t_1 = T$ (the argument for Player 2 is symmetric). Given $t_1 = T$, Player 2's message count is either $t_2 = T - 1$ or $t_2 = T$. The conditional probability is:

$$p := \Pr(t_2 = T - 1 \mid t_1 = T) = \frac{\varepsilon}{\varepsilon + (1-\varepsilon)\varepsilon} = \frac{1}{2-\varepsilon} > \frac{1}{2}.$$

This is because, given $t_1 = T$: with probability $\varepsilon$ the next message ($P_1 \to P_2$) is lost, yielding $t_2 = T - 1$; with probability $(1-\varepsilon)\varepsilon$ the message reaches $P_2$ (so $t_2 \geq T$) but the return is lost, yielding $t_2 = T$.

By the induction hypothesis, $\sigma_2(T-1) = N$. Even in the best case where Player 2 plays $A$ when $t_2 = T$, if Player 1 plays $A$:

$$\text{Payoff to } P_1 \leq p \times (-3) + (1-p) \times 2 = \frac{-3 + 2(1-\varepsilon)}{2-\varepsilon} = \frac{-1-2\varepsilon}{2-\varepsilon} < 0.$$

The $-3$ arises because when $t_2 = T - 1$, Player 2 plays $N$ (by induction), so Player 1 attacks alone. The 2 is the best-case payoff if Player 2 also attacks. Since $p > \frac{1}{2}$, the expected payoff from attacking is strictly negative. Therefore $\sigma_1(T) = N$.

By symmetric reasoning applied to Player 2, $\sigma_2(T) = N$. This completes the induction.

### 10.3.4 Interpretation

The result is striking. The following two scenarios yield completely different equilibrium outcomes:

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| **Information structure** | **Equilibrium outcome** |
|---|---|
| $G$ is **common knowledge** | $(A, A)$ is a NE |
| $G$ is "almost" common knowledge | $(N, N)$ is the *unique* NE |

No matter how small $\varepsilon > 0$ is, and no matter how many thousands of confirmations each player has received, the *unique* equilibrium is for both to play $N$. The problem is not that the players lack confidence that $G$ is true—after many messages, they are nearly certain. The problem is that each player is never quite sure the *other* player is sure that the other player is sure… that the state is $G$.

**Remark (Common $p$-Belief).**

The key quantitative insight is the conditional probability $p = \frac{1}{2-\varepsilon}$. Even when $\varepsilon$ is tiny, $p$ is close to $\frac{1}{2}$—far from 1. This means that no matter how many messages Player $i$ has received, Player $i$ always assigns probability at least $\frac{1}{2}$ to the event that the other player received one fewer message. The information structure "self-replicates" at every level: the uncertainty never shrinks as we go deeper in the epistemic hierarchy.

This connects to the concept of *common $p$-belief* (Monderer and Samet, 1989): an event is common $p$-belief if everyone believes it with probability at least $p$, everyone believes with probability at least $p$ that everyone believes it with probability at least $p$, and so on. In the electronic mail game, the state $G$ is common $p$-belief only for $p \leq \frac{1}{2-\varepsilon} \approx \frac{1}{2}$, which is far from the $p = 1$ required for common knowledge. Coordination requires common knowledge (or at least common $p$-belief with $p$ sufficiently large); "almost common knowledge" is not enough.

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# Chapter 11

# Dynamic Games

## 11.1 Repeated Games

### 11.1.1 Setup

Consider a **stage game** $G = (S_i, u_i)_{i=1}^n$, where $S_i$ is the action set of player $i$ and $u_i : S \to \mathbb{R}$ is the payoff function, with $S = \prod_{i=1}^n S_i$. The **infinitely repeated game** $G^\delta(\infty)$ consists of playing $G$ in every period $t = 1, 2, \ldots$, with players discounting future payoffs by a common discount factor $\delta < 1$. We assume **perfect monitoring**: after each period, all players observe the realized action profile.

**Definition 11.1: Strategy in $G^\delta(\infty)$**

A **strategy** for player $i$ in the infinitely repeated game is a sequence $\sigma_i = (\sigma_i^1, \sigma_i^2, \ldots)$, where:

- $\sigma_i^1 \in S_i$ specifies the action in period 1;
- For $t > 1$, $\sigma_i^t : S^{t-1} \to S_i$ maps the history of play $h^{t-1} = (s^1, s^2, \ldots, s^{t-1})$ to an action in period $t$.

**[TODO: to confirm: so basically we are assuming that we only consider pure strategies?]**

Given a strategy profile $\sigma = (\sigma_1, \ldots, \sigma_n)$, the induced action profile in period $t$ is $s^t(\sigma)$. Player $i$'s **average discounted payoff** in $G^\delta(\infty)$ is:

$$U_i(\sigma) = (1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} \, u_i\big(s^t(\sigma)\big) \, .$$

The $(1 - \delta)$ normalization ensures that perpetual play of a constant action profile $s$ yields payoff $u_i(s)$, making payoffs in the repeated game directly comparable to stage-game payoffs.

### 11.1.2 Subgame Perfect Equilibrium

**Definition 11.2: Subgame Perfect Equilibrium (SPE) of $G^\delta(\infty)$**

A strategy profile $\sigma^* = (\sigma_1^*, \sigma_2^*, \ldots, \sigma_n^*)$ is a **subgame perfect equilibrium** if:

(i) $\sigma^*$ is a Nash equilibrium of $G^\delta(\infty)$: for all $i$ and all alternative strategies $\sigma_i$,

$$(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}\,u_i\big(s^t(\sigma^*)\big) \geq (1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}\,u_i\big(s^t(\sigma_i,\sigma_{-i}^*)\big)\,.$$

**[TODO: i think it might be more straightforward if we could write the discounted term as $\sum_{t=0}^{\infty}\delta^t$, do you agree?]**

(ii) For every period $t$ and every history $h^{t-1} = (s^1, s^2, \ldots, s^{t-1})$, the **continuation strategy** $\sigma^*|_{h^{t-1}}$ is also a Nash equilibrium of $G^\delta(\infty)$.**[TODO: is there any strict definition of continuation strategy? better if you illustrate it with a graph or diagram?]**

Condition (ii) is what distinguishes SPE from NE: not only must the strategy be optimal on the equilibrium path, but it must also be optimal after *every* possible history, including histories that arise from deviations. **[TODO: so i think the intuition is that, every subgame is essentially the same as the original game, execpt that some history has been generated. so the strategy profile must be a NE in every subgame, which is what condition (ii) requires. is that right?]**

**Example (Cooperative Outcomes in the Prisoner's Dilemma).**

Consider the following stage game (Prisoner's Dilemma):

| | $C$ | $D$ |
|---|---|---|
| $C$ | $2,\ 2$ | $-1,\ 3$ |
| $D$ | $3,\ -1$ | $0,\ 0$ |

In the stage game, $D$ is strictly dominant for both players, so the unique Nash equilibrium is $(D, D)$ with payoff $(0, 0)$. The cooperative outcome $(C, C)$ with payoff $(2, 2)$ is Pareto superior but not a Nash equilibrium.

**Can cooperation be sustained in $G^\delta(\infty)$?** Consider the following *grim trigger* strategy for each player $i$:

$$\sigma_i^1 = C, \qquad \sigma_i^t = \begin{cases} C & \text{if } s^\tau = (C,C) \text{ for all } \tau < t, \\ D & \text{otherwise.} \end{cases}$$

Under this strategy, both players cooperate in every period as long as no deviation has ever occurred. If any player deviates, both players switch to $D$ forever (the "punishment").

**On-path payoff.** If both players follow $\sigma^*$, the outcome is $(C, C)$ in every period. The

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average discounted payoff is:

$$(1-\delta)\sum_{t=1}^{\infty}\delta^{t-1}\cdot 2 = 2(1-\delta)\cdot\frac{1}{1-\delta} = 2.$$

**Deviation payoff.** Suppose player $i$ deviates to $D$ in period 1. They get 3 in period 1 (playing $D$ against $C$), but trigger permanent punishment: $(D, D)$ with payoff 0 in every subsequent period. The average discounted payoff from deviating is:

$$(1-\delta)\left[3+\sum_{t=2}^{\infty}\delta^{t-1}\cdot 0\right] = 3(1-\delta).$$

**No-deviation condition.** Cooperation is sustained if the on-path payoff exceeds the deviation payoff:

$$2 \geq 3(1-\delta) \quad \Longleftrightarrow \quad \delta \geq \frac{1}{3}.$$

So for $\delta \geq \frac{1}{3}$, the grim trigger strategy profile constitutes an SPE that sustains the cooperative outcome $(C, C)$ in every period. When players are sufficiently patient, the threat of permanent punishment is credible and severe enough to deter short-run deviations.

### 11.1.3 The One-Deviation Principle

Verifying that a strategy profile is an SPE requires checking that no player can profit from deviation following *any* history. In an infinitely repeated game, a deviation could in principle involve changing actions in infinitely many periods. The following result greatly simplifies the verification.

**Theorem 11.3: One-Deviation Principle**

To check that $\sigma^*$ is an SPE, it is sufficient to check that, following any history, no player $i$ can benefit by deviating once and then returning to $\sigma_i^*$.

**Remark (Why It Works).**

The intuition behind the one-deviation principle is the following. Suppose, for contradiction, that there exists an infinite (multi-period) deviation that is profitable. Then there must exist a finite deviation (say, deviating in $N$ periods) that is also profitable: with discounting $\delta < 1$, the contribution of periods beyond some finite $N$ to the total payoff becomes arbitrarily small, so any profitable infinite deviation can be approximated by a profitable finite one. Among all profitable finite deviations, consider the latest period in which a deviation occurs. At that period, the player must benefit from deviating once and then returning to $\sigma^*$—which is exactly a one-shot deviation. Hence, if no profitable one-shot deviation exists, no profitable multi-period deviation can exist either.

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## 11.2 The Folk Theorem

We now turn to the central question of repeated games: which payoff profiles can be sustained as SPE outcomes when players are sufficiently patient? The *Folk Theorem* provides a remarkably broad answer.

### 11.2.1 Feasible and Individually Rational Payoffs

Fix a stage game $G = (S_i, u_i)_{i=1}^n$. The image of the payoff function $u(S) \subseteq \mathbb{R}^n$ collects all payoff profiles achievable by some pure action profile.

**Definition 11.4: Feasible Set of Payoffs**

The **feasible set** of payoffs is the convex hull of the pure-action payoff profiles:

$$F = \operatorname{co} u(S) \subseteq \mathbb{R}^n.$$

Any $u \in F$ can be achieved by appropriate randomization over (or time-averaging across) pure action profiles.

To define the lower bound on what players can be forced to accept, we introduce the *minmax payoff*.

**Definition 11.5: Minmax Payoff**

Player $i$'s **minmax payoff** is:

$$v_i = \min_{s_{-i}} \max_{s_i} u_i(s_i, s_{-i}).$$

This is the lowest payoff that the other players can force player $i$ to accept, given that player $i$ best-responds. Equivalently, letting $m_{-i}$ denote the minmaxing action profile by $-i$:

$$v_i = \min_{s_{-i}} u_i(\mathrm{BR}_i(s_{-i}), s_{-i}) = u_i(\mathrm{BR}_i(m_{-i}), m_{-i}),$$

and for all $s_i$, $u_i(s_i, m_{-i}) \leq v_i$.

**Example (Computing Minmax Payoffs).**

Consider the stage game:

| | $L$ | $R$ |
|---|---|---|
| $U$ | $3, 2$ | $5, 1$ |
| $D$ | $4, 0$ | $1, -1$ |

To compute $v_1$: player 2 chooses $s_2$ to minimize the maximum payoff player 1 can guarantee. If 2 plays $L$, player 1's best response gives $\max\{3, 4\} = 4$. If 2 plays $R$, player 1 gets $\max\{5, 1\} = 5$. So $v_1 = \min\{4, 5\} = 4$, with $m_2 = L$. Similarly, $v_2 = 0$.

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**Definition 11.6: Feasible and Individually Rational Set**

The **feasible and individually rational set** is:

$$F^* = \{u \in F \mid u_i \geq v_i \ \ \forall i\}.$$

These are the payoff profiles that are both achievable and weakly preferred by every player to their minmax payoff.

### 11.2.2 Necessity: NE Payoffs Lie in $F^*$

**Proposition 11.7: NE Payoffs Are Individually Rational**

In any Nash equilibrium of $G^\delta(\infty)$, the discounted average payoff profile lies in $F^*$.

***Proof for Proposition.***

Two parts:

(i) **NE payoffs lie in $F$.** The average discounted payoff in any strategy profile is a convex combination of stage-game payoff vectors $u(s)$ for $s \in S$. Hence the average payoff lies in $F = \operatorname{co} u(S)$.

(ii) **NE payoffs satisfy $u_i \geq v_i$.** Player $i$ can always guarantee at least $v_i$ in every period by best-responding to whatever the others play. Specifically, no matter what strategy the opponents use, player $i$ can achieve at least $v_i$ in each period by playing $\mathrm{BR}_i$ to the realized $s^t_{-i}$. So in any NE, player $i$'s payoff must satisfy $u_i \geq v_i$.

Combining, NE payoffs $\in F^*$.

### 11.2.3 Sufficiency: The Folk Theorem

The converse—every payoff in $F^*$ can be sustained as an SPE for sufficiently patient players—is the celebrated **Folk Theorem**.

**Theorem 11.8: Folk Theorem**

Let $u \in F^*$. Then there exists $\bar{\delta} < 1$ such that for all $\delta > \bar{\delta}$, there exists an SPE of $G^\delta(\infty)$ whose average discounted payoff profile is $u$.

***Proof for Theorem***

We prove the result for $n = 2$ and assume that there exists a pure action profile $s$ with $u(s) = u$. (The general case requires public randomization or time-averaging across action profiles; the construction below extends naturally.)

Let $m_i$ denote player $i$'s minmaxing action against player $j \neq i$ (i.e., $m_i$ is the action that minmaxes player $j$).

Consider the following strategy profile $\sigma^*$ for each player $i$:

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1. **Period 1:** Play $s_i$.
2. **On-path play:** Play $s_i$ in period $t$ if the history is $h^{t-1} = (s, s, \ldots, s)$ (i.e., $s$ was played in every previous period).
3. **Punishment phase:** If $s$ was not played in period $t-1$ (i.e., some player deviated), play $m_i$ for $k$ periods.
4. **Resume:** After the $k$-period punishment, return to phases (1)–(2). If any player deviates from the punishment in phase (3), restart phase (3).

We will show that there exist $k$ and $\bar{\delta} < 1$ such that for all $\delta > \bar{\delta}$, the strategy profile $\sigma^*$ is an SPE.

By the one-deviation principle, it suffices to check that no player can profit from a one-shot deviation, in either the cooperative phase or the punishment phase.

**Cooperative phase.** On the equilibrium path, player $i$ receives $u_i$ in every period. The best one-shot deviation gives player $i$ a payoff of at most $\bar{u}_i := \max_{s_i'} u_i(s_i', s_j)$ in the deviation period, followed by $k$ periods of minmax payoff $v_i$, and then a return to $u_i$ forever. The deviation is unprofitable if:

$$u_i \;\geq\; (1-\delta)\bar{u}_i + (1-\delta)\sum_{\tau=1}^{k} \delta^\tau v_i + \delta^{k+1} u_i.$$

Since $u_i > v_i$ (as $u \in F^*$ and we can choose $u$ in the interior; the boundary case requires a small perturbation), one can choose $k$ large enough and $\delta$ close enough to 1 so that this inequality holds.

**Punishment phase.** During punishment, player $i$ is playing $m_i$ (minmaxing the other player). Deviating from $m_i$ does not help, because the punishment phase only restarts—player $i$ would face the same punishment period again. Formally, the continuation value from following the prescription weakly exceeds the value from deviating.

Choosing $k$ and $\bar{\delta}$ appropriately yields an SPE with average payoff $u$.

**Remark (The Folk Theorem in Words).**

The Folk Theorem states that *anything that is feasible and individually rational can be sustained as an SPE*, provided players are patient enough. This is both a celebration and a caution: while cooperation, fair sharing, and many other socially desirable outcomes can be supported, so can a vast multitude of other outcomes—including very inefficient or asymmetric ones. The theorem highlights the fundamental **indeterminacy** of repeated games: the equilibrium concept alone does not pin down a unique prediction, and selection among the many SPE requires additional considerations (such as renegotiation-proofness, evolutionary stability, or focal points).

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